Osmo360 reconstruction and viewer review

Why the latest result is blurry, lacks a dollhouse view, and feels worse to control than iPhone Fast / Lite.

Prepared for Gemini Argon (Architect)
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Audited release: 20261031-osmo-clean-v1
Website: https://rumah.shizuku.rest/

Current product verdict: rejected by the user

The delivered Osmo experience has not met the requested standard. Successful loading, retaining nearly three million Gaussians, and removing an earlier bathroom obstruction did not establish acceptable sharpness, dollhouse viewing or control quality.



Figure 1. The latest user-reported result
The user supplied this 1891 x 1240 screenshot with the complaint about blur, missing dollhouse view and frustrating controls. The original image is embedded without AI enhancement. It has no visible diagnostics, so its exact camera pose, device, LoD state and viewport scale are unknown. [E01]

This is one self-contained PDF. It includes the quality comparisons, performed processing steps, key numerical results, relevant code, an issue ledger, proposed corrective work, and questions for architectural review. No reconstruction or live-site change was made while preparing this report.

Contents and reading map

Read the findings and product comparison first. The pipeline and diagnosis sections explain what the evidence establishes. Proposed work is explicitly separated from work already performed. Code, per-view scores and provenance are included in the appendices.

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1. Executive finding and responsibility

The user is right to reject the current experience. The latest Osmo model is visibly soft, and the viewer does not offer the whole-apartment dollhouse interaction available in iPhone Fast, the current name for the preferred Lite option. These are separate failures and require separate remedies.

I changed the Osmo delivery from a textured triangle mesh to a native Gaussian radiance-field view to bypass damaging mesh extraction and support filtering. That helped preserve visible room coverage and removed an opaque obstruction at a difficult bathroom camera. I then made the Gaussian branch start inside the room, disabled mesh-specific cutaway controls, and treated preservation of an interior viewpoint as a navigation success. The resulting product lost an interaction the user expected. [E03; E19]

Requirement	Current evidence	Verdict
Overall sharpness	The supplied screenshot is blurry. Source-matched images also show detail loss before web compression.	Not accepted
Dollhouse / whole-room overview	Top view and See inside are hidden for Gaussians; native auto-fitting is bypassed.	Missing feature
Controls comparable to Lite	Same labels lead to different starting modes, orbit targets and framing behavior.	Product regression
Continuous interior coverage	Seven saved poses are more coherent and the bathroom obstruction is gone; extreme views remain rough.	Partial improvement
Download and startup	45.18 MB file; public desktop Chrome reached a visible room in 12.45 s in one test.	Technical check passed
Fold 7 / iPhone 17	Desktop viewport checks exist; no physical-device confirmation.	Unverified

Where my acceptance process failed

I placed too much weight on loading successfully, keeping the source splat count, numerical transport fidelity and fixing one severe bathroom failure. I did not require a usable dollhouse overview, mode/control parity, a full-viewport sharpness review, or reproduction of the user's freely navigated complaint view. My release record says those narrower checks passed; it is not evidence that the user experience was satisfactory. [E22; E21; E20]

> **The immediate direction**
>
> Restore the intended viewing experience and diagnose the blur with controlled comparisons. Increasing file size, point count or training time without isolating the cause is not a justified next step.

2. What is actually being compared

The website has three choices but only two capture families: iPhone Fast and iPhone Full are derivatives of the same iPhone reconstruction; Osmo is a different recording and a different representation. iPhone is the correct product-experience baseline. These captures do not form a controlled camera-hardware benchmark. Their camera positions, room coverage, exposure, reconstruction recipes and rendering representations differ. [E02; E23; E24]

Property	iPhone Fast / Lite	iPhone Full	Current Osmo
Representation	Textured GLB mesh	Textured GLB mesh	SOG Gaussian radiance field
File size	21.14 MB	435.61 MB	45.18 MB
Expected wire size	17.85 MB with gzip	302.56 MB with gzip	45.18 MB; already compressed
Geometry / primitives	1,603,164 triangles	11,891,853 triangles	2,997,908 source Gaussians
Colour representation	Two 2048 x 2048 atlases	Two 8192 x 8192 atlases	Per-Gaussian colour, SH degree 2
Initial interaction	Auto-fitted Orbit; see-through exterior	Interior Walk by default	Interior Walk by default
Overview tools	Top, See inside, fit-to-model Orbit	Top, See inside, fit-to-model Orbit	Top / See inside hidden; local Orbit
Main coverage	Kitchen, living / corridor / bedroom	Same source reconstruction	Living / dining and bathroom capture

A triangle count and a Gaussian count are not interchangeable quality measures. Keeping every source Gaussian in the download does not imply photographic detail, a watertight surface, or that every primitive is drawn on screen. The public Gaussian branch selects an LoD representation within a rendering budget. [E04]

The iPhone Lite reference is useful but not perfect

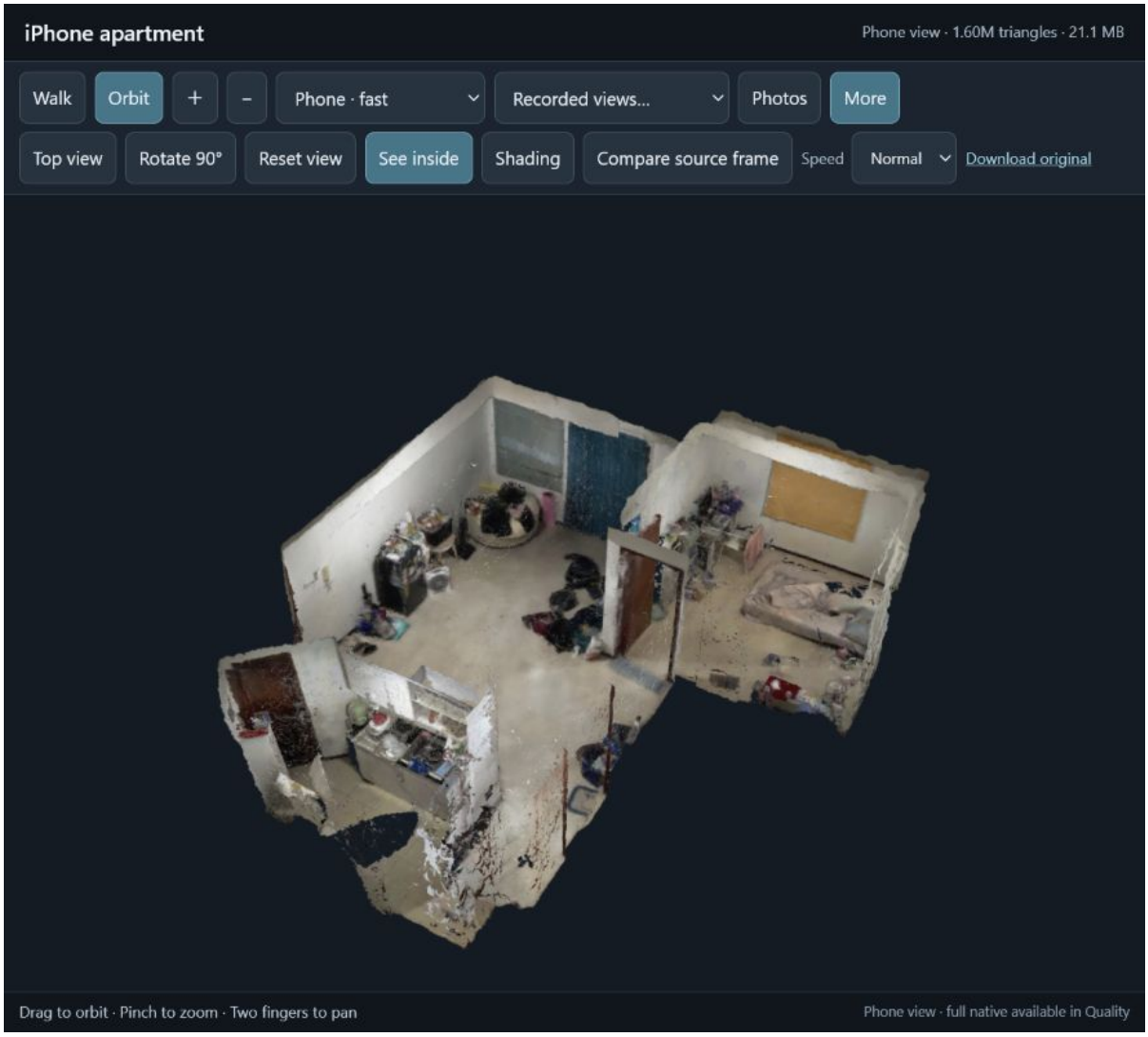


Figure 2. The preferred iPhone Fast interaction
Retained 29 October 2026 browser capture. The visible UI identifies the 21.1 MB / 1.60M-triangle phone model, with Orbit, Top view and See inside. This model still has holes and imperfect geometry. Its advantage here is usable overview and familiar interaction. The current model bytes have the same verified identity; the capture is historical, not a new physical-phone test. [E32; E26; E23]

3. Visual evidence: blur exists before compression

At the exact saved camera, the source photograph resolves stickers, packaging edges and appliance details more clearly than the reconstructed Gaussian. The original PLY render and the compressed SOG render look very similar to each other. This places substantial appearance loss upstream of web-file packing. It does not isolate which training or preprocessing choice caused that loss. [E16; E18]

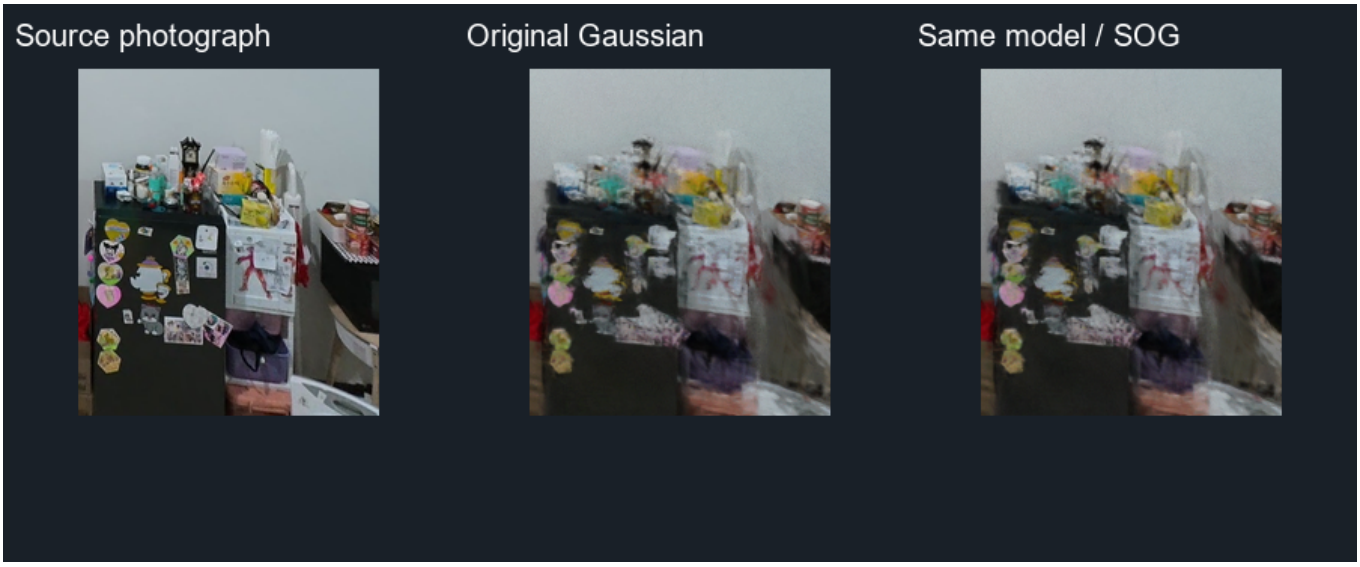


Figure 3. Matched fridge-detail comparison
Diagnostic crops use the same 600 x 600 reference grid and exact saved source camera. The 1200-pixel source crop was resized to that grid before the same rectangle was extracted. Crops are not sharpened or AI repaired. Left: source photograph. Centre: final original PLY, all-source drawing. Right: the SOG file, also all-source drawing.



Figure 4. Matched fan, floor and small-object detail
The blur and rough object boundaries remain visible in the original Gaussian, not just the compressed transport. These are same-camera appearance diagnostics; the source capture also participates in SfM pose estimation.

The latest user screenshot is a larger, freely navigated view, not one of the seven documented 600-pixel captures. Its camera and renderer state were not supplied. The controlled figures explain part of the complaint, but cannot quantify the exact error in that screenshot or attribute it to its device. [E01; E16]

4. What the quality numbers prove - and do not prove

Measurement	Observed result	Interpretation
All 188 held-out virtual views	Mean PSNR 24.10 dB; median 24.02 dB; minimum 15.57 dB.	There is real appearance error and a wide quality range. This is not a sharpness guarantee.
All valid pixels, pooled	PSNR 23.22 dB; RMSE 17.60 on a 0-255 scale.	Weighted by the number of valid pixels, unlike averaging per-view dB values.
GT/source identity check	Unique 188-to-188 image mapping; maximum full-resolution RMSE 0.234/255.	Avoids assuming eval file numbering is chronological or corresponds to a guessed camera.
Original vs SOG at seven poses	Minimum 40.06 dB; mean RGB RMSE 2.38/255.	Small packing loss in the tested renderer configuration. It does not certify all public LoD or free-view states.
Public desktop startup	12.45 s to a visible room in one Chrome / RTX 5080 session.	Delivery evidence only; not image-quality acceptance or a mobile benchmark.

The native trainer's unmasked whole-face average was about 11.74 dB. Many virtual faces contain black lens padding or excluded operator regions, while the renderer can predict the background there. The later audit applies the exact binary keep-mask and measures all 188 held-out views consistently. The masks are not chosen from pixel error. This explains why the valid-pixel result differs; neither score substitutes for the user's visual judgement. [E17]

Seven documented source cameras

Time	Appearance split	Original vs source dB	SOG vs source dB	SOG vs original dB
0.90s	train	23.98	23.97	40.64
11.91s	train	26.46	26.42	41.08
20.02s	train	28.18	28.12	41.12
25.03s	eval	25.19	25.13	40.06
39.94s	train	25.51	25.43	40.19
51.95s	eval	27.85	27.74	41.00
60.06s	train	25.84	25.79	40.28

Two of these times are held out from Gaussian appearance training. SfM uses the same capture to estimate their cameras, so these are not independent survey or geometric-ground-truth tests. A mostly plain wall can score reasonably well while providing little evidence about fine object detail. [E16]

> **An important renderer mismatch**
>
> The seven compression comparisons used a 600-square canvas, LoD disabled, extSplats enabled, linear depth sorting and SH2. The public viewer uses LoD, extSplats disabled, radial sorting and a different viewport. Therefore the 40 dB packing figure does not bound the public viewer's total visual error. [E03; E16]

5. Why the dollhouse view is missing

This is a confirmed viewer implementation gap. It is not established that the Osmo footage or Gaussian splatting inherently prevents a dollhouse presentation. The current application simply does not implement an equivalent Gaussian cutaway and whole-scene fitting path.

Exact code behavior	User-visible consequence
controlsEnabled(): hides shade, see-inside and top whenever nativeGaussian is true.	The Top view and See inside functions disappear for Osmo.
updateSurface(): the Gaussian branch forces seeInside=false and returns.	The mesh back-face-culling toggle has no Gaussian counterpart.
setMode(): native Orbit targets camera position + forward x 2.	Orbit pivots around a nearby interior point instead of the apartment centre.
fittedOrbitRadius(): immediately returns the existing radius for Gaussians.	The mesh fit-to-model framing algorithm is deliberately bypassed.
loadAsset(): auto-Orbit is enabled only when the asset is not Gaussian.	iPhone Fast opens as an overview; Osmo opens inside the room.

Source locations: room-viewer.js lines 43-49, 71-101, 268 and 386. The code comments explicitly describe preserving the interior viewpoint. That was my implementation decision. It should not have been treated as satisfying the user's whole-apartment viewing requirement. [E03]

Why mesh culling is not a drop-in Gaussian cutaway

For the iPhone mesh, See inside uses front-face rendering to hide exterior back faces. It does not apply a blanket horizontal height cut, so tall cabinets and door/window proportions are retained. Gaussian ellipsoids are alpha-composited primitives, and the mesh material-side switch is not an equivalent control. Osmo needs a deliberate, reversible visibility/cutaway design. [E03; E26]

Spark documents per-splat RGBA edits using signed-distance shapes, including planes and boxes. This gives a plausible renderer-side route to test a cutaway. It is not implemented here. The current SplatMesh is created with editable:false, and a correct design must verify edit scope, LoD interaction and structural preservation with the pinned SDK. A single global height plane risks repeating the cabinet-cutting problem. (R1) [E03]

6. Why the controls feel inconsistent

Action	iPhone Fast / Lite	Osmo
Open model	Orbit fitted to sampled mesh geometry, FOV 42 degrees.	Walk from the first interior camera, FOV 75 degrees.
Select Orbit	Centred on model bounds; oblique fitted overview.	Keeps the current interior position and uses a local target two units ahead.
Drag in initial mode	Orbits around the apartment.	Changes yaw/pitch while staying inside.
Wheel / pinch in Walk	Moves the camera through the scene.	Same Walk semantics; it is not an overview zoom.
Top / See inside	Available.	Hidden / disabled.
Reset or R	Returns to interior Walk home; it does not preserve the initial Lite overview.	Also returns to interior Walk home.
Recorded view then drag	Leaves calibrated framing for free-view FOV.	Also changes from square calibrated framing to free-view FOV 75.

Both branches share many event handlers, but their defaults and camera targets change the meaning of the same gestures. A functional movement button is not evidence of a good interaction model. The saved 350 ms button test checked delivery, motion and release in a desktop narrow viewport; it did not assess real phone gestures, comfort or discoverability. [E03; E21]

Additional control weaknesses

The camera can freely cross walls because this viewer has no collision model. Recorded views are unlocked on the first input: clearRecorded() removes reference framing and changes the FOV, which can cause an apparent zoom/framing jump. Reset is not mode-aware. These are code-confirmed behaviors or shared limitations, not measured proof of the exact action that frustrated the user. [E03]

Scene framing metadata also needs correction

The Osmo bounds were exported as the minimum and maximum of all finite sparse SfM points. Their vertical span is about 33.03 normalized scene units, while the saved interior review-camera heights include about 0.46 to 1.54. This is a poor starting point for a future whole-room fit and needs robust occupied-scene bounds. Current native auto-fit is bypassed, however, and walking speed is a fixed numeric value. The oversized bounds are not proven to be the cause of current walking speed. [E06; E05; E03]

7. Pipeline comparison: iPhone and Osmo

The iPhone options are not independent reconstructions. Fast / Lite is a delivery derivative of the full iPhone model. Its lower file size comes from geometry simplification, texture resizing and packing. The user's preferred dollhouse behavior comes from the mesh viewer branch. [E23; E24]

iPhone reconstruction and Lite delivery

01 Perspective source video

1080 x 1920 portrait source; about 156.9 seconds.
Retained source record: 4705 decoded frames.

↓

02 Calibrated reconstruction components

501 original camera poses; two original SfM components.
Kitchen and shared living / corridor / bedroom frame.

↓

03 Recovery geometry and training

Fresh depth/normal inputs and corrected dense seeds.
40k kitchen pilot; 50k living component, then all-usable-view correction.

↓

04 Native mesh extraction and assembly

Selected 0.5 merge / default grid 512 extraction recipe.
One consistent similarity per original component; shared living/bedroom.

↓

05 Full native delivery

11,891,853 triangles, 435.6 MB, two 8K textures.
No post-extraction simplification in the selected full delivery.

↓

06 Fast / Lite derivative

1,603,164 triangles; two 2K textures; 21.1 MB GLB.
Geometry quantization: position 16, normal 8, UV 12 bits.

↓

07 Mesh viewer

Whole-model Orbit fitting, Top, back-face See inside.
A familiar architectural overview, with remaining source/model defects.

The iPhone recovery also has limitations: approximate rather than surveyed scale, uncertain inter-component alignment evidence, and local mesh defects. Its comparison photos were not all held out from its final all-usable-view training. Therefore its reports cannot be used as a fair held-out PSNR ranking against Osmo. [E24; E25; E27]

Current Osmo pipeline

01 Original dual-fisheye OSV

Two 3840 x 3840 HEVC 10-bit streams, 1955 frames per lens.
65.2318 seconds at 30000/1001 fps; original OSV retained.

02 Adaptive selection and masks

226 synchronized pairs / 452 native JPEG images.
Per-frame operator masks and whole-mirror exclusions.

03 Fresh native-resolution SfM

Native fisheye features / matching; one connected component.
452 registered images; one +Y-up normalization.

04 Materialized calibrated pinhole views

2219 virtual 1200-square views; 90-degree pinhole fields.
2031 train / 188 eval; splits grouped across lens and timestamp.

05 Validated learned geometry and seed

768-square inferred depth/normals; positive scale, zero offset.
439 cross-view-qualified maps; 1,381,699-point dense seed.

06 Cold-start Gaussian optimization

50k steps, 3M cap, Mip primitive, SH2.
Conservative position/noise updates and strong floater suppression.

07 SOG and interior viewer

491.66 MB PLY -> 45.18 MB SOG, source count retained.
Public LoD budgets 2M desktop / 750k phone; no dollhouse path.

This branch directly delivers Gaussians; it does not run the old mesh carving/support filter. Bypassing that stage avoided one class of destructive mesh holes, but did not deliver the architectural surface and cutaway experience the user expected. [E08; E03]

8. Video, selection and resolution: exact current state

Stage	Actual resolution or count	Limit / interpretation
Original recording	2 x 3840 x 3840, HEVC 10-bit, 65.2318 s.	Source is preserved. LRF preview media is not the reconstruction input.
Working source JPEGs	452 images at 3840 x 3840.	8-bit JPEG decode; not lossless 10-bit appearance training.
Selection	652 candidate pairs, 226 selected pairs.	Adaptive sharpness and optical image motion; not measured translation or guaranteed parallax.
Temporal density	Average about 3.46 selected pairs/s; largest gap 0.7007 s.	Not uniform sampling; 226 distinct times, not 2219 independent viewpoints.
Gaussian RGB training	1200 x 1200 virtual views, focal length 600 pixels.	Native-resolution SfM does not mean native-resolution Gaussian RGB supervision.
Learned maps	Actual stored depth/normal images: 768 x 768.	Geometry guidance resolution; RGB training remains 1200 square.
Matched visual QA	600 x 600 all-source renders.	Useful controlled evidence, but insufficient alone for a large-viewport sharpness judgement.

The selection compares joint lens sharpness and image motion and retains synchronized pairs. About 11.6% of source times were used. A high candidate count, more virtual faces or more pixels processed does not create new physical baseline. Doorway overlap, stable calibration and actual translation remain important. [E14; E10]

A more honest way to describe the resampling

The current front pinhole spans 90 degrees across 1200 pixels. With the recorded native fisheye calibration, the same horizontal angular interval spans about 1704 native image pixels, and the local centre sampling ratio is about 1.75 native pixels per virtual pixel. The ratio varies over the fisheye projection. It is incorrect to compare 3840 and 1200 as if they covered the same rectangular field of view, but there is still meaningful resampling of appearance information. [E10]

This is a confirmed input-resolution choice, and a plausible contributor to loss of small detail. It is not a proven single cause of all the blur. The original PLY is also affected by learned geometry, optimization, primitive size, per-view appearance modelling and coverage. No resolution-only ablation was completed.

9. Masks, calibration and sparse reconstruction

The rebuild used true per-selected-frame neural segmentation of operator regions, additional hand/body prompts and reviewed manual corrections. Whole mirror-glass polygons exclude reflected people and virtual geometry. Source binary masks use 255 for keep and 0 for exclude. The native-mask builder applies an operator margin of 32 pixels and a mirror margin of 16 pixels. Reprojection into virtual views uses nearest-neighbour sampling. [E15; E10]

Mask readiness covers 452 native images. Earlier review work included 44 manual pinhole repairs, rejected static-scene false detections and mirror annotations across both bathroom visits. This did not establish perfect hair or body-boundary exclusion. The lower-scoring native eval views still show rough fringe regions; some source views have only a small amount of valid background. A blanket larger mask can remove legitimate background support and is not automatically an improvement.

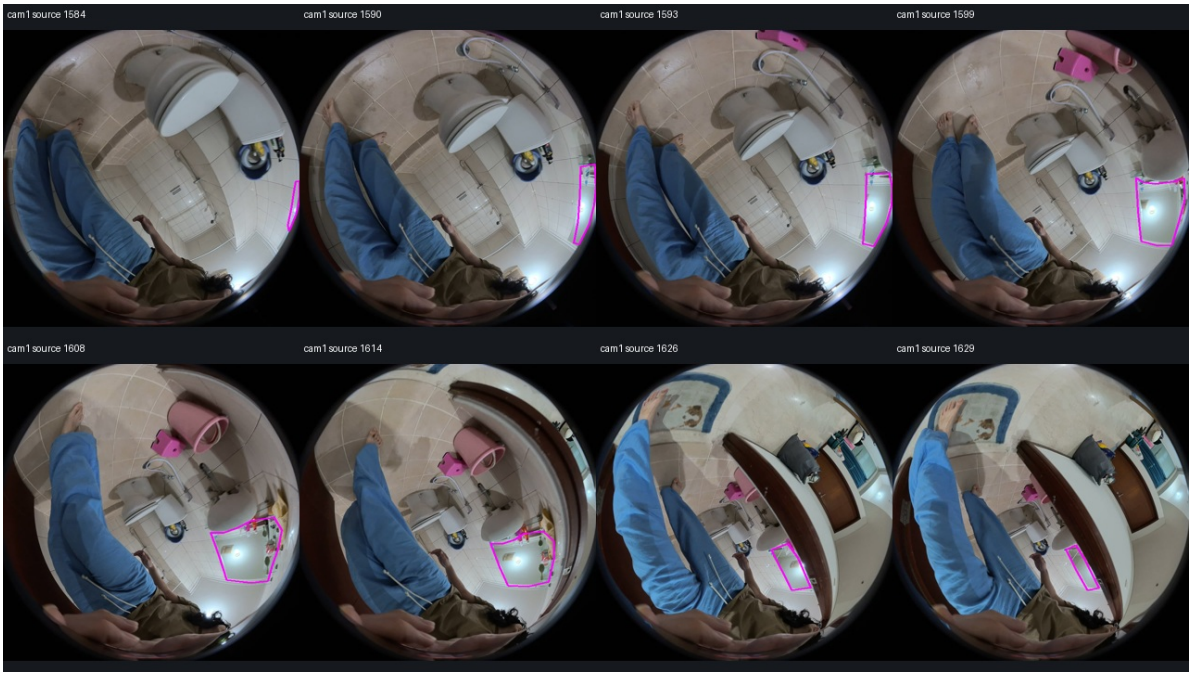


Figure 5. Retained mirror-mask review evidence
This is a dated processing diagnostic, not a final-model quality certificate. Mirror and operator exclusions protect pose/geometry estimation, but do not reconstruct unseen reflected content or guarantee every boundary is clean.

Calibration and registration findings

The current runtime is the isolated Spirula 2026.10.30 build. Factory KB5 calibration was approximated by the native KB4 fisheye model with recorded native-pixel residuals around 1.21 / 1.09 maximum and 0.254 / 0.228 RMS over the audited domain. The optimized relative lens translation is nonzero, about 0.02097 normalized scene units. It is not a surveyed physical lens separation. The initial zero translation was an initialization, not proof that the final rig had zero baseline.

Fresh ALIKED / LightGlue matching used the masks at native 3840-square resolution. A 3-pixel PnP threshold fragmented registration. Increasing the native-pixel threshold to 6, with strong inlier requirements and cached features/matches, produced one component containing all 452 images and 226 synchronized pairs. The selected model has 552,160 sparse points. Independently recomputed mean source reprojection error is about 1.405 pixels, with 95th percentile 3.625 pixels. [E13]

A single +Y-up similarity was then applied to points and cameras. The transform scale is about 2.98175, and it establishes a consistent normalized frame, not measured metres. Explicit pinhole reprojection later checked about 1.59 million observations with mean error about 0.967 virtual pixels. These checks establish camera/coordinate consistency; they do not certify photographic sharpness or all reconstructed surfaces. [E13; E10]

10. Depth validation, dense seeding and optimization

How relative learned depth was accepted

MoGe-2 inference ran on the explicit pinhole images. Per training view, a positive scale s maps relative Euclidean ray range into the normalized SfM frame: depth_scene = s x depth_raw. The additive offset is exactly zero. Qualified SfM anchors require observations across at least three distinct training timestamps, stored error at most 1.5 native pixels and actual virtual reprojection at most 2.5 pixels. Scale fitting requires at least 40 observations, 70% within a 5% residual band, median residual at most 3%, 90th percentile at most 10%, and support in at least five cells of a 4 x 4 image grid. [E11]

440 views passed the scale gate. A target pixel then survives only if it agrees within 5% with depth from at least two other distinct training timestamps. Evaluation images, depths and colours do not participate in this guidance/seed stage. The outcome was 439 trusted maps and 119,266,081 trusted pixels. Normal guidance was restricted to these validated pixels. [E12]

The depth PNG convention is uint16 / 1000 = normalized scene units. The factor 0.001 is an encoding scale, not a claim that this Osmo reconstruction was measured in metres. Combining sparse train-only seeds with validated dense candidates produced 1,381,699 seed points after voxel reduction. The recorded seed voxel size is 0.0219893 scene units; it is an initialization choice, not the final model's guaranteed spatial resolution. [E12]

The exact current training choices

Setting	Executed value
Iterations / refinement stop	50000 / 35000
Splat cap	3,000,000
Initialization	Fresh dense seed; no resume checkpoint and no old init PLY
RGB views	2031 explicit 1200-square train views
Primitive / appearance	Mip / spherical harmonics degree 2
Depth guidance	Weight 0.05; trusted ray depth; encoded-unit factor 0.001
Normal guidance	Weight 0.01; only trusted pixels
Surface regularizers	Normal 0.04; depth distortion 0.01; normal distortion 0.01
Floater / size policy	strong suppression; maximum world size 0.12
Position learning	means LR 0.000064 -> 0.0000008
Noise schedule	16 -> 0.16
Frame / resolution	No additional runtime pinhole warp; no progressive resolution; no automatic pose scaling

Training completed 50,000 steps in 30 minutes 46.928 seconds, producing 2,997,908 exported Gaussians. The conservative recipe was chosen to avoid floaters and an opaque shell. Its effect on fine-detail recovery has not been isolated. Strong surface regularization, conservative position updates, seed density and Mip filtering are plausible trade-offs to test, not established individual root causes. Mip filtering also controls aliasing; simply disabling it may make the result noisier. (R4) [E07; E08]

11. Export, compression and the actual display path

The final 491,657,919-byte floating-point PLY was packed to a 45,177,102-byte SOG file using @playcanvas/splat-transform 3.10.3. A temporary header-only Mip declaration was added for packing; the source PLY was not modified. Export checks verify the Gaussian count, SH degree and antialiased model tag. SOG is a quantized transport, not byte-for-byte preservation of every float attribute. [E18]

Layer	Private fidelity review	Public website
Input	Original PLY and SOG separately	SOG only
Source count	2,997,908 in both	2,997,908 in file
LoD	Disabled; all-source draw	Enabled; 2M desktop / 750k phone target
Extended splat storage	extSplats:true	extSplats:false
Sorting	Linear depth	Radial
Blur options	preBlurAmount 0; blurAmount 0.3	Same nominal blur values
Image size	600 x 600; pixel ratio 1	Responsive canvas; DPR capped 1 phone / 1.25 desktop
Starting view	Calibrated reference camera	Interior Walk home; free-view FOV 75

The phone draw budget is about 25% of the file's primitive count and the desktop budget about 67%, but LoD selection is not a uniform permanent deletion of that percentage. It can choose coarser representations. The whole SOG is still downloaded and an LoD tree is prepared in the browser. Therefore this policy does not by itself bound transient decode/build memory. [E04; E21]

Spark documents a memory/precision trade-off between packed and extended splat storage, and offers prebuilt LoD trees and paged RAD streaming. These are relevant prototype options, not completed improvements. Their total bytes, retained appearance and real-device peak memory must be measured. The current phone budget was manually chosen and is more restrictive than some SDK platform defaults; no physical-device quality test justified it. (R2)

12. What the rebuild improved, and what it did not

The earlier fresh normals-only full run reached 50k steps but produced an opaque obstruction at the original 51.9519-second bathroom camera. The same issue was independently visible in its native evaluation output, so it was not only a browser problem. A controlled removal of 319 visibility candidates did not materially improve it and was rejected. Those results were not published as the final clean reconstruction.

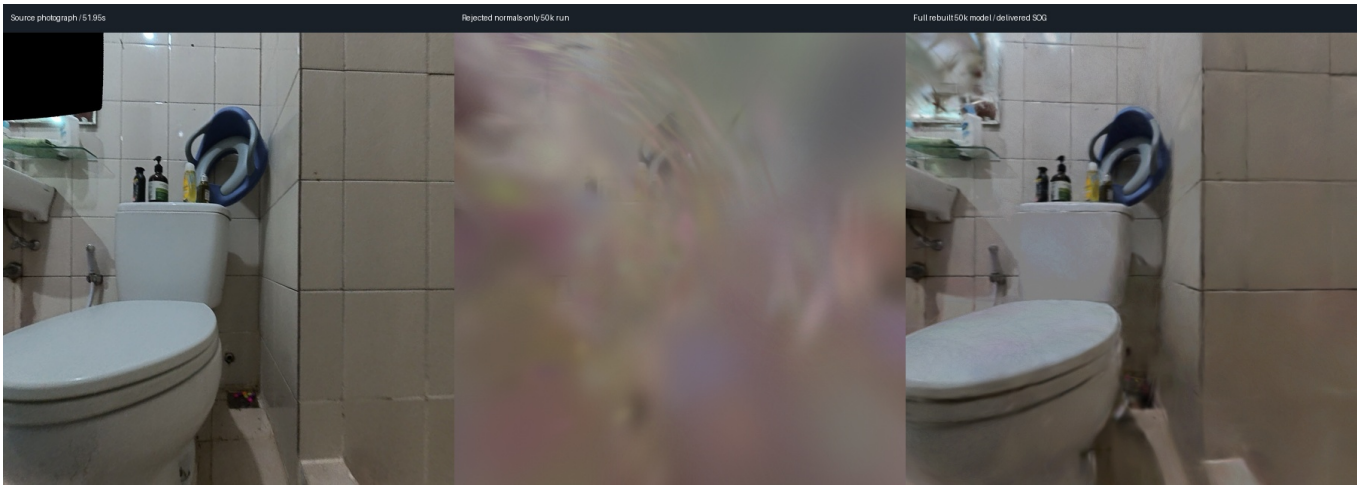


Figure 6. A genuine local improvement with a limited scope
Left: source photograph. Centre: rejected 50k normals-only rebuild. Right: the new 50k full rebuild shown from the same original camera. The obstruction is removed. That is meaningful progress, but the new result still has softer detail than the photograph and this camera does not test dollhouse usability. [E30; E31; E16]

An explicit-pinhole, scale-only, cross-view-validated bathroom pilot passed before the full rebuild. The full capture then used that combined workflow from step zero. Materializing camera views, validating geometry, densifying the seed, restricting normals and changing optimization were combined interventions. I did not run a single-variable causal study. It would be incorrect to assign all improvement, or all remaining blur, to one of them.

The new workflow did not implement Gaussian cutaway/overview, a clean architectural surface proxy, a display-fidelity audit across LoD and packed-storage variants, or a user-accepted navigation review. It also did not fix the old C++ mask-drop/per-vertex mesh-carving behavior; it bypassed mesh extraction for this delivery. [E03; E19]

13. Issue ledger: confirmed facts and hypotheses

Issue	Evidence / confidence	Practical consequence
Missing dollhouse	Confirmed in viewer branches and hidden controls.	Viewer implementation work is required.
Original model is soft	Confirmed in matched original-Ply/source images.	Web packing alone cannot recover the missing detail.
Extra public rendering reduction	Confirmed LoD, packed storage and DPR caps. Exact contribution unmeasured.	Need same-pose, same-viewport display ablation.
Gaussian RGB resampling	Confirmed 1200-square supervision; native fisheye data is denser near the front-view centre.	A plausible fine-detail bottleneck, requiring an isolated test.
Regularization / seed / position trade-off	Confirmed settings; causal blur attribution remains a hypothesis.	Do not blindly increase iterations or disable every constraint.
Operator/mirror boundaries	Reviewed masks exist; rough fringe regions remain in some views.	Inspect leaks and false exclusions with coverage checks.
SfM framing outliers	Confirmed raw sparse min/max bounds. Current fit is bypassed.	Repair bounds before adding global auto-fit.
Off-path consistency	Latest screenshot pose unknown; extreme native views are weaker.	Reproduce the complaint view and assess a short camera path.
Phone compatibility / memory	Actual Fold 7 / iPhone 17 tests are absent.	Desktop/narrow-layout success is insufficient.

Other unresolved candidates include lens-rim calibration error, remaining rolling-shutter/stabilization effects, varying exposure and per-view appearance optimization. The retained evidence does not establish any one of these as the dominant current failure. They should not become unsupported explanations that displace the confirmed viewer defects or source-resolution choices.

14. Recommendations and a bounded plan

Stage A - restore the expected product behavior

Make dollhouse / overview the Osmo entry experience, matching the preferred iPhone Lite interaction. Use robust occupied-room bounds and an explicit global orbit target. Provide Top, fit/reset, rotate and an interior Walk option with consistent meanings across models. Make Reset mode-aware and avoid an abrupt FOV jump when leaving a calibrated preset. These are proposed changes; this report does not deploy them.

Prototype reversible Gaussian ceiling/exterior visibility using the renderer's supported edit mechanisms, or a lightweight architectural mesh proxy with the Gaussian used for interior appearance. Preserve the full-height cabinet, window and door geometry. Select structural ceiling regions by validated geometry/room regions; do not revive a blanket low horizontal cut. Test the clipping behavior under LoD because a coarse primitive can span a cut boundary. (R1)

Stage B - isolate display loss before another training run

Controlled comparison	Keep fixed	Question answered
All-source original PLY vs all-source SOG	Exact camera, viewport, SH, sorting and splat storage.	Transport loss at the actual complaint scale.
SOG with extSplats on / off	LoD off; same camera and viewport.	Runtime packed-storage error separate from SOG packing.
All-source vs 2M vs 750k LoD	Same file, camera, DPR and sort.	Detail cost of the deployed draw budgets.
DPR cap / viewport change	Match field of view and crop location.	Display-resolution loss, not camera movement.
Sorting / filter option change	Same points and projection.	Rendering contribution; avoid disguising geometry failure.

These tests should use the latest screenshot's reproduced camera if available, plus the fridge, floor/fan, bathroom and doorway views. Save full-viewport images and consistent detail crops. Keep a controlled metric for comparison, but require a visible improvement rather than a primitive-count certificate.

Stage C - a small, causal processing experiment

Freeze the audited masks and cameras for the first comparison. On a representative living/bathroom subset, compare 1200-square training with a higher angular sampling target, while holding the primitive budget and optimization recipe constant. Confirm that the higher-resolution input actually supplies additional source detail, rather than upsampling the same data. Only then vary the strongest regularization/position constraints one group at a time. Use a fresh initialization for each variant; preserve the baseline and exact commands.

Inspect mask boundaries and lens-rim views separately. A larger operator mask should be judged by both reduced foreground leakage and retained background support. Evaluate views excluded by time, all tested paths and the latest user view. Stop the full-scene rerun cycle until a small pilot gives a clear visual gain.

Stage D - choose the right architectural representation

If a Gaussian cutaway alone cannot give a coherent overview, test a surface-oriented or hybrid representation. The 2DGS research explicitly targets surface consistency and mesh reconstruction. It is a candidate for a controlled prototype, not a method already run here or a guarantee of success on this footage. A validated-depth mesh/planar proxy also needs foreground-aware fusion and hole/occlusion checks. The old circular distance-to-holed-probe filter must not become its acceptance rule. (R3)

Stage E - real-device sign-off

Test the selected final viewer on the actual Fold 7 and iPhone 17, including cold loading, visible progress, decode memory, touch orbit/pinch/pan, cancel/retry, orientation changes and background/return. Prebuilt/paged LoD is worth evaluating if it improves startup and memory while respecting the total download budget. Physical-device results remain pending; the user previously said they would check later. This report does not ask them to repeat that answer. (R2)

15. Proposed acceptance criteria

These are proposed engineering gates for the next implementation. They are not claims about completed work, and numerical limits should support, not replace, the user's acceptance of the experience.

Requirement	Concrete review	Pass condition
Dollhouse first	Open Osmo; Top; orbit; rotate; reset.	A coherent whole-room overview is immediately available; controls match the Lite baseline.
Structural proportions	Kitchen cabinet, door/window frames and ceiling transition.	Cutaway removes the obstructing shell without cutting tall furniture or shortening openings.
Sharpness	Reproduced complaint view and fixed full-viewport detail crops.	Visible source-to-render gap is reduced; no merely larger file or misleading small screenshot.
Navigation continuity	Switch Walk / Orbit; leave preset; zoom/pinch; reset.	No unexplained framing jump; predictable target and speed; no accidental stuck input.
Coverage	Both bathroom visits plus a continuous path, floors and oblique doorway views.	No return of the opaque shell, large false sheets or new unsupported holes.
Delivery / quota	Bare URL, all three choices, cancel and failed response.	No 3D resources before Open; visible failure/recovery; no automatic fallback download.
Budget	Measure actual packaged data and optional proxy/LoD files together.	Approximately 50 MB target is respected, or a concrete exception is explained before delivery.
Phones	Actual Fold 7 and iPhone 17.	Measured cold loading and controls pass on those devices; desktop emulation is labelled separately.

16. Questions for Gemini Argon / Architect

> **Requirements already resolved by the user**
>
> The user wants an experience comparable to iPhone Lite: sharper output, a dollhouse overview and usable controls. Around 50 MB is acceptable. The task is to choose the implementation, not to ask whether those requirements matter.

Architectural question	Evidence / decision needed
Cutaway strategy	Renderer-side structural opacity edits, a surface proxy, or a hybrid? Define ceiling/wall selection and furniture preservation, including LoD behavior.
Framing and shared controls	Specify robust occupied bounds, global orbit target, overview camera and mode-aware reset. Preserve pose/projection continuity.
Primary blur experiment	Which minimum experiment best separates 1200-pixel angular sampling, seed constraints, regularization, appearance optimization and runtime LoD/packing?
Geometry backend	Can current validated depths support a coherent surface proxy, or should a surface-oriented Gaussian method be piloted? How will unobserved surfaces and masks be handled?
Mobile delivery	Would prebuilt or paged LoD improve real-device behavior without unacceptable total bytes or loss of the overview? Define a quality budget based on physical tests.
Evidence still needed	Recover the complaint screenshot's camera and renderer settings; define independent withheld-time/path views and an appropriate appearance/geometry check.

My recommended order is viewer parity first, matched display ablations second, a small processing experiment third, and full-scene rebuilding only after that pilot demonstrates the intended quality. This reduces the risk of spending another long run on the wrong failure mode.

17. Historical Nikon and Samsung comparison

The original three-camera comparison is retained here because it explains the user's expectation. The Nikon entry is specifically the Codex 034 version, not another person's reconstruction. Nikon and Samsung are no longer active projects; these are retained, dated evidence snapshots, not newly loaded models or current-site choices. [E28]

Historical model	Recorded result	What it can establish
Nikon Z8 - Codex 034	576,595 triangles; 41,979,020 bytes; local cooker/counter scope. Source working frames recorded as 8256 x 4644 RAW developments.	It was itself rejected for fragmentation and holes. It cannot be presented as an unqualified successful Nikon reference.
Samsung accepted release	Roofless model: 8,569,710 triangles / about 419.8 MB. Source recorded as 2160 x 3840, 7,964 frames / 132.91 s.	Useful coherent-room / dollhouse reference, but much larger, different coverage and processing; defects remained.
Current iPhone Fast	1,603,164 triangles / 21.1 MB, derived from the later recovered iPhone model.	The preferred low-bandwidth product baseline. It still contains local reconstruction defects.

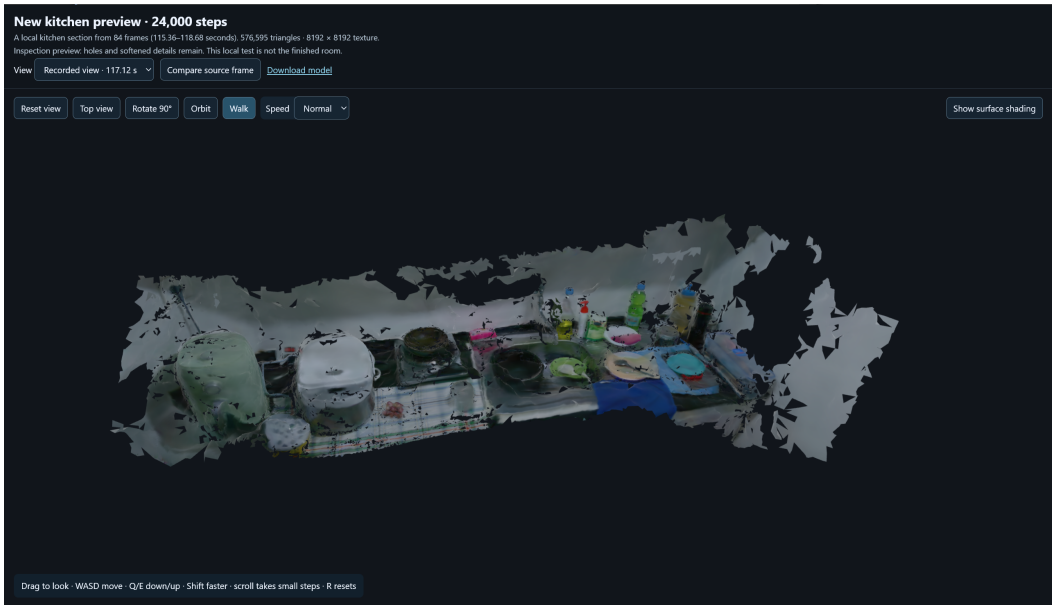


Figure 7. The Codex Nikon 034 reconstruction
Retained original report image. This is the requested Codex version and only a local kitchen section. Its identity and limitations are preserved; it is not substituted with somebody else's better Nikon result. [E35]

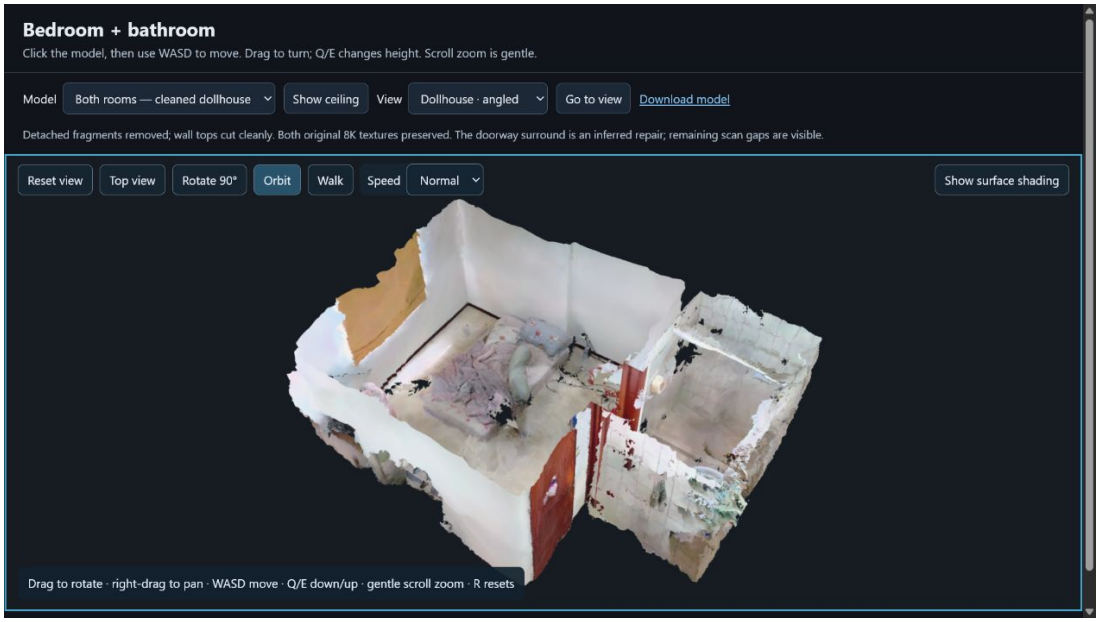


Figure 8. The retained Samsung dollhouse reference
Accepted historical view with more retained mesh geometry. Image/date/model differences prevent an equal-budget, same-scene camera ranking. The comparison is useful for expected overview behavior. [E36]

18. Delivery status versus product acceptance

The release is technically live at rumah.shizuku.rest in Proxmox LXC 710. Model selection remains explicit, with iPhone Fast, iPhone Full and Osmo. The public pre-choice access-log check recorded only the landing request; one 45,177,102-byte Osmo GET followed the explicit Open action. The iPhone raw/gzip hashes and inherited inodes were preserved. [E20]

A first promotion automatically rolled back because Cloudflare appended its managed robots.txt preamble beyond a strict authored-byte budget. A robots-only exact-prefix verifier fix passed six checks; unchanged site/model files were rehashed and re-promoted. That resolved a deployment-check mismatch. It is unrelated to reconstruction blur or the missing dollhouse feature.

26 viewer tests and 21 packaging/deployment tests passed before publication. They checked quota gating, format decoding, timeouts, first draw, cancellation and file preservation. They did not require good dollhouse composition, comparable control feel or acceptable large-viewport sharpness. The user's rejection supersedes the earlier product-quality implication of the release sign-off. [E22; E21; E20]

> **Honest current status**
>
> The model is deployed and loads. The product experience is not accepted. The next action should be a targeted viewer/design and image-quality correction, using the evidence and proposed gates in this PDF.

Appendix A. All seven source-view comparisons

These are retained real browser renders of the final SOG, with all-source drawing in the controlled review renderer. The source photographs are actual calibrated crops with operator/mirror exclusions. They are not generated images. The public viewer uses a different LoD/storage configuration, detailed in Section 11. [E16]

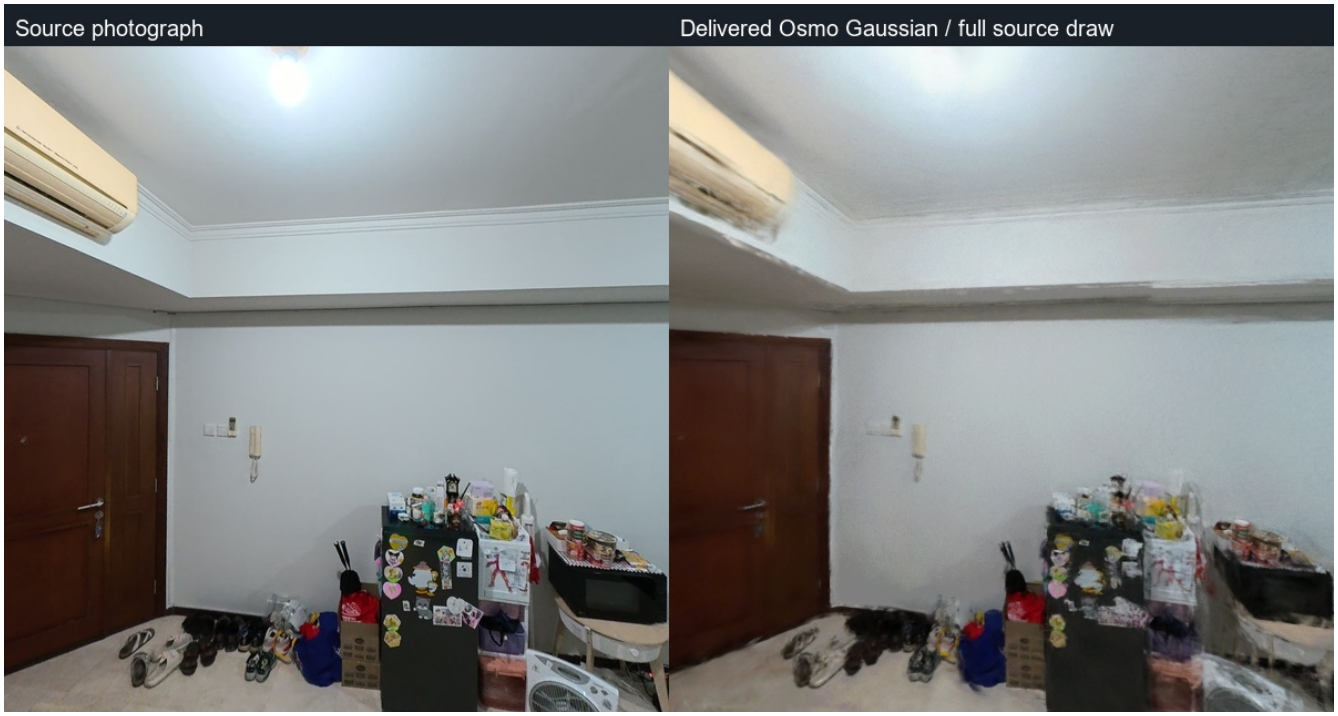


Figure 9. View 1: 0.90 seconds / train
SOG vs source: 23.97 dB. SOG vs original Gaussian: 40.64 dB. The photographs and renders are aligned at the same registered source camera; this is same-capture appearance evidence, not surveyed geometry.

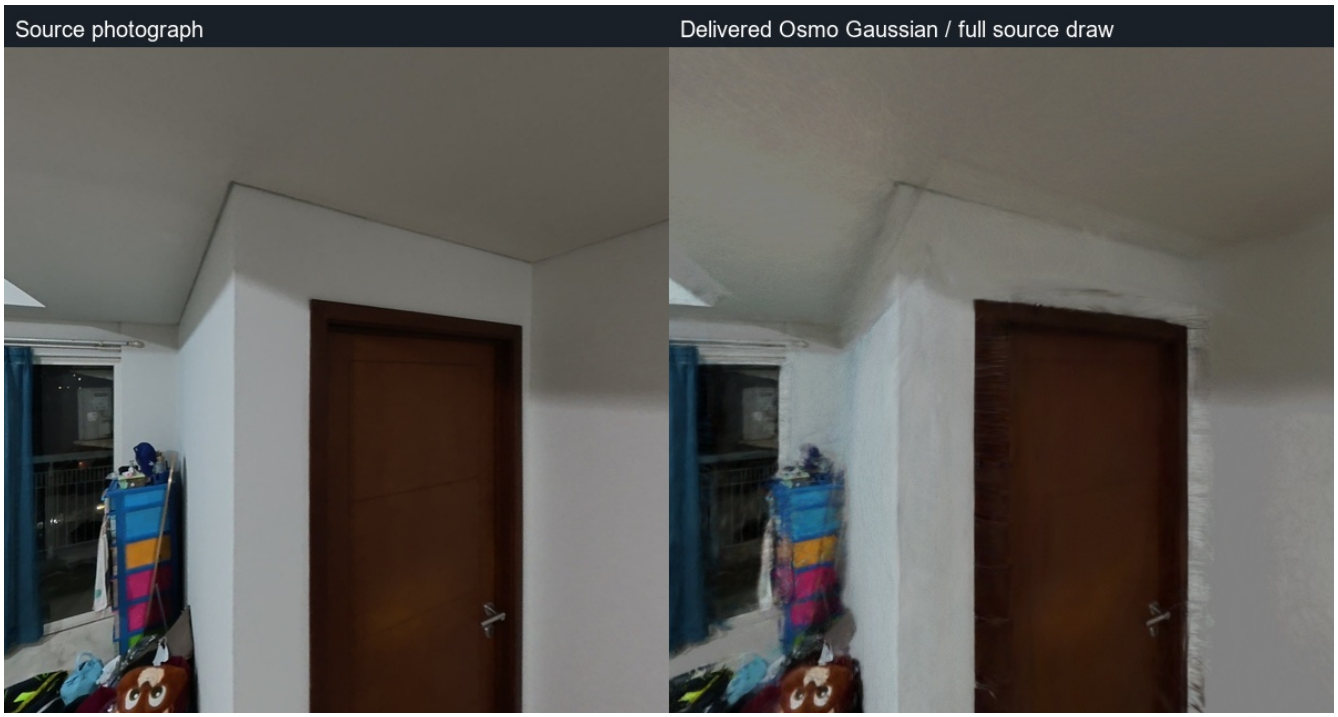


Figure 10. View 2: 11.91 seconds / train
SOG vs source: 26.42 dB. SOG vs original Gaussian: 41.08 dB. The photographs and renders are aligned at the same registered source camera; this is same-capture appearance evidence, not surveyed geometry.



Figure 11. View 3: 20.02 seconds / train
SOG vs source: 28.12 dB. SOG vs original Gaussian: 41.12 dB. The photographs and renders are aligned at the same registered source camera; this is same-capture appearance evidence, not surveyed geometry.

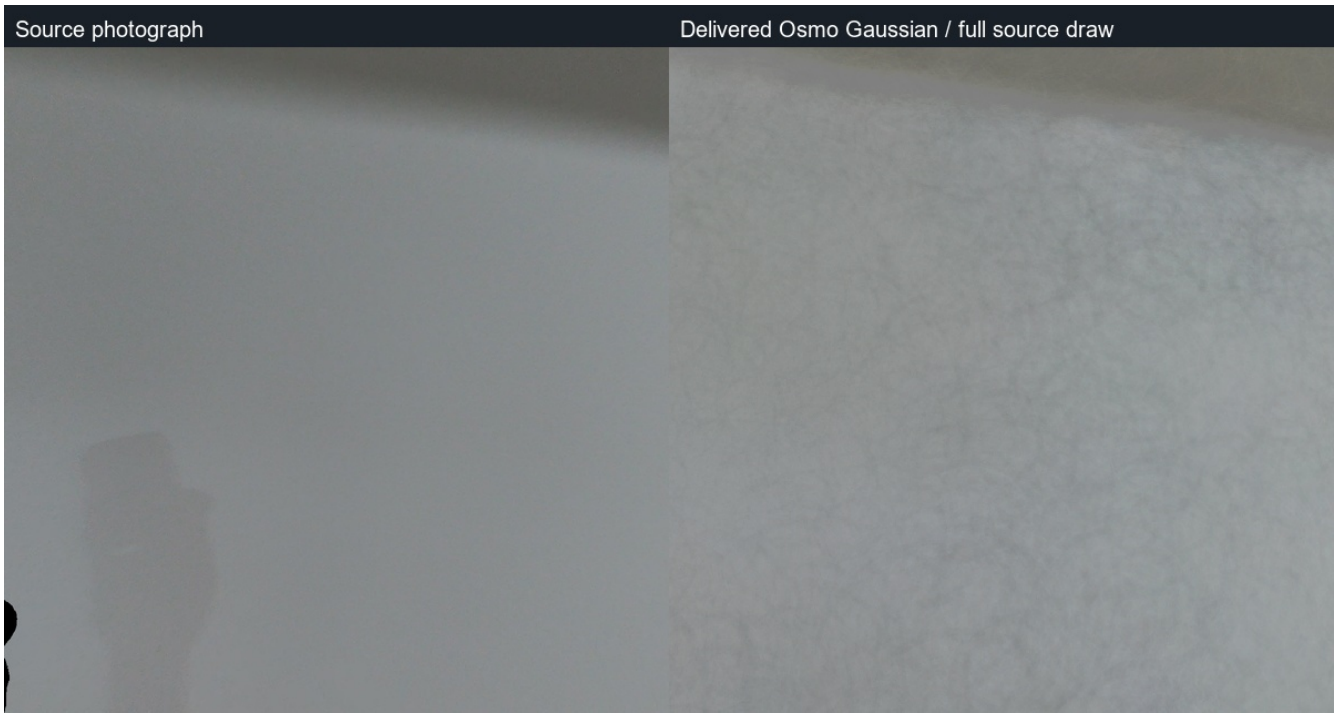


Figure 12. View 4: 25.03 seconds / eval
SOG vs source: 25.13 dB. SOG vs original Gaussian: 40.06 dB. The photographs and renders are aligned at the same registered source camera; this is same-capture appearance evidence, not surveyed geometry.

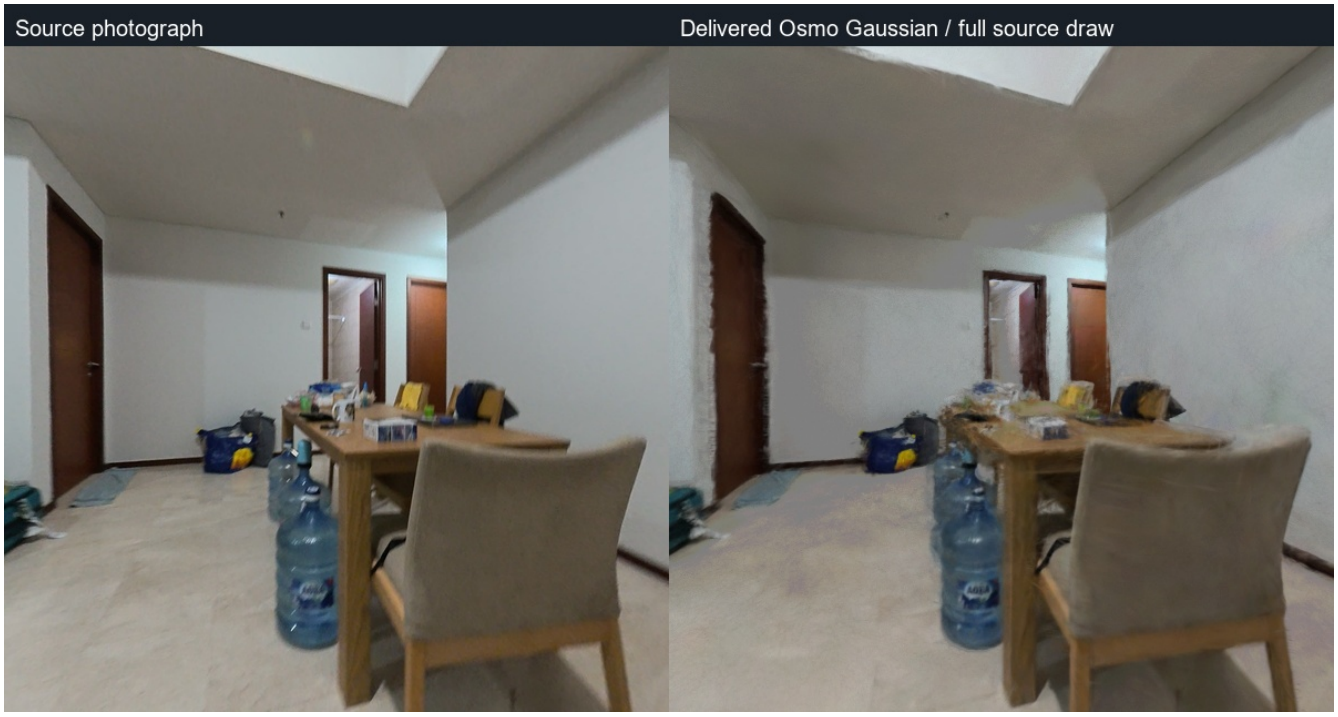


Figure 13. View 5: 39.94 seconds / train
SOG vs source: 25.43 dB. SOG vs original Gaussian: 40.19 dB. The photographs and renders are aligned at the same registered source camera; this is same-capture appearance evidence, not surveyed geometry.

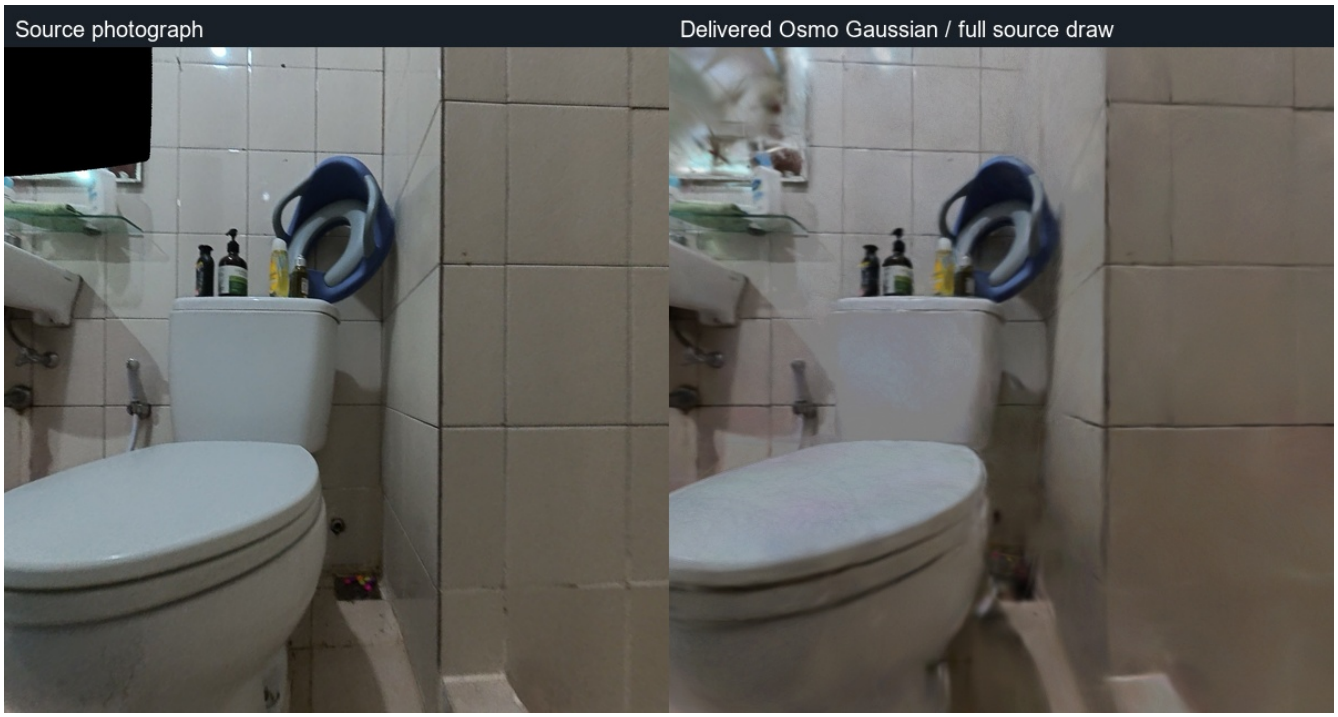


Figure 14. View 6: 51.95 seconds / eval
SOG vs source: 27.74 dB. SOG vs original Gaussian: 41.00 dB. The photographs and renders are aligned at the same registered source camera; this is same-capture appearance evidence, not surveyed geometry.



Figure 15. View 7: 60.06 seconds / train
SOG vs source: 25.79 dB. SOG vs original Gaussian: 40.28 dB. The photographs and renders are aligned at the same registered source camera; this is same-capture appearance evidence, not surveyed geometry.

Appendix B. Lowest-scoring held-out views

The six lowest valid-pixel PSNR results were selected automatically from all 188 held-out virtual views, then visually reviewed. This avoids showing only the clearest saved cameras. Many low-score views are oblique, near the operator or have limited visible background. Predicted content in black excluded regions is not scored. [E17]

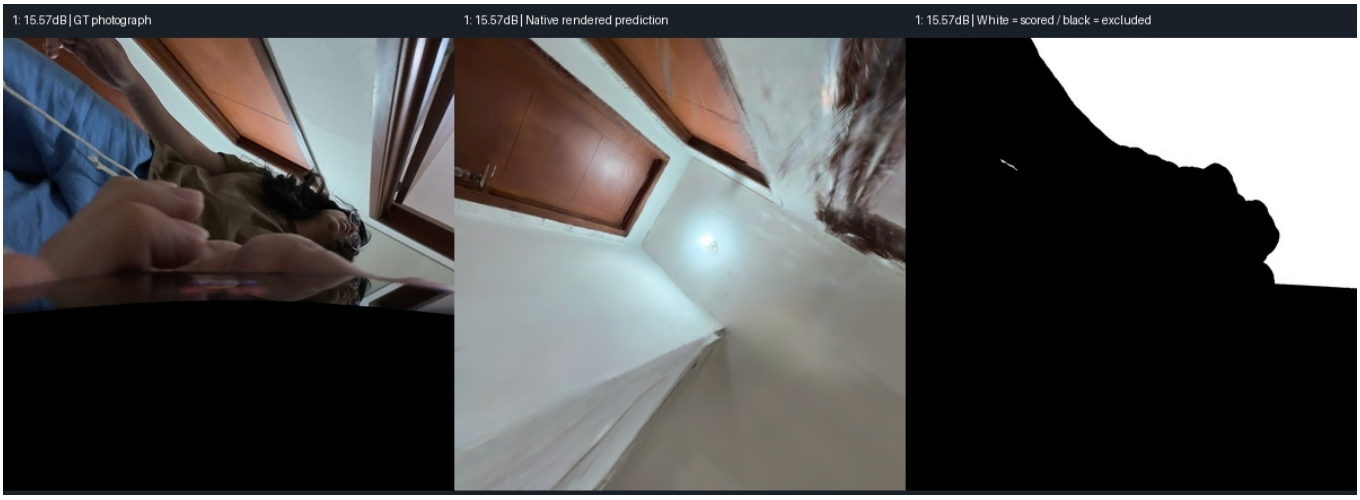


Figure 16. Native eval 1: 15.57 dB
cam1-n001467-f3_eval.jpg; valid image fraction 22.9%. Left: full GT photograph. Centre: native prediction. Right: the exact scoring keep-mask. No scored pixels were selected by their reconstruction error.

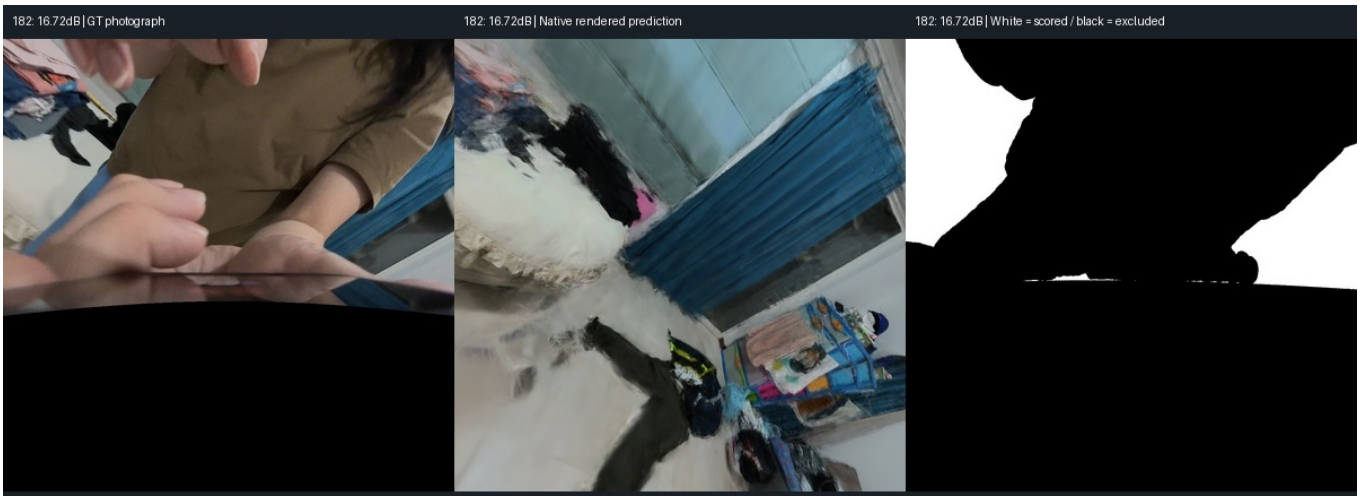


Figure 17. Native eval 182: 16.72 dB
cam1-n001926-f3_eval.jpg; valid image fraction 13.8%. Left: full GT photograph. Centre: native prediction. Right: the exact scoring keep-mask. No scored pixels were selected by their reconstruction error.

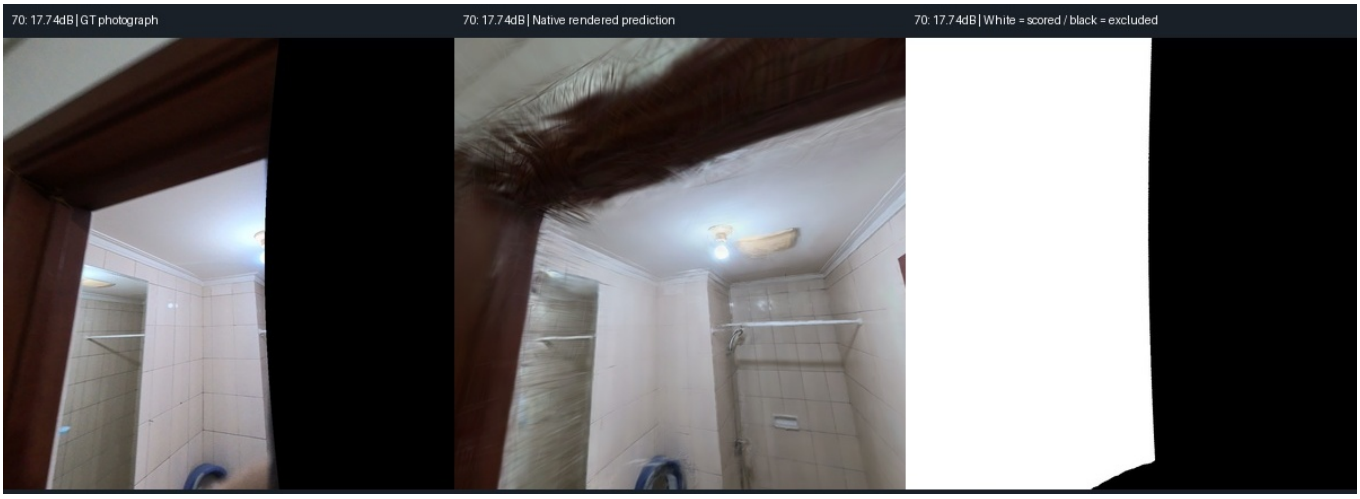


Figure 18. Native eval 70: 17.74 dB
cam0-n000552-f1_eval.jpg; valid image fraction 53.8%. Left: full GT photograph. Centre: native prediction. Right: the exact scoring keep-mask. No scored pixels were selected by their reconstruction error.

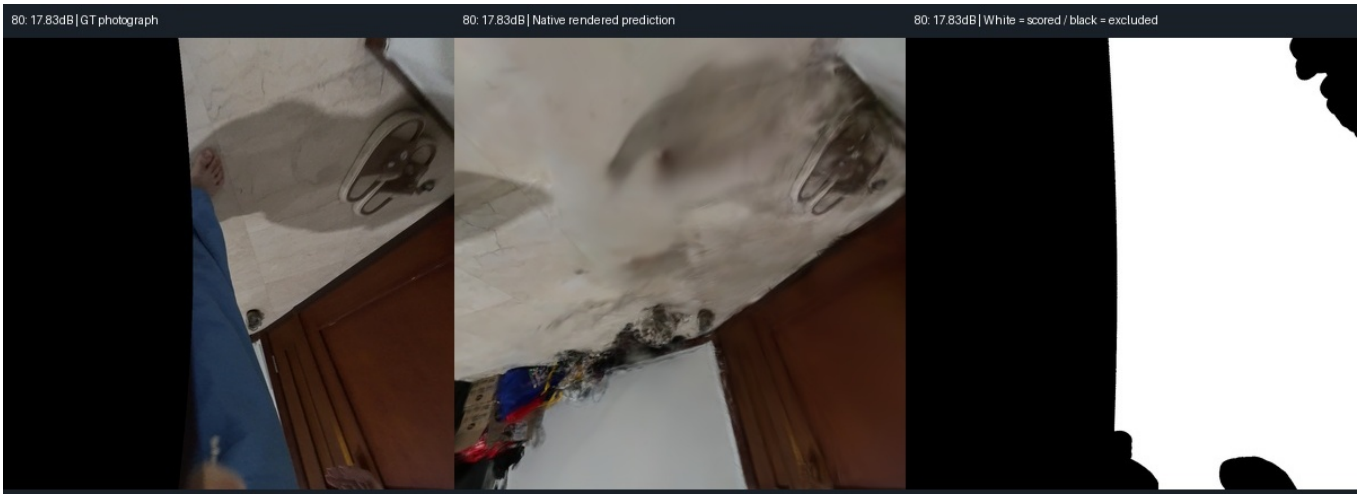


Figure 19. Native eval 80: 17.83 dB
cam1-n001284-f2_eval.jpg; valid image fraction 50.0%. Left: full GT photograph. Centre: native prediction. Right: the exact scoring keep-mask. No scored pixels were selected by their reconstruction error.

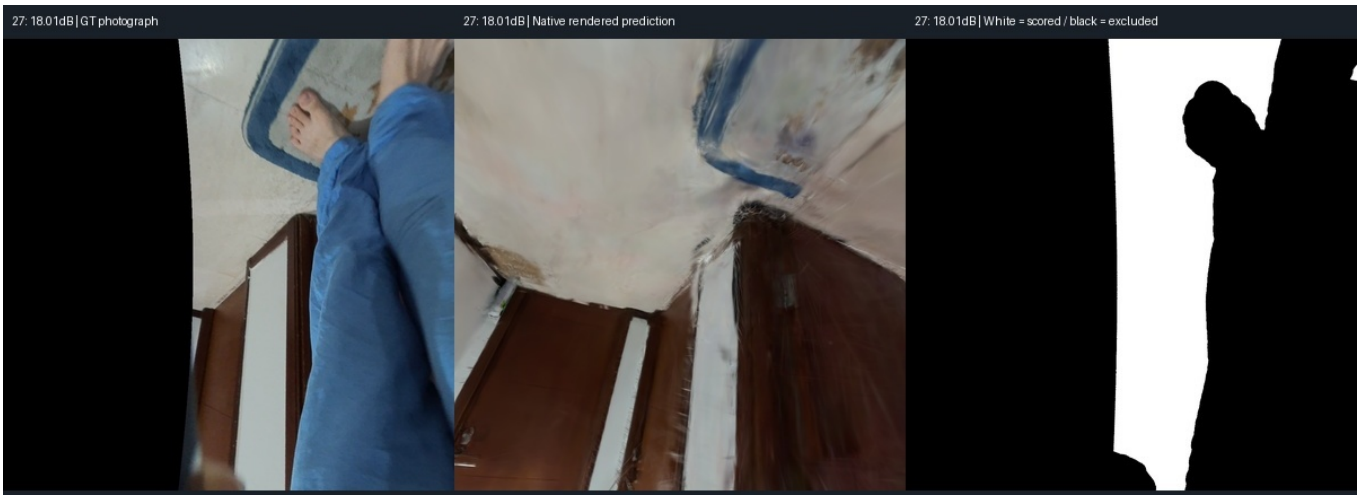


Figure 20. Native eval 27: 18.01 dB
cam1-n001641-f2_eval.jpg; valid image fraction 20.8%. Left: full GT photograph. Centre: native prediction. Right: the exact scoring keep-mask. No scored pixels were selected by their reconstruction error.

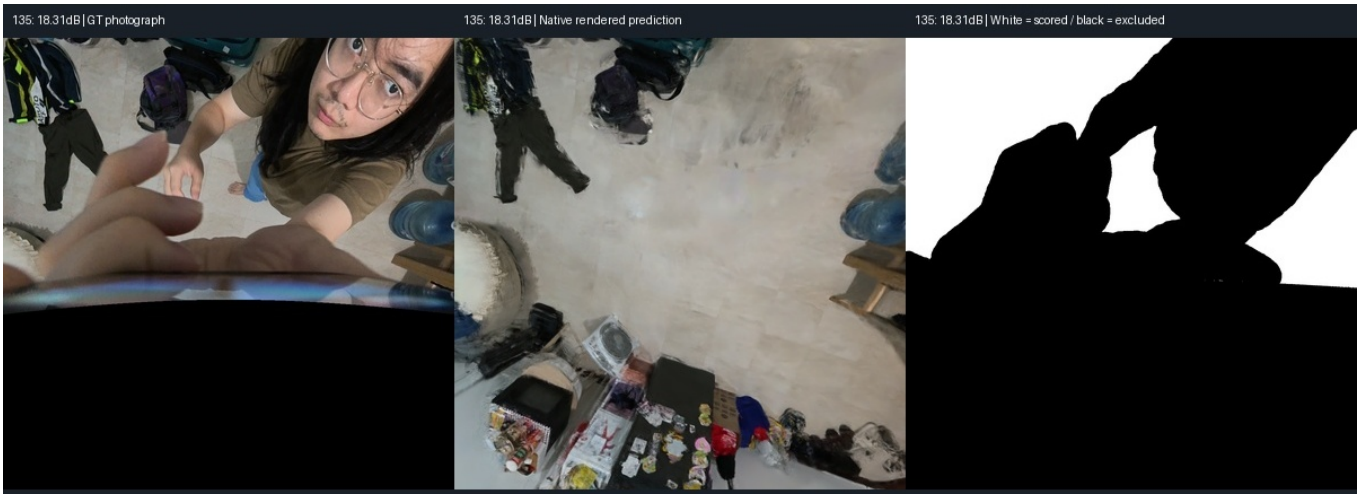


Figure 21. Native eval 135: 18.31 dB
cam1-n000066-f3_eval.jpg; valid image fraction 21.3%. Left: full GT photograph. Centre: native prediction. Right: the exact scoring keep-mask. No scored pixels were selected by their reconstruction error.

Appendix C. Complete held-out score table

All 188 native evaluation images were matched to source images by RGB similarity, with a unique bijection and full-resolution confirmation. The GT/source maximum RMSE was 0.234/255. Valid pixels use unchanged binary masks. These scores are in the native renderer and do not include the public LoD downgrade. [E17]

Eval	Source virtual view	Keep %	PSNR dB	RMSE /255
0	cam1-n000201-f4_eval.jpg	54.6	28.08	10.06
1	cam1-n001467-f3_eval.jpg	22.9	15.57	42.46
2	cam1-n001557-f4_eval.jpg	54.1	31.07	7.13
3	cam0-n000882-f1_eval.jpg	53.6	21.29	21.97
4	cam1-n001641-f0_eval.jpg	72.6	23.97	16.14
5	cam0-n001179-f4_eval.jpg	53.2	27.10	11.26
6	cam1-n000318-f1_eval.jpg	47.9	24.28	15.57
7	cam1-n000552-f3_eval.jpg	29.8	23.16	17.72
8	cam0-n000750-f2_eval.jpg	44.2	21.56	21.32
9	cam1-n000987-f1_eval.jpg	21.8	25.64	13.33
10	cam0-n000066-f2_eval.jpg	53.6	25.35	13.77
11	cam0-n001467-f4_eval.jpg	53.2	25.40	13.70
12	cam1-n001842-f2_eval.jpg	30.7	22.33	19.51
13	cam0-n000318-f4_eval.jpg	53.2	26.38	12.23
14	cam1-n001749-f3_eval.jpg	16.9	23.95	16.18
15	cam1-n001377-f4_eval.jpg	51.3	28.04	10.10
16	cam0-n001179-f1_eval.jpg	53.7	28.94	9.11
17	cam1-n001749-f4_eval.jpg	54.1	29.92	8.14
18	cam0-n001284-f1_eval.jpg	30.1	23.32	17.41
19	cam1-n001467-f2_eval.jpg	22.0	22.14	19.94
20	cam0-n000201-f0_eval.jpg	100.0	25.13	14.12
21	cam0-n000645-f2_eval.jpg	53.6	21.67	21.03
22	cam0-n001377-f2_eval.jpg	53.6	21.92	20.45
23	cam1-n001641-f4_eval.jpg	52.7	20.44	24.23
24	cam1-n000318-f2_eval.jpg	53.3	26.10	12.63
25	cam1-n001086-f0_eval.jpg	100.0	25.89	12.94
26	cam1-n001467-f4_eval.jpg	54.6	23.40	17.23

Eval	Source virtual view	Keep %	PSNR dB	RMSE /255
27	cam1-n001641-f2_eval.jpg	20.8	18.01	32.06
28	cam0-n000453-f3_eval.jpg	39.1	21.06	22.58
29	cam0-n001926-f0_eval.jpg	100.0	22.83	18.42
30	cam1-n001179-f0_eval.jpg	100.0	20.33	24.55
31	cam1-n000201-f3_eval.jpg	28.7	19.06	28.43
32	cam1-n000201-f1_eval.jpg	47.6	24.26	15.62
33	cam1-n000318-f3_eval.jpg	28.5	24.37	15.42
34	cam0-n001557-f2_eval.jpg	53.6	27.96	10.20
35	cam1-n001557-f0_eval.jpg	87.8	25.30	13.85
36	cam1-n001284-f4_eval.jpg	46.9	27.26	11.05
37	cam1-n000453-f1_eval.jpg	49.1	22.43	19.28
38	cam1-n001641-f1_eval.jpg	54.8	26.97	11.43
39	cam0-n000201-f1_eval.jpg	53.8	22.11	20.01
40	cam0-n000453-f2_eval.jpg	53.6	29.51	8.53
41	cam0-n000066-f3_eval.jpg	41.3	23.50	17.05
42	cam1-n000882-f0_eval.jpg	100.0	22.65	18.80
43	cam1-n000750-f1_eval.jpg	44.6	22.57	18.98
44	cam0-n001284-f3_eval.jpg	35.0	24.91	14.49
45	cam1-n000645-f2_eval.jpg	44.0	22.22	19.75
46	cam1-n001284-f3_eval.jpg	33.2	21.30	21.97
47	cam0-n001749-f0_eval.jpg	100.0	21.78	20.77
48	cam0-n000750-f0_eval.jpg	100.0	22.22	19.75
49	cam0-n001179-f2_eval.jpg	44.9	25.40	13.70
50	cam1-n001842-f0_eval.jpg	86.5	21.36	21.80
51	cam0-n001179-f0_eval.jpg	100.0	24.21	15.71
52	cam1-n000066-f4_eval.jpg	54.6	29.90	8.16
53	cam1-n001467-f0_eval.jpg	98.4	21.45	21.58
54	cam0-n000201-f2_eval.jpg	53.6	24.32	15.50
55	cam0-n000318-f0_eval.jpg	98.1	22.76	18.56
56	cam0-n000882-f0_eval.jpg	100.0	22.42	19.29

Eval	Source virtual view	Keep %	PSNR dB	RMSE /255
57	cam1-n000750-f4_eval.jpg	54.6	22.88	18.31
58	cam0-n001557-f3_eval.jpg	31.3	20.63	23.73
59	cam0-n000882-f2_eval.jpg	53.6	24.00	16.09
60	cam0-n001557-f4_eval.jpg	53.2	27.41	10.87
61	cam1-n000453-f0_eval.jpg	100.0	25.35	13.78
62	cam0-n001377-f1_eval.jpg	49.7	23.78	16.50
63	cam1-n001557-f3_eval.jpg	20.9	20.50	24.07
64	cam1-n000066-f1_eval.jpg	52.4	22.87	18.33
65	cam1-n001377-f0_eval.jpg	94.1	24.24	15.65
66	cam1-n000750-f2_eval.jpg	53.3	19.87	25.87
67	cam1-n000645-f0_eval.jpg	99.9	21.66	21.07
68	cam0-n000552-f2_eval.jpg	53.6	25.86	12.99
69	cam0-n000066-f0_eval.jpg	100.0	24.91	14.49
70	cam0-n000552-f1_eval.jpg	53.8	17.74	33.09
71	cam0-n001086-f2_eval.jpg	43.5	22.48	19.17
72	cam0-n001086-f1_eval.jpg	53.5	25.84	13.02
73	cam1-n000318-f4_eval.jpg	54.6	25.79	13.09
74	cam1-n001179-f3_eval.jpg	28.2	21.64	21.12
75	cam1-n001926-f4_eval.jpg	54.6	18.76	29.40
76	cam1-n000882-f4_eval.jpg	54.6	25.83	13.03
77	cam0-n001284-f2_eval.jpg	53.6	22.86	18.34
78	cam1-n001377-f3_eval.jpg	23.4	22.41	19.31
79	cam1-n001557-f2_eval.jpg	26.3	26.42	12.18
80	cam1-n001284-f2_eval.jpg	50.0	17.83	32.75
81	cam0-n000318-f1_eval.jpg	53.8	23.97	16.15
82	cam0-n001086-f0_eval.jpg	100.0	21.91	20.47
83	cam1-n000552-f4_eval.jpg	54.6	20.71	23.49
84	cam1-n000552-f0_eval.jpg	100.0	21.22	22.16
85	cam1-n000882-f1_eval.jpg	45.8	22.54	19.04
86	cam1-n001179-f1_eval.jpg	28.8	25.87	12.97

Eval	Source virtual view	Keep %	PSNR dB	RMSE /255
87	cam0-n000987-f1_eval.jpg	48.8	24.61	15.00
88	cam1-n000453-f2_eval.jpg	53.2	29.61	8.44
89	cam0-n001842-f3_eval.jpg	36.9	24.03	16.03
90	cam0-n000645-f3_eval.jpg	39.5	21.56	21.30
91	cam0-n000645-f4_eval.jpg	53.2	24.75	14.75
92	cam0-n000987-f2_eval.jpg	51.6	25.44	13.63
93	cam1-n001641-f3_eval.jpg	16.3	21.26	22.07
94	cam1-n001749-f1_eval.jpg	54.8	23.35	17.33
95	cam0-n000645-f1_eval.jpg	40.3	24.27	15.59
96	cam1-n001284-f1_eval.jpg	54.8	28.71	9.35
97	cam0-n001377-f3_eval.jpg	39.9	24.30	15.53
98	cam1-n001842-f4_eval.jpg	53.4	22.72	18.63
99	cam0-n001842-f4_eval.jpg	53.2	23.02	18.00
100	cam0-n001467-f0_eval.jpg	100.0	21.79	20.75
101	cam0-n001641-f3_eval.jpg	38.1	26.79	11.67
102	cam1-n000066-f0_eval.jpg	93.4	21.29	21.99
103	cam1-n000987-f0_eval.jpg	94.9	26.63	11.89
104	cam1-n001926-f2_eval.jpg	53.1	19.24	27.83
105	cam1-n001086-f2_eval.jpg	47.1	25.54	13.48
106	cam1-n000645-f4_eval.jpg	54.6	18.84	29.13
107	cam0-n001377-f4_eval.jpg	53.2	27.62	10.61
108	cam0-n001179-f3_eval.jpg	34.7	25.27	13.90
109	cam0-n000066-f4_eval.jpg	53.2	28.66	9.41
110	cam0-n001926-f1_eval.jpg	50.1	22.83	18.41
111	cam1-n000552-f1_eval.jpg	47.8	29.88	8.18
112	cam0-n000453-f4_eval.jpg	53.2	34.01	5.08
113	cam0-n001557-f1_eval.jpg	43.8	29.31	8.73
114	cam1-n000453-f4_eval.jpg	54.6	33.56	5.35
115	cam1-n000645-f3_eval.jpg	26.6	20.78	23.30
116	cam0-n001926-f2_eval.jpg	53.6	26.51	12.06

Eval	Source virtual view	Keep %	PSNR dB	RMSE /255
117	cam1-n000201-f0_eval.jpg	100.0	25.00	14.33
118	cam0-n001842-f1_eval.jpg	53.8	22.77	18.54
119	cam0-n000552-f0_eval.jpg	100.0	24.60	15.01
120	cam1-n001467-f1_eval.jpg	54.8	22.20	19.80
121	cam0-n001641-f1_eval.jpg	53.8	24.30	15.55
122	cam0-n001467-f3_eval.jpg	40.2	22.91	18.24
123	cam0-n001641-f2_eval.jpg	53.6	25.02	14.31
124	cam0-n001641-f0_eval.jpg	100.0	26.37	12.25
125	cam0-n001467-f2_eval.jpg	53.4	27.92	10.24
126	cam1-n001926-f1_eval.jpg	51.6	24.06	15.97
127	cam1-n000453-f3_eval.jpg	30.3	26.22	12.46
128	cam0-n000750-f4_eval.jpg	53.2	20.16	25.05
129	cam1-n000987-f4_eval.jpg	54.6	27.19	11.15
130	cam0-n000552-f4_eval.jpg	53.2	25.55	13.47
131	cam1-n001086-f3_eval.jpg	20.7	21.40	21.71
132	cam0-n000750-f1_eval.jpg	53.6	22.59	18.93
133	cam0-n001641-f4_eval.jpg	53.2	28.53	9.56
134	cam0-n000987-f3_eval.jpg	30.8	32.67	5.93
135	cam1-n000066-f3_eval.jpg	21.3	18.31	30.96
136	cam0-n000318-f3_eval.jpg	33.2	20.63	23.70
137	cam1-n001842-f1_eval.jpg	54.8	24.25	15.63
138	cam0-n001086-f3_eval.jpg	34.7	22.81	18.45
139	cam1-n001284-f0_eval.jpg	61.5	30.91	7.26
140	cam1-n001179-f2_eval.jpg	53.1	25.64	13.32
141	cam1-n001179-f4_eval.jpg	54.6	26.17	12.54
142	cam1-n001377-f1_eval.jpg	54.8	28.40	9.69
143	cam1-n001749-f0_eval.jpg	80.5	24.54	15.13
144	cam0-n001284-f0_eval.jpg	100.0	21.66	21.07
145	cam1-n000882-f3_eval.jpg	30.7	22.46	19.21
146	cam1-n001557-f1_eval.jpg	54.8	28.25	9.87

Eval	Source virtual view	Keep %	PSNR dB	RMSE /255
147	cam0-n001749-f2_eval.jpg	53.6	25.97	12.82
148	cam1-n001086-f1_eval.jpg	32.9	24.18	15.75
149	cam0-n000987-f0_eval.jpg	99.9	22.71	18.67
150	cam0-n001926-f4_eval.jpg	53.2	19.56	26.83
151	cam1-n000750-f0_eval.jpg	100.0	20.90	22.99
152	cam1-n000750-f3_eval.jpg	28.8	19.72	26.33
153	cam0-n000453-f1_eval.jpg	53.7	23.43	17.18
154	cam0-n000987-f4_eval.jpg	53.2	29.74	8.31
155	cam0-n001467-f1_eval.jpg	47.8	24.81	14.65
156	cam1-n000987-f3_eval.jpg	30.2	23.26	17.53
157	cam1-n000645-f1_eval.jpg	45.8	20.65	23.66
158	cam1-n000882-f2_eval.jpg	53.2	23.29	17.46
159	cam0-n000318-f2_eval.jpg	53.6	24.76	14.75
160	cam1-n001926-f0_eval.jpg	75.0	24.60	15.02
161	cam0-n000552-f3_eval.jpg	39.4	23.78	16.51
162	cam1-n000318-f0_eval.jpg	100.0	25.09	14.19
163	cam0-n000645-f0_eval.jpg	100.0	23.13	17.79
164	cam0-n001377-f0_eval.jpg	100.0	25.47	13.58
165	cam1-n000987-f2_eval.jpg	53.0	24.33	15.49
166	cam0-n001086-f4_eval.jpg	53.2	28.26	9.85
167	cam0-n000066-f1_eval.jpg	54.0	22.74	18.59
168	cam1-n000552-f2_eval.jpg	53.2	21.59	21.24
169	cam0-n001749-f1_eval.jpg	53.9	19.35	27.50
170	cam0-n001842-f2_eval.jpg	53.6	24.27	15.60
171	cam0-n000201-f3_eval.jpg	40.8	22.80	18.48
172	cam1-n001086-f4_eval.jpg	54.6	25.83	13.04
173	cam0-n001749-f3_eval.jpg	38.8	28.01	10.15
174	cam0-n000882-f4_eval.jpg	53.2	28.77	9.29
175	cam0-n001749-f4_eval.jpg	53.2	27.01	11.38
176	cam0-n000750-f3_eval.jpg	35.0	24.95	14.42

Eval	Source virtual view	Keep %	PSNR dB	RMSE /255
177	cam0-n000201-f4_eval.jpg	53.2	27.44	10.83
178	cam0-n001926-f3_eval.jpg	37.5	19.07	28.39
179	cam1-n001749-f2_eval.jpg	13.0	22.52	19.08
180	cam0-n001284-f4_eval.jpg	53.2	27.99	10.17
181	cam0-n000453-f0_eval.jpg	100.0	19.92	25.74
182	cam1-n001926-f3_eval.jpg	13.8	16.72	37.19
183	cam1-n000066-f2_eval.jpg	52.2	23.69	16.67
184	cam0-n001557-f0_eval.jpg	100.0	29.77	8.28
185	cam1-n000201-f2_eval.jpg	53.2	25.64	13.32
186	cam0-n000882-f3_eval.jpg	39.6	19.69	26.43
187	cam0-n001842-f0_eval.jpg	100.0	21.82	20.69

Appendix D. Executed settings and reproduction contracts

The following settings are historical execution evidence. They are included for architectural review, not an instruction to launch compute. Local paths are identified so the exact inputs can be found. No credentials or access tokens are included.

Exact full-scene Gaussian training argv

Recorded argv; paths and values as executed

```
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-2026
1031\runtime\spirula-2026.10.30\spirula.exe train 3dgs
--data C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\202
61031_OSMO360\clean-v3\explicit-full-v1\dataset
--output-dir-prefix C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruc
tion\data\20261031_OSMO360\clean-v3\explicit-full-v1\training
--output-dir-name osmo-explicit-depth-v1
--num-iterations 50000
--refine-stop-iter 35000
--quality high
--cap-max 3000000
--seed-pointcloud C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstructi
on\data\20261031_OSMO360\clean-v3\explicit-full-v1\dataset\seed-dense-train.ply
--warp-to-pinhole 0
--warp-spherical-to-pinhole 0
--train-resolution-divisor 1
--progressive-resolution 0
--progressive-splat-budget 0
--cache-images disk
--load-masks 1
--apply-loss-for-mask 0
--mask-boundary-offset -0.002
--load-depths 1
--depth-dir depths-trusted
--depth-unit-scale-factor 0.001
--input-depth-is-ray-depth 1
--depth-supervision-weight 0.05
--load-normals 1
--normal-dir normals-trusted
--normal-supervision-weight 0.01
--normal-reg-weight 0.04
--depth-distortion-reg 0.01
--normal-distortion-reg 0.01
--distraction-robustness mild
--floater-suppression strong
--max-world-size 0.12
--means-lr 0.000064
--means-lr-final 0.0000008
--noise-lr 16
--noise-lr-final 0.16
--primitive mip
--sh-degree 2
--scene-center none
--train-frame points
--orientation-method none
--center-method none
--auto-scale-poses 0
--eval-mode filename
--validation-fraction 0
--save-eval-images 1
--eval-lpips 0
--steps-per-save 5000
--save-only-latest-checkpoint 1
--save-full-checkpoint 1
```

Recorded argv; paths and values as executed (continued)

```
--disable-viewer 1
--keep-viewer-alive 0
--device 0
```

Geometry inference argv

Recorded MoGe inference argv

```
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-2026
1031\runtime\spirula-2026.10.30\spirula.exe
geometry
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_O
SMO360\clean-v3\explicit-full-v1\dataset
--model
C:/Users/bombe/OneDrive/Documents/ChatGPT/Ruang Kamare/roomscan-reconstruction/data/20261019_0
95800/runtime/spirula/models/moge2-vitl-normal.onnx
--split
no
--max-size
768
--face-res
source
--num-tokens
2400
--normal-format
png
--depth
--depth-units
relative
--ray-depth
yes
--device
0
```

Resolution and coordinate contracts

Contract	Required interpretation
Native source	3840-square fisheye JPEGs and keep-masks use the same calibration grid.
Virtual views	Five proper face rotations per source, masked by physical FOV and foreground; frames below the keep-fraction threshold are omitted. All 452 sources are represented.
Frame identity	Both lenses and every virtual view at the same timestamp share train/eval assignment.
World identity	The explicit views preserve the audited normalized world; no independent component transform is substituted.
Depth	Euclidean ray range; positive scale only; offset exactly zero; uint16 / 1000 in normalized units.
Normals	Only pixels supported by validated depth remain normal-valid in trusted guidance.
Seed	Training colours only; sparse anchors retained with higher priority; dense candidates need cross-time support.
Transport	Original float PLY retained; SOG is quantized; source count and SH degree are preserved.

Appendix E. Processing code and viewer branches

Complete critical processing files are printed below, followed by exact viewer excerpts. Line numbers refer to the preserved source file; a + marks a wrapped display continuation. Third-party libraries, model weights and credentials are excluded. The viewer listing is its pre-bundle source template: build.mjs replaces __MANIFEST__ and the provisional release marker before bundling. Those code-generation tokens are not unresolved placeholders in the live site.

E1. Adaptive native-frame selection - complete

select_frames.py | Complete file
SHA-256 36dda82a9ab3646bddd5032354fe181b7c1fb3218b7f9950f41981e163f54e76

select_frames.py

```
"""Select synchronized, sharper native-resolution frames; never alter the source."""
from pathlib import Path
import cv2, json, subprocess, time, numpy as np

WORK=Path(__file__).resolve().parent
CAP=Path(r'C:/Users/bombe/OneDrive/Documents/ChatGPT/Ruang Kamare/roomscan-reconstruction/data/20261031_OSMO360')
RUN=CAP/'clean-v3'; DS=RUN/'dataset'
SOURCE=CAP/'originals/CAM_20261031005909_0004_D.OSV'
FFMPEG=Path(r'C:/Program Files/ffmpeg/bin/ffmpeg.exe')
FPS=30000/1001
for p in [WORK/'thumbnails/cam0',WORK/'thumbnails/cam1',RUN/'logs',RUN/'analysis',DS/'images/cam0',DS/'images/cam1']:
    p.mkdir(parents=True,exist_ok=True)

def run_decode(filters,outputs,logname):
    cmd=[str(FFMPEG),'-hide_banner','-nostdin','-n','-threads','8','-i',str(SOURCE),'-filter_complex',filters]
    for i,out in enumerate(outputs):cmd+=['-map',f'[c{i}]','-q:v','1','-fps_mode','passthrough',str(out)]
    (RUN/f'analysis/{logname}-command.json').write_text(json.dumps(cmd,indent=2))
    with (RUN/f'logs/{logname}.log').open('wb') as f:subprocess.run(cmd,stdout=f,stderr=subprocess.STDOUT,check=True)

thumbs=[WORK/f'thumbnails/cam{i}/sample_%06d.jpg' for i in range(2)]
if not all(len(list(p.parent.glob('sample_*.jpg'))) == 652 for p in thumbs):
    if any(list(p.parent.glob('sample_*.jpg')) for p in thumbs):raise RuntimeError('Partial thumbnails exist; inspect, do not overwrite.')
    run_decode(';'.join(f'[0:v:{i}]select=not(mod(n\\,3)),scale=640:640:flags=lanczos,format=yuvj420p[c{i}]' for i in range(2)),thumbs,'thumbnail-d
ecode')
print('Thumbnails complete',flush=True)

yy,xx=np.mgrid[:640,:640]
# Optical interior, excluding black borders and much of the carried-camera body region.
scoremask=((xx-320)**2+(yy-320)**2 < 240**2)&(yy<460)
previous=[None,None];records=[]
for j in range(652):
    sharp=[];motion=[]
    for lens in range(2):
```

select_frames.py (continued)

```
        im=cv2.imread(str(thumbs[lens].parent/f'sample_{j+1:06d}.jpg'),cv2.IMREAD_GRAYSCALE)
        if im is None:raise RuntimeError(f'Missing thumbnail {lens} {j}')
        lap=cv2.Laplacian(im,cv2.CV_32F)
        sharp.append(float(np.var(lap[scoremask])))
        small=cv2.resize(im,(320,320),interpolation=cv2.INTER_AREA)
        if previous[lens] is None:motion.append(0.)
        else:
            flow=cv2.calcOpticalFlowFarneback(previous[lens],small,None,.5,3,15,3,5,1.2,0)
            motion.append(float(np.median(np.linalg.norm(flow,axis=2)[scoremask[::2,::2]])))
        previous[lens]=small
    records.append({'source_frame':j*3,'time_seconds':j*3/FPS,'filename':f'frame_{j*3:06d}.jpg','sharpness':sharp,'image_motion_px_320':motion})
quality=np.array([np.sqrt(max(r['sharpness'][0],.001)*max(r['sharpness'][1],.001)) for r in records])
motion=np.array([np.median(r['image_motion_px_320']) for r in records])
moving=float(np.percentile(motion[1:],80));stationary=float(np.percentile(motion[1:],20))
selected=[];start=0
while start<len(records):
    speed=float(np.median(motion[start:min(start+4,len(records))]))
    width=2 if speed>moving else (4 if speed<stationary else 3)
    end=min(start+width,len(records))
    # Mild center preference prevents choosing a boundary frame for negligible sharpness gain.
    scores=quality[start:end]*(1-.04*np.abs(np.arange(start,end)-(start+end-1)/2))
    k=start+int(np.argmax(scores));selected.append(records[k]);start=end
manifest={'source':str(SOURCE),'fps_numerator':30000,'fps_denominator':1001,'source_frames_per_lens':1955,
    'candidate_stride_frames':3,'selected_pair_count':len(selected),'native_size':[3840,3840],
    'note':'Optical flow is image motion, not verified camera translation. Joint lens sharpness selection; motion-adaptive 2/3/4 candidate windows.
 Original source preserved. JPEG q1 is an 8-bit training decode; not a claim of lossless 10-bit preservation.',
    'max_selected_gap_seconds':max(b['time_seconds']-a['time_seconds'] for a,b in zip(selected,selected[1:])),
    'selected':selected,'candidates':records}
(WORK/'frame-selection.json').write_text(json.dumps(manifest,indent=2))
(RUN/'analysis/frame-selection.json').write_text(json.dumps(manifest,indent=2))
print(f'Selected {len(selected)} synchronized pairs; native extraction starting',flush=True)
expression='+'.join(f'eq(n\\,{r["source_frame"]})' for r in selected)
outputs=[DS/f'images/cam{i}/selected_%06d.jpg' for i in range(2)]
if any(list(p.parent.glob('*.jpg')) for p in outputs):raise RuntimeError('Native output is not empty; inspect before proceeding.')
run_decode(';'.join(f'[0:v:{i}]select={expression},format=yuvj420p[c{i}]' for i in range(2)),outputs,'native-decode')
for lens in range(2):
    for k,r in enumerate(selected,1):
        old=outputs[lens].parent/f'selected_{k:06d}.jpg';new=outputs[lens].parent/r['filename']
        if new.exists():raise RuntimeError('Refusing destination overwrite')
        old.rename(new)
        im=cv2.imread(str(new))
        if im.shape[:2]!=(3840,3840):raise RuntimeError(f'Wrong dimensions {new}')
manifest['native_decode_complete']=True
(WORK/'frame-selection.json').write_text(json.dumps(manifest,indent=2))
(RUN/'analysis/frame-selection.json').write_text(json.dumps(manifest,indent=2))
```

select_frames.py (continued)

```
77 print(json.dumps({'complete':True,'pairs':len(selected),'image_dir':str(DS/'images')}),flush=True)
```

E2. Timestamp-grouped train/eval split - complete

assign_split.py | Complete file
SHA-256 041abd4cd6dd411c008fa07d6b89346ba82b01ee23d3cc19858d6281833f432c

assign_split.py

```
 1 from pathlib import Path
 2 import json
 3
 4 WORK=Path(__file__).resolve().parent
 5 RUN=Path(r'C:/Users/bombe/OneDrive/Documents/ChatGPT/Ruang Kamare/roomscan-reconstruction/data/20261031_OSMO360/clean-v3')
 6 p=WORK/'frame-selection.json';m=json.loads(p.read_text())
 7 assert m['native_decode_complete']
 8 assert not m.get('chronological_names'), 'Already assigned; do not repeat'
 9 for i,r in enumerate(m['selected']):
10     split='eval' if i%12==6 else 'train'
11     old=r['filename'];new=f'frame_{r["source_frame"]:06d}_{split}.jpg'
12     for lens in range(2):
13         parent=RUN/f'dataset/images/cam{lens}'
14         src=parent/old;dst=parent/new
15         if not src.is_file() or dst.exists():raise RuntimeError(f'Rename guard failed {src}')
16         src.rename(dst)
17     r.update(filename=new,split=split)
18 m['split_assigned']=True
19 m['chronological_names']=True
20 m['split_note']='Every twelfth selected timestamp offset six held out; both lenses share split before SfM/warping. Camera poses estimated by SfM wi
 + th all images; training seed will use only train observations/colors. Validation is novel-timestamp rendering within the same video, not independen
 + t capture ground truth.'
21 m['train_pairs']=sum(r['split']=='train' for r in m['selected']);m['eval_pairs']=sum(r['split']=='eval' for r in m['selected'])
22 for dst in [p,RUN/'analysis/frame-selection.json']:dst.write_text(json.dumps(m,indent=2))
23 print(json.dumps({k:m[k] for k in ['selected_pair_count','train_pairs','eval_pairs','split_assigned']}))
```

E3. Explicit calibrated pinhole materialization - complete

build_explicit_pinhole.py | Complete file
SHA-256 9c610a58a0fe75156cfdb4c9e2d74e4f896c9528ab4a49906bb0b8ec4d07bd5c

build_explicit_pinhole.py

```
"""Materialize calibrated pinhole RGB/mask views with independently checked poses.

The global fresh SfM frame is preserved. Source timestamps retain their train/eval
split across both lenses and every virtual view. No SfM rerun, old depth map,
learned image synthesis, or previous Gaussian checkpoint is used.
"""
from pathlib import Path
from concurrent.futures import ThreadPoolExecutor
import argparse,hashlib,json,sys,time
import cv2,numpy as np
W=Path(__file__).resolve().parent;sys.path.insert(0,str(W.parent/'osmo-reconstruct-20261031'))
from seed_fisheye import read_model,write_model_text,write_seed_ply,Fisheye,qvec_to_rotmat,rotmat_to_qvec
P=argparse.ArgumentParser();P.add_argument('--source',type=Path,required=True);P.add_argument('--output',type=Path,required=True);P.add_argument('--size',type=int,default=1024);P.add_argument('--subset',choices=['bathroom','all'],default='bathroom');a=P.parse_args();started=time.monotonic();cv2.setNumThreads(1)
if a.output.exists():raise FileExistsError(a.output)
if not 512<=a.size<=2048:raise ValueError('Invalid face size')
selection=json.loads((W/'frame-selection.json').read_text());by_name={r['filename']:r for r in selection['selected']}
cameras,images,points=read_model(a.source/'sparse/0')
def selected(im):
    row=by_name[Path(im['name']).name];n=row['source_frame'];return a.subset=='all' or 540<=n<=720 or 1470<=n<=1650
inputs=sorted((im for im in images.values() if selected(im)),key=lambda im:(by_name[Path(im['name']).name]['source_frame'],im['name']))
faces=[np.eye(3),np.array([[0,0,-1],[0,1,0],[1,0,0]]),np.array([[0,0,1],[0,1,0],[-1,0,0]]),np.array([[1,0,0],[0,0,-1],[0,1,0]]),np.array([[1,0,0],[0,0,1],[0,-1,0]])]
for face in faces:
    if not np.allclose(face@face.T,np.eye(3)) or not np.isclose(np.linalg.det(face),1):raise ValueError('Invalid proper face rotation')
S=a.size;f=S/2;xx,yy=np.meshgrid(np.arange(S),np.arange(S));virtual=np.stack([(xx-f)/f,(yy-f)/f,np.ones_like(xx)],axis=-1).reshape(-1,3)
maps={};models={}
for cid,camera in cameras.items():
    models[cid]=Fisheye(camera,max_theta_degrees=99)
    for j,face in enumerate(faces):
        uv,valid=models[cid].project(virtual@face);valid&=(uv[:,0]>=0)&(uv[:,0]<camera['width'])&(uv[:,1]>=0)&(uv[:,1]<camera['height'])
        maps[cid,j]=(uv[:,0].reshape(S,S).astype(np.float32),uv[:,1].reshape(S,S).astype(np.float32),valid.reshape(S,S))
(a.output/'images').mkdir(parents=True);(a.output/'masks').mkdir()
def process(im):
    row=by_name[Path(im['name']).name];source=cv2.imread(str(a.source/'images'/im['name']));mask_path=(a.source/'masks'/im['name']).with_suffix('.png');mask=cv2.imread(str(mask_path),0)
    if source is None or mask is None:raise ValueError('Missing source RGB/mask '+im['name'])
    camera=cameras[im['camera_id']]
    if source.shape[:2]!=(camera['height'],camera['width']) or mask.shape!=source.shape[:2]:raise ValueError('Native RGB/mask/calibration mismatch')
    R=qvec_to_rotmat(im['qvec']);rays,ok=models[im['camera_id']].unproject(im['points2D_xy']);ids=im['point3D_ids'];ok&=ids>=0
    indices=np.flatnonzero(ok);xy=im['points2D_xy'][indices];inside=(xy[:,0]>=0)&(xy[:,0]<camera['width'])&(xy[:,1]>=0)&(xy[:,1]<camera['height']);indices=indices[inside];xy=xy[inside]
    keep=mask[np.clip(xy[:,1].astype(int),0,camera['height']-1),np.clip(xy[:,0].astype(int),0,camera['width']-1)]>127;indices=indices[keep];xy=xy[k
```

build_explicit_pinhole.py (continued)

```
eep]
    rgb=source[xy[:,1].astype(int),xy[:,0].astype(int),::-1];colour=[(int(ids[i]),c) for i,c in zip(indices,rgb)] if row['split']=='train' else []
    results=[];error_values=[]
    for j,face in enumerate(faces):
        mx,my,valid=maps[im['camera_id'],j];face_mask=cv2.remap(mask,mx,my,cv2.INTER_NEAREST,borderMode=cv2.BORDER_CONSTANT,borderValue=0);face_mas
k=np.where(valid&(face_mask>127),255,0).astype(np.uint8)
        if np.mean(face_mask>0)<.12:continue
        # Keep real RGB under the separate operator/mirror mask for monocular
        # inference context. Only rays outside the physical lens are black.
        photo=cv2.remap(source,mx,my,cv2.INTER_LINEAR,borderMode=cv2.BORDER_CONSTANT,borderValue=0);photo[~valid]=0
        stem=f'{im["name"].split("/")[0]}-n{row["source_frame"]:06d}-f{j}_{row["split"]}';name=stem+'.jpg'
        cv2.imwrite(str(a.output/'images'/name),photo,[cv2.IMWRITE_JPEG_QUALITY,96]);cv2.imwrite(str(a.output/'masks'/(stem+'.png')),face_mask)
        Rv=face@R;tv=face@im['tvec'];direction=rays[indices]@face.T;z=direction[:,2];uv=direction[:,:2]/np.maximum(z[:,None],1e-15)*f+f
        usable=(z>0)&(uv[:,0]>=3)&(uv[:,0]<S-3)&(uv[:,1]>=3)&(uv[:,1]<S-3);use=indices[usable];uv=uv[usable];pids=ids[use]
        foreground=face_mask[uv[:,1].astype(int),uv[:,0].astype(int)]>0;uv=uv[foreground];pids=pids[foreground]
        have=np.array([int(pid) in points for pid in pids],dtype=bool);uv=uv[have];pids=pids[have]
        if len(pids):
            world=np.array([points[int(pid)]['xyz'] for pid in pids]);xc=world@Rv.T+tv;projected=xc[:,:2]/xc[:,2,None]*f+f;errors=np.linalg.norm(pr
ojected-uv,axis=1);error_values.extend(errors.tolist())
            if not np.isfinite(errors).all():raise ValueError('Invalid independent reprojection')
        # Verify a world ray from each camera/face stays the same after conversion.
        test=virtual[::10007];original_world=(test@face)@R;virtual_world=test@Rv
        if np.max(np.abs(original_world-virtual_world))>1e-12:raise ValueError('World-ray identity failed')
        results.append({'name':name,'qvec':rotmat_to_qvec(Rv),'tvec':tv,'camera_id':1,'points2D_xy':uv,'point3D_ids':pids,'sourceName':im['name'],'
sourceFrame':row['source_frame'],'sourceSplit':row['split'],'face':j,'keepFraction':float(np.mean(face_mask>0))})
    return results,colour,error_values
output_images={};tracks={};train_times={};colours={};errors=[];provenance=[]
with ThreadPoolExecutor(max_workers=4) as pool:
    for k,(views,samples,errs) in enumerate(pool.map(process,inputs),1):
        for view in views:
            iid=len(output_images)+1;view['id']=iid;output_images[iid]=view
            provenance.append({key:view[key] for key in ['name','sourceName','sourceFrame','sourceSplit','face','keepFraction']})
            for j,pid in enumerate(view['point3D_ids']):
                pid=int(pid);tracks.setdefault(pid,[]).append((iid,j))
                if view['sourceSplit']=='train':train_times.setdefault(pid,set()).add(view['sourceFrame'])
        for pid,c in samples:
            if pid not in colours:colours[pid]=[np.zeros(3,np.float64),0]
            colours[pid][0]+=c;colours[pid][1]+=1
        errors.extend(errs)
        if k%20==0:print(f'Explicit pinhole {k}/{len(inputs)} inputs, {len(output_images)} output views',flush=True)
output_points={}
for pid,track in tracks.items():
    if len(track)<2:continue
    point=dict(points[pid]);point['image_ids']=np.array([v[0] for v in track]);point['point2D_idxs']=np.array([v[1] for v in track]);point['track_l
en']=len(track);output_points[pid]=point
```

build_explicit_pinhole.py (continued)

```
for im in output_images.values():im['point3D_ids']=np.array([pid if int(pid) in output_points else -1 for pid in im['point3D_ids']])
output_cameras={1:{'id':1,'model':'PINHOLE','width':S,'height':S,'params':np.array([f,f,f,f])}}
write_model_text(a.output/'sparse/0',output_cameras,output_images,output_points)
seed_ids=[pid for pid in output_points if len(train_times.get(pid,set()))>=2 and pid in colours]
seed_xyz=np.array([points[pid]['xyz'] for pid in seed_ids]);seed_rgb=np.array([np.clip(np.rint(colours[pid][0]/colours[pid][1]),0,255) for pid in s
+eed_ids],np.uint8)
write_seed_ply(a.output/'seed-sparse-train.ply',seed_xyz,seed_rgb,np.zeros_like(seed_xyz))
report={'purpose':__doc__,'sourceDataset':str(a.source.resolve()),'outputDataset':str(a.output.resolve()),'subset':a.subset,'size':S,'sourceInputs'
+:len(inputs),'virtualViews':len(output_images),'trainViews':sum(v['sourceSplit']=='train' for v in output_images.values()),'evalViews':sum(v['sourc
+eSplit']=='eval' for v in output_images.values()),'seedPoints':len(seed_ids),'seedColours':'Only unmasked source training images; at least two dist
+inct training timestamps per seed point.','worldCoordinateFrame':'+Y up; identical to audited fresh SfM, no new similarity','perFaceWorldRayIdentit
+yTolerance':1e-12,'reprojection':{'observations':len(errors),'meanPixels':float(np.mean(errors)),'medianPixels':float(np.median(errors)),'p95Pixels
+':float(np.percentile(errors,95)),'maxPixels':float(np.max(errors))},'newTrainingOrDepthUsed':False,'maskConvention':'255 keep / 0 exclude; source
+operator and mirror masks reprojected by nearest neighbour','imagesSynthesized':False,'sourceFilesModified':False,'seconds':time.monotonic()-starte
+d,'views':provenance,'sha256':{name:hashlib.sha256((a.output/name).read_bytes()).hexdigest() for name in ['sparse/0/cameras.txt','sparse/0/images.t
+xt','sparse/0/points3D.txt','seed-sparse-train.ply']}}
(a.output/'explicit-pinhole-audit.json').write_text(json.dumps(report,indent=2));print(json.dumps({k:v for k,v in report.items() if k not in ['purp
+ose','views','sha256']},indent=2))
```

E4. Positive scale-only depth alignment - complete

fit_explicit_depth.py | Complete file
SHA-256 dd52bb5d91e29bef42a91e55c220ffe67ef48cd49ac65fa63bc8324d524f7924

fit_explicit_depth.py

```
"""Fit positive, zero-offset depth scales against qualified training SfM anchors.

This only audits predicted depth. Accepted frames must later pass multi-view
pixel consistency before their depth is used to seed or supervise training.
"""
from pathlib import Path
import argparse,json,sys,time
import cv2,numpy as np
W=Path(__file__).resolve().parent;sys.path.insert(0,str(W.parent/'osmo-reconstruct-20261031'))
from seed_fisheye import read_model,qvec_to_rotmat
P=argparse.ArgumentParser();P.add_argument('--dataset',type=Path,required=True);a=P.parse_args();started=time.monotonic();cv2.setNumThreads(2)
report_path=a.dataset/'depth-scale-audit.json'
if report_path.exists():raise FileExistsError(report_path)
cameras,images,points=read_model(a.dataset/'sparse/0');provenance=json.loads((a.dataset/'explicit-pinhole-audit.json').read_text());by_name={v['nam
+e']:v for v in provenance['views']}
train_times={pid:{by_name[images[int(i)]['name']]['sourceFrame'] for i in point['image_ids'] if by_name[images[int(i)]['name']]['sourceSplit']=='tr
+ain'} for pid,point in points.items()}
rows=[]
```

fit_explicit_depth.py (continued)

```
for number,im in enumerate(images.values(),1):
    meta=by_name[im['name']]
    if meta['sourceSplit']!='train':continue
    row={'name':im['name'],'sourceFrame':meta['sourceFrame'],'sourceName':meta['sourceName'],'acceptedScale':False,'offset':0.0}
    try:
        raw=cv2.imread(str((a.dataset/'depths'/im['name']).with_suffix('.png')),cv2.IMREAD_UNCHANGED)
        if raw is None or raw.dtype!=np.uint16:raise ValueError('Expected raw relative uint16 depth')
        H,DW=raw.shape;camera=cameras[im['camera_id']];fx,fy,cx,cy=camera['params'];mask=cv2.imread(str((a.dataset/'masks'/im['name']).with_suffix(
'.png')),0);mask=cv2.resize(mask,(DW,H),interpolation=cv2.INTER_NEAREST);mask=cv2.erode(mask,np.ones((7,7),np.uint8))
        idx=[j for j,pid in enumerate(im['point3D_ids']) if int(pid) in points and len(train_times[int(pid)])>=3 and points[int(pid)]['error']<=1.5
]
        if len(idx)<40:raise ValueError('Fewer than 40 qualified multi-timestamp anchors')
        ids=im['point3D_ids'][idx];observed=im['points2D_xy'][idx];xyz=np.array([points[int(pid)]['xyz'] for pid in ids]);R=qvec_to_rotmat(im['qvec
']);xc=xyz@R.T+im['tvec'];ranges=np.linalg.norm(xc,axis=1);uv=xc[:,:2]/xc[:,2,None]*[fx,fy]+[cx,cy]
        xy=observed*[DW/camera['width'],H/camera['height']];mx=xy[:,0].astype(np.float32).reshape(-1,1);my=xy[:,1].astype(np.float32).reshape(-1,1)
        depth=cv2.remap(raw.astype(np.float32),mx,my,cv2.INTER_LINEAR,borderMode=cv2.BORDER_CONSTANT,borderValue=0).ravel();valid=cv2.remap(mask,mx
,my,cv2.INTER_NEAREST,borderMode=cv2.BORDER_CONSTANT,borderValue=0).ravel()>127
        valid&=(xc[:,2]>.02)&(np.linalg.norm(uv-observed,axis=1)<=2.5)&(depth>0)&np.isfinite(ranges)
        depth=depth[valid];ranges=ranges[valid];xy=xy[valid]
        if len(depth)<40:raise ValueError('Fewer than 40 unmasked accurate anchor observations')
        ratios=ranges/depth;scale=float(np.median(ratios))
        for _ in range(4):
            inside=np.abs(scale/ratios-1)<=.05
            if inside.sum()<10:break
            scale=float(np.median(ratios[inside]))
        residual=np.abs(scale*depth-ranges)/ranges;inside=residual<=.05
        cells=np.floor(xy[inside]/[DW/4,H/4]).astype(int);coverage=len(np.unique(cells,axis=0));median=float(np.median(residual));p90=float(np.perc
entile(residual,90));fraction=float(inside.mean())
        row.update(scale=scale,offset=0.0,observations=len(depth),inliersAt5Percent=int(inside.sum()),inlierFractionAt5Percent=fraction,medianRelat
iveResidual=median,p90RelativeResidual=p90,coveredGridCells4x4=coverage,depthShape=[H,DW])
        reasons=[]
        if not np.isfinite(scale) or scale<=0:reasons.append('Invalid positive scale')
        if fraction<.70:reasons.append('Fewer than 70% of anchors agree within 5%')
        if median>.03:reasons.append('Median relative error exceeds 3%')
        if p90>.10:reasons.append('90th percentile relative error exceeds 10%')
        if coverage<5:reasons.append('Anchor support covers fewer than five 4x4 cells')
        row['reasons']=reasons;row['acceptedScale']=not reasons
    except Exception as error:row['reason']=str(error)
    rows.append(row)
    if number%50==0:print(f'Depth scale checks {number}/{len(images)}; accepted {sum(r["acceptedScale"] for r in rows)}',flush=True)
report={'method':__doc__,'dataset':str(a.dataset.resolve()),'depthConvention':'Euclidean ray range; raw PNG relative levels -> audited fresh SfM sc
ene units using one positive scale, offset exactly zero','framesAudited':len(rows),'acceptedScales':sum(r['acceptedScale'] for r in rows),'evalDept
hUsed':False,'oldDepthOrCheckpointOpened':False,'notYetApprovedForTraining':True,'gates':{'pointTrackDistinctTrainingTimestamps':3,'maximumPointSto
redErrorNativePixels':1.5,'actualVirtualReprojectionPixels':2.5,'minimumObservations':40,'anchorRelativeInlierTolerance':.05,'minimumInlierFraction
':.70,'maximumMedianRelativeResidual':.03,'maximumP90RelativeResidual':.10,'minimumCovered4x4Cells':5},'frames':rows,'seconds':time.monotonic()-sta
```

fit_explicit_depth.py (continued)

```
    + rted}
   52 report_path.write_text(json.dumps(report,indent=2));print(json.dumps({k:v for k,v in report.items() if k not in ['method','frames']},indent=2))
```

E5. Cross-time depth validation and dense seed - complete

validate_explicit_depth.py | Complete file
SHA-256 38025d8d850bc56247c09f3699dfda430828229d95dd3e609ea2b953a7b3bbf6

validate_explicit_depth.py

```
    1 """Validate scale-only pinhole depths across distinct training timestamps.
    2
    3 Only pixels agreeing within 5% with at least TWO other camera timestamps become
    4 depth targets or dense seeds. Eval images/depths never participate. The sparse
    5 train-only seed is retained at higher priority during seed voxel reduction.
    6 """
    7 import os,faulthandler
    8 os.environ['OPENBLAS_NUM_THREADS']='1';os.environ['OMP_NUM_THREADS']='1';os.environ['MKL_NUM_THREADS']='1'
    9 faulthandler.enable();faulthandler.dump_traceback_later(45,repeat=True)
   10 from pathlib import Path
   11 from concurrent.futures import ThreadPoolExecutor
   12 import argparse,hashlib,json,sys,time
   13 import cv2,numpy as np
   14 W=Path(__file__).resolve().parent;sys.path.insert(0,str(W.parent/'osmo-reconstruct-20261031'))
   15 from seed_fisheye import read_model,qvec_to_rotmat,voxel_reduce,write_seed_ply
   16 P=argparse.ArgumentParser();P.add_argument('--dataset',type=Path,required=True);P.add_argument('--max-seed',type=int,default=450000);a=P.parse_args
    + ();started=time.monotonic();cv2.setNumThreads(1)
   17 out=a.dataset/'depths-trusted'
   18 if out.exists() and any(out.iterdir()):raise FileExistsError(out)
   19 out.mkdir(exist_ok=True);cameras,images,points=read_model(a.dataset/'sparse/0');by_name={im['name']:im for im in images.values()};scale_audit=json.
    + loads((a.dataset/'depth-scale-audit.json').read_text());views=[]
   20 for row in scale_audit['frames']:
   21     if not row['acceptedScale']:continue
   22     if '_eval' in row['name']:raise ValueError('Eval depth leaked into training validation')
   23     im=by_name[row['name']];cam=cameras[im['camera_id']];R=qvec_to_rotmat(im['qvec']).astype(np.float32);t=im['tvec'].astype(np.float32);centre=-t@
    + R;fx,fy,cx,cy=cam['params']
   24     raw=cv2.imread(str((a.dataset/'depths'/row['name']).with_suffix('.png')),cv2.IMREAD_UNCHANGED);depth=raw.astype(np.float32)*row['scale'];H,DW=d
    + epth.shape
   25     mask=cv2.imread(str((a.dataset/'masks'/row['name']).with_suffix('.png')),0);mask=cv2.resize(mask,(DW,H),interpolation=cv2.INTER_NEAREST);mask=c
    + v2.erode(mask,np.ones((7,7),np.uint8))>127
   26     mask&=np.isfinite(depth)&(depth>.02)&(depth<60)
   27     k=np.array([fx*DW/cam['width'],fy*H/cam['height'],cx*DW/cam['width'],cy*H/cam['height']],np.float32)
   28     forward=R[2];median=float(np.median(depth[mask]));target=centre+forward*median*.85
   29     views.append({**row,'R':R,'t':t,'centre':centre,'depth':depth,'mask':mask,'k':k,'forward':forward,'target':target})
```

validate_explicit_depth.py (continued)

```
if not views:raise ValueError('No scale-qualified depth frames')
shapes={v['depth'].shape for v in views}
if len(shapes)!=1:raise ValueError('Expected uniform explicit pinhole depth maps')
H,DW=next(iter(shapes));fx,fy,cx,cy=views[0]['k'];x,y=np.meshgrid(np.arange(DW,dtype=np.float32),np.arange(H,dtype=np.float32));rays=np.stack([(x-c
x)/fx,(y-cy)/fy,np.ones_like(x)],axis=-1);rays/=np.linalg.norm(rays,axis=-1,keepdims=True)
print(f'Loaded {len(views)} accepted depth views for consistency validation',flush=True)
def sample_image(image,mx,my,interpolation):
    # OpenCV remap dimensions are bounded by SHRT_MAX. Pack long point lists
    # into a small 2D grid instead of a hundreds-of-thousands-row N-by-1 map.
    n=len(mx);cols=1024;rows=(n+cols-1)//cols
    xmap=np.full(rows*cols,-1,np.float32);ymap=np.full_like(xmap,-1);xmap[:n]=mx;ymap[:n]=my
    result=cv2.remap(image,xmap.reshape(rows,cols),ymap.reshape(rows,cols),interpolation,borderMode=cv2.BORDER_CONSTANT,borderValue=0)
    return result.reshape(-1,image.shape[2])[:n] if image.ndim==3 else result.ravel()[:n]
def process(index):
    view=views[index];target=view['target'];candidates=[]
    for j,other in enumerate(views):
        if other['sourceFrame']==view['sourceFrame'] or abs(other['sourceFrame']-view['sourceFrame'])<6:continue
        baseline=float(np.linalg.norm(other['centre']-view['centre']))
        if not .03<=baseline<=3:continue
        local=target@other['R'].T+other['t']
        if local[2]<=.02:continue
        off=np.max(np.abs(local[:2]/local[2]))
        if off>1.6:continue
        score=baseline+.5*float(off);candidates.append((score,j))
    # One neighbour per timestamp; include available opposite-visit evidence.
    opposite=sorted((item for item in candidates if (views[item[1]]['sourceFrame']<1000)!=(view['sourceFrame']<1000)))
    ordered=opposite[:3]+sorted(candidates);chosen=[];stamps=set()
    for _,j in ordered:
        stamp=views[j]['sourceFrame']
        if stamp in stamps:continue
        stamps.add(stamp);chosen.append(j)
        if len(chosen)==14:break
    valid_indices=np.flatnonzero(view['mask']);depth=view['depth'].ravel()[valid_indices];local_points=rays.reshape(-1,3)[valid_indices]*depth[:,No
ne];world=(local_points-view['t'])@view['R'];votes=np.zeros(len(world),np.uint8);cross=np.zeros(len(world),bool)
    for j in chosen:
        other=views[j];xc=world@other['R'].T+other['t'];distance=np.linalg.norm(xc,axis=1);z=xc[:,2];fx,fy,cx,cy=other['k'];uv=xc[:,:2]/np.maximum(
z[:,None],1e-12)*[fx,fy]+[cx,cy];mx=uv[:,0].astype(np.float32).reshape(-1,1);my=uv[:,1].astype(np.float32).reshape(-1,1)
        observed=sample_image(other['depth'],mx.ravel(),my.ravel(),cv2.INTER_NEAREST);valid=sample_image(other['mask'].astype(np.uint8),mx.ravel(),
my.ravel(),cv2.INTER_NEAREST)>0
        agree=valid&(z>.02)&(np.abs(observed-distance)<=.05*np.maximum(distance,1e-5));votes+=agree
        if (other['sourceFrame']<1000)!=(view['sourceFrame']<1000):cross|=agree
    trusted=votes>=2;trusted_map=np.zeros((H,DW),bool);trusted_map.ravel()[valid_indices]=trusted
    # Remove depth-boundary pixels; zero remains the explicit 'no depth' sentinel.
    trusted_map=cv2.erode(trusted_map.astype(np.uint8),np.ones((3,3),np.uint8))>0
    count=int(trusted_map.sum());row={'name':view['name'],'sourceFrame':view['sourceFrame'],'neighbourTimestamps':[views[j]['sourceFrame'] for j in
```

validate_explicit_depth.py (continued)

```
 chosen],'candidatePixels':len(valid_indices),'trustedPixels':count,'crossVisitSupportedPixelsBeforeErosion':int(np.sum(trusted&cross)),'written':count>=500}
    if count<500:return row,None
    mm=np.rint(view['depth']*1000);encoded=np.where(trusted_map&(mm>0)&(mm<65535),mm,0).astype(np.uint16);path=(out/view['name']).with_suffix('.png');cv2.imwrite(str(path),encoded);row['sha256']=hashlib.sha256(path.read_bytes()).hexdigest()
    # Depth-derived normals provide seed orientation, independent of normal-map conventions.
    pc=rays*view['depth'][...,None];dx=np.zeros_like(pc);dy=np.zeros_like(pc);dx[:,1:-1]=pc[:,2:]-pc[:,:-2];dy[1:-1]=pc[2:]-pc[:-2];normal=np.cross(dx,dy);length=np.linalg.norm(normal,axis=2);normal/=np.maximum(length[...,None],1e-12);normal[np.sum(normal*pc,axis=2)>0]*=-1
    sparse_grid=np.zeros((H,DW),bool);sparse_grid[::4,::4]=True;sample=trusted_map&sparse_grid&(length>1e-10)
    points_camera=pc[sample];world_seed=(points_camera-view['t'])@view['R'];normals=normal[sample]@view['R']
    image=cv2.imread(str(a.dataset/'images'/view['name']));ys,xs=np.nonzero(sample);rgb=sample_image(image,xs.astype(np.float32)*image.shape[1]/DW,ys.astype(np.float32)*image.shape[0]/H,cv2.INTER_LINEAR)[:,::-1]
    scores=np.ones(len(rgb),np.float32)
    # Record stronger cross-visit evidence in priorities without inventing geometry.
    cross_map=np.zeros((H,DW),bool);cross_map.ravel()[valid_indices]=cross;scores+=cross_map[sample]
    return row,(world_seed,rgb,normals,scores)
rows=[];seed_parts=[]
for i,index in enumerate(range(len(views)),1):
    row,seed=process(index);rows.append(row)
    if seed is not None:seed_parts.append(seed)
    if i%10==0:print(f'Multi-view depths {i}/{len(views)}; targets {sum(r["written"] for r in rows)}',flush=True)
if not seed_parts:raise ValueError('No multi-view-consistent depth pixels')
with (a.dataset/'seed-sparse-train.ply').open('rb') as f:
    header=b''
    while not header.endswith(b'end_header\n'):header+=f.readline()
    n=int(next(line.split()[-1] for line in header.decode().splitlines() if line.startswith('element vertex ')))
    dtype=np.dtype([(k,'<f4') for k in ['x','y','z','nx','ny','nz']]+[(k,'u1') for k in ['red','green','blue']]);sparse=np.fromfile(f,dtype=dtype,count=n)
seed_parts.insert(0,(np.column_stack([sparse[k] for k in ['x','y','z']]),np.column_stack([sparse[k] for k in ['red','green','blue']]),np.column_stack([sparse[k] for k in ['nx','ny','nz']]),np.full(n,10,np.float32)))
xyz,colour,normal,score=[np.concatenate([part[i] for part in seed_parts]) for i in range(4)];before=len(xyz);xyz,colour,normal,voxel=voxel_reduce(xyz,colour,normal,score,.004,a.max_seed);seed_path=a.dataset/'seed-dense-train.ply';write_seed_ply(seed_path,xyz,colour,normal)
report={'method':__doc__,'dataset':str(a.dataset.resolve()),'positiveScaleOnly':True,'offset':0.0,'relativeConsistencyTolerance':.05,'minimumOtherDistinctTrainingTimestamps':2,'scaleAcceptedFrames':len(views),'trustedDepthFrames':sum(r['written'] for r in rows),'trustedPixels':sum(r['trustedPixels'] for r in rows),'crossVisitSupportedPixels':sum(r['crossVisitSupportedPixelsBeforeErosion'] for r in rows),'sourceSparseSeedPoints':n,'seedCandidatesBeforeVoxel':before,'finalSeedPoints':len(xyz),'seedVoxelSizeSceneUnits':voxel,'seedSha256':hashlib.sha256(seed_path.read_bytes()).hexdigest(),'depthNumericUnits':'uint16 /1000 = original audited normalized SfM scene units, NOT measured metres','coordinateFrame':'+Y up, same fresh global SfM frame','oldGaussianOrDepthOpened':False,'evalInputsUsed':False,'scaleAuditSha256':hashlib.sha256((a.dataset/'depth-scale-audit.json').read_bytes()).hexdigest(),'explicitCameraAuditSha256':hashlib.sha256((a.dataset/'explicit-pinhole-audit.json').read_bytes()).hexdigest(),'frames':rows,'seconds':time.monotonic()-started}
(a.dataset/'trusted-depth-and-seed-audit.json').write_text(json.dumps(report,indent=2));faulthandler.cancel_dump_traceback_later();print(json.dumps({k:v for k,v in report.items() if k not in ['method','frames']},indent=2))
```

E6. Cold-start full reconstruction controller - complete

continue_explicit_full.py | Complete file
SHA-256 f75e8dae7b45e2ba29e0107fc7b117af63d9babfd2b1a3accc9f98532cb1c67a

continue_explicit_full.py

```
"""Continue the verified explicit-pinhole workflow; stop before visual release QA."""
from pathlib import Path
import hashlib,json,os,subprocess,sys,time
import cv2,numpy as np
W=Path(__file__).resolve().parent;ROOT=Path('C:/Users/bombe/OneDrive/Documents/ChatGPT/Ruang Kamare/roomscan-reconstruction/data/20261031_OSMO360/clean-v3/explicit-full-v1');DS=ROOT/'dataset';STATE=ROOT/'analysis/continuation.json'
if STATE.exists():raise FileExistsError('Continuation already has state; inspect before starting a second owner')
STATE.parent.mkdir(parents=True,exist_ok=True);state={'pid':os.getpid(),'started_unix':time.time(),'status':'running','stage':'waiting_geometry'}
def save(stage=None,**extra):
    if stage:state['stage']=stage
    state.update(extra,updated_unix=time.time());temporary=STATE.with_suffix('.json.partial');temporary.write_text(json.dumps(state,indent=2));temporary.replace(STATE)
def sha(path):
    with path.open('rb') as f:return hashlib.file_digest(f,'sha256').hexdigest()
def run(stage,script,*args):
    save(stage);result=subprocess.run([sys.executable,str(W/script),*map(str,args)],cwd=W)
    if result.returncode:raise RuntimeError(f'{stage} exited {result.returncode}')
save()
try:
    pilot_path=W/'review/bath-pilot-review.json';pilot=json.loads(pilot_path.read_text())
    if pilot['status']!='pilot_passed_for_full_scene_processing' or not pilot['firstVisitAndReturnViewReviewed']:raise ValueError('Bathroom pilot has not passed review')
    deadline=time.monotonic()+3600
    while True:
        geometry=json.loads((ROOT/'analysis/geometry.json').read_text())
        if geometry['status']=='finished' and geometry.get('exit_code')==0:break
        if geometry['status']=='failed':raise RuntimeError('Full geometry inference failed')
        if time.monotonic()>deadline:raise TimeoutError('Geometry stage exceeded one hour; inspect owned process')
        time.sleep(5)
    builder_path=DS/'explicit-pinhole-audit.json';builder=json.loads(builder_path.read_text())
    if builder['sourceInputs']!=452 or len({v['sourceName'] for v in builder['views']})!=452 or builder['subset']!='all':raise ValueError('Full physical-source coverage failed')
    for relative,expected in builder['sha256'].items():
        if sha(DS/relative)!=expected:raise ValueError('Explicit camera/seed input changed: '+relative)
    if len(list((DS/'depths').glob('*.png')))!=builder['virtualViews'] or len(list((DS/'normals').glob('*.png')))!=builder['virtualViews']:raise ValueError('Incomplete raw geometry maps')
    run('scale_only_depth_checks','fit_explicit_depth.py','--dataset',DS)
    scale=json.loads((DS/'depth-scale-audit.json').read_text())
    if scale['acceptedScales']<100:raise ValueError('Too few independently scale-qualified full-scene views')
    run('multi_timestamp_depth_checks','validate_explicit_depth.py','--dataset',DS,'--max-seed','1500000')
```

continue_explicit_full.py (continued)

```
    trusted_path=DS/'trusted-depth-and-seed-audit.json';trusted=json.loads(trusted_path.read_text())
    if trusted['trustedDepthFrames']<100 or trusted['finalSeedPoints']<500000 or trusted['evalInputsUsed'] or trusted['offset']!=0:raise ValueError
('Full-scene trusted depth/seed gate failed')
    first=sum(r['written'] for r in trusted['frames'] if 540<=r['sourceFrame']<=720);second=sum(r['written'] for r in trusted['frames'] if 1470<=r[
'sourceFrame']<=1650)
    if min(first,second)<5:raise ValueError('One bathroom visit lacks trusted geometry')
    save('restricting_normal_guidance_to_validated_pixels')
    normal_dir=DS/'normals-trusted'
    if normal_dir.exists():raise FileExistsError(normal_dir)
    normal_dir.mkdir();normal_hashes={}
    for row in trusted['frames']:
        if not row['written']:continue
        name=Path(row['name']).with_suffix('.png');depth=cv2.imread(str(DS/'depths-trusted'/name),cv2.IMREAD_UNCHANGED);normal=cv2.imread(str(DS/'n
ormals'/name),cv2.IMREAD_COLOR)
        if normal is None or depth is None or depth.shape!=normal.shape[:2]:raise ValueError('Trusted depth/normal shape mismatch')
        normal[depth==0]=0;target=normal_dir/name
        if not cv2.imwrite(str(target),normal):raise IOError(target)
        normal_hashes[name.as_posix()]=sha(target)
    (DS/'trusted-normal-manifest.json').write_text(json.dumps({'normalFiles':normal_hashes,'mask':'Only multi-view-consistent depth pixels remain n
ormal-valid; encoded black is invalid','sourceNormalsModified':False},indent=2))
    template=json.loads((W/'explicit-bath-train-job.json').read_text());command=template['command'][:]
    def option(flag,value):
        index=command.index(flag);command[index+1]=str(value)
    for flag,value in {'--data':DS,'--output-dir-prefix':ROOT/'training','--output-dir-name':'osmo-explicit-depth-v1','--num-iterations':50000,'--r
efine-stop-iter':35000,'--cap-max':3000000,'--seed-pointcloud':DS/'seed-dense-train.ply','--steps-per-save':5000}.items():option(flag,value)
    job={'stage':'explicit-full-cold-depth-rebuild','command':command,'cwd':str(W),'log':str(ROOT/'logs/train.log'),'status':str(ROOT/'analysis/tra
in.json'),'provenance':{'coldStart':True,'oldCheckpointUsed':False,'physicalSourceImages':452,'registeredSourcePairs':226,'virtualTrainViews':build
er['trainViews'],'virtualEvalViews':builder['evalViews'],'positiveDepthScaleOnly':True,'depthOffset':0,'depthAndSeedAuditSha256':sha(trusted_path),
'seedSha256':trusted['seedSha256'],'explicitCameraAuditSha256':sha(builder_path),'trustedNormalManifestSha256':sha(DS/'trusted-normal-manifest.json
'),'pilotReviewSha256':sha(pilot_path),'scope':'Full Osmo capture; same audited fresh +Y-up world coordinates; native Gaussian delivery planned'}}
    job_path=W/'explicit-full-train-job.json';job_path.write_text(json.dumps(job,indent=2))
    run('training_from_step_zero','run_logged_job.py',job_path)
    save('training_complete_pending_visual_review',status='finished',fullCheckpoint=str(ROOT/'training/osmo-explicit-depth-v1/step-000050000.ckpt/s
plat.ply'),publicDeploymentApproved=False)
except BaseException as error:
    save(status='failed',error=repr(error));raise
```

E7. SOG packing and preservation audit - complete

review/pack-sog.py | Complete file
SHA-256 e25b39c70fa08ab3c32530e3c632899addf09a9767c6beec7828e55faba1898a

pack-sog.py

```
"""Reproducible official SOG packing with an explicit Mip tag; CPU by default."""
from pathlib import Path
import argparse, hashlib, json, subprocess, time, zipfile
HERE=Path(__file__).resolve().parent
parser=argparse.ArgumentParser();parser.add_argument('--source',type=Path,required=True);parser.add_argument('--output',type=Path,required=True);pa
rser.add_argument('--gpu',default='cpu');parser.add_argument('--gpu-backend',choices=['vulkan','d3d12','metal']);args=parser.parse_args()
source=args.source.resolve();output=args.output.resolve();tool=HERE/'converter/node_modules/@playcanvas/splat-transform/bin/cli.mjs'
if output.exists():raise FileExistsError(output)
scratch=HERE/(output.stem+'.mip-tagged-input.ply')
if scratch.exists():raise FileExistsError(scratch)
start=time.monotonic()
source_hash=hashlib.sha256()
# Spirula's PLY lacks its config's primitive tag. Add only a standard metadata
# comment to a temporary copy; preserve every binary Gaussian attribute byte.
with source.open('rb') as f:
    header=b''
    while not header.endswith(b'end_header\n'):
        line=f.readline();header+=line
        if not line or len(header)>100000:raise ValueError('Invalid PLY header')
    if b'format binary_little_endian 1.0' not in header:raise ValueError('Expected binary little-endian PLY')
    if b'SplatRenderMode:' in header and b'SplatRenderMode: mip' not in header:raise ValueError('Source declares a different primitive; do not over
ride it')
    properties=[line.split()[-1] for line in header.decode().splitlines() if line.startswith('property ')]
    rest=[name for name in properties if name.startswith('f_rest_')]
    degree_by_rest_count={0:0,9:1,24:2,45:3}
    if len(rest) not in degree_by_rest_count or set(rest)!={f'f_rest_{i}' for i in range(len(rest))}:raise ValueError('Unexpected SH coefficient la
yout')
    source_degree=degree_by_rest_count[len(rest)]
    tagged=header.replace(b'end_header\n',b'comment SplatRenderMode: mip\nend_header\n')
    source_hash.update(header)
    with scratch.open('xb') as dst:
        dst.write(tagged)
        while chunk:=f.read(1024*1024):source_hash.update(chunk);dst.write(chunk)
command=['node',str(tool),'-g',args.gpu,'--max-workers','2']
if args.gpu_backend:command+=['--gpu-backend',args.gpu_backend]
command += [str(scratch),str(output)]
log=output.with_suffix('.sog.log')
with log.open('w',encoding='utf-8') as f:result=subprocess.run(command,stdout=f,stderr=subprocess.STDOUT,cwd=HERE)
if result.returncode:raise RuntimeError(f'Converter exit {result.returncode}; inspect {log}')
with zipfile.ZipFile(output) as z:meta=json.loads(z.read('meta.json'));members=[{'name':x.filename,'bytes':x.file_size} for x in z.infolist()]
count=int(next(x.split()[-1] for x in header.decode().splitlines() if x.startswith('element vertex ')))
if meta['count']!=count or meta.get('model')!='antialiased':raise AssertionError('Count/antialias tag changed')
output_degree=meta.get('shN',{}).get('bands',0)
if output_degree!=source_degree:raise AssertionError('SH degree changed')
with output.open('rb') as f:digest=hashlib.file_digest(f,'sha256').hexdigest()
```

pack-sog.py (continued)

```
report={'source':str(source),'output':str(output),'sourceSha256':source_hash.hexdigest(),'sourceBytes':source.stat().st_size,'bytes':output.stat().
+ st_size,'sha256':digest,'splats':count,'shDegree':output_degree,'sourceShDegree':source_degree,'model':meta.get('model'),'sourceCountPreserved':Tru
+ e,'sourceShDegreePreserved':True,'under50MB':output.stat().st_size<=50000000,'format':'sog','tool':'@playcanvas/splat-transform 3.10.3','command':c
+ ommand,'primitiveDeclaration':'Mip supplied by wrapper, consistent with the audited old v4 and planned fresh mip training. Verify training config b
+ efore using on another scene.','sourceModification':'Temporary header-only Mip comment added; all binary splat bytes unchanged. Original file unmod
+ ified.','members':members,'seconds':time.monotonic()-start,'validation':'Count/SH-degree/model metadata only. Visual fidelity must be checked separ
+ ately against original Gaussian renders; not a phone loading benchmark.'}
output.with_suffix('.sog.json').write_text(json.dumps(report,indent=2))
# This exact disposable copy was created above, exclusively inside this review.
assert scratch.parent==HERE and scratch.name==output.stem+'.mip-tagged-input.ply';scratch.unlink()
print(json.dumps({k:v for k,v in report.items() if k!='members'},indent=2))
```

E8. Native format, budget and nonempty-draw policy - complete

review/production/native-delivery.js | Complete file
SHA-256 85a419a301c77e1e3a236f0e39de09c198c1d391727c570426fefef7dde68fa0

native-delivery.js

```
export const NATIVE_LIMITS=Object.freeze({fileBytes:50_000_000,baseSplats:3_100_000,phoneRenderedSplats:750_000,desktopRenderedSplats:2_000_000});
export function nativeSparkFileType(format){
 // Spark's API calls a zipped .sog file "pcsogszip". Passing the filename
 // extension as an explicit type bypasses auto-detection and cannot decode it.
 if(format==='sog')return 'pcsogszip';
 if(format==='spz')return 'spz';
 throw Error('Unsupported native room format.');
}
export function nativePolicy(asset,phone=false){
 if(asset.representation!=='gaussian'||!['spz','sog'].includes(asset.format))throw Error('Unsupported native room format.');
 if(!Number.isSafeInteger(asset.fileBytes)||asset.fileBytes<12||asset.fileBytes>NATIVE_LIMITS.fileBytes)throw Error('Native room exceeds its downlo
+ ad budget.');
 if(!Number.isSafeInteger(asset.splats)||asset.splats<1||asset.splats>NATIVE_LIMITS.baseSplats)throw Error('Native room exceeds its decoded splat b
+ udget.');
 return {lodSplatCount:phone?NATIVE_LIMITS.phoneRenderedSplats:NATIVE_LIMITS.desktopRenderedSplats,baseSplats:asset.splats,lod:true,nonLod:false,ex
+ tSplats:false};
}
export function validateModelHeader(bytes,format='glb'){
 const view=new DataView(bytes.buffer,bytes.byteOffset,bytes.byteLength);
 if(format==='glb')return view.getUint32(0,true)===0x46546c67&&view.getUint32(4,true)===2&&view.getUint32(8,true)===bytes.byteLength;
 if(format==='spz')return (bytes[0]===0x1f&&bytes[1]===0x8b)||view.getUint32(0,true)===0x5053474e;
 if(format==='sog')return view.getUint32(0,true)===0x04034b50;
 return false;
}

```

native-delivery.js (continued)

```
// Decoding the file is not sufficient: Spark's first sort may still be pending.
// Keep the loading state until an actual draw has a nonzero active splat count.
export async function waitForNativeDraw({update,draw,activeSplats,isCurrent=()=>true,signal,timeoutMs=30000}){
 let stopped=false,timer,rejectAbort;
 const abortError=()=>new DOMException('Room preparation cancelled','AbortError');
 const check=()=>{if(stopped||signal?.aborted||!isCurrent())throw abortError();};
 const aborted=new Promise((_,reject)=>{rejectAbort=reject;});
 const abort=()=>{stopped=true;rejectAbort(abortError());};
 signal?.addEventListener('abort',abort,{once:true});
 const work=(async()=>{
  for(;;){
   check();await update();check();draw();check();
   const count=activeSplats();
   if(Number.isFinite(count)&&count>0)return count;
   await new Promise(resolve=>setTimeout(resolve,16));
  }
 })();
 const deadline=new Promise((_,reject)=>{timer=setTimeout(()=>{
  stopped=true;reject(new Error('The browser could not draw the room. Choose a model to retry, or use Photos.'));
 },timeoutMs);});
 try{return await Promise.race([work,deadline,aborted]);}
 finally{stopped=true;clearTimeout(timer);signal?.removeEventListener('abort',abort);}
}
```

E9. Native keep-mask composition

masking/build_masks.py | Exact source excerpt, lines 150-208
SHA-256 668650fbdff04025f9abd29078a5770318bfa805519a1a56ccee7f409e321cf4

build_masks.py

```
    h, w = shape
    out = np.zeros(shape, np.uint8)
    for polygon in polygons:
        xy = np.asarray(polygon, dtype=float)
        if xy.ndim != 2 or xy.shape[1] != 2 or len(xy) < 3 or not np.isfinite(xy).all():
            raise ValueError('Polygon must contain at least 3 finite normalized XY points')
        if xy.min() < 0 or xy.max() > 1:
            raise ValueError('Polygon must use normalized source-image coordinates')
        cv2.fillPoly(out, [np.round(xy * [w, h]).astype(np.int32)], 255)
    return out


def finalize(args):
```

build_masks.py (continued)

```
    records = selected(args)
    if args.source_frames:
        records = [r for r in records if r['source_frame'] in args.source_frames]
    annotations = json.loads(args.annotations.read_text(encoding='utf-8-sig'))
    circles = annotations.get('optical_circle', {cam:[.5,.5,.485] for cam in args.cameras})
    reviewed = annotations['frames']
    audit_path = args.work / ('candidate-audit.json' if args.candidate else 'mask-audit.json')
    replaced = {(cam, r['source_frame']) for cam in args.cameras for r in records}
    rows = [r for r in json.loads(audit_path.read_text())['frames']
            if (r['camera'], r['source_frame']) not in replaced] if args.source_frames and audit_path.exists() else []
    for cam in args.cameras:
        final_dir = (args.work / 'candidate-final' if args.candidate else args.dataset / 'masks') / cam
        final_dir.mkdir(parents=True, exist_ok=True)
        for r in records:
            key = cam + '/' + r['filename']
            if key not in reviewed or (not args.candidate and not reviewed[key].get('reviewed')):
                raise ValueError(f'Visual review/mirror decision absent: {key}')
            raw_path = args.work / args.operator_layer / cam / Path(r['filename']).with_suffix('.png')
            mask = np.asarray(Image.open(raw_path).convert('L'))
            if mask.shape != (3840,3840):
                raise ValueError(f'Invalid raw dimensions: {raw_path}')
            remove = (mask < 128).astype(np.uint8)
            radius = args.dilation
            if radius:
                kernel = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (2*radius+1, 2*radius+1))
                remove = cv2.dilate(remove, kernel)
            review = reviewed[key]
            mirror = polygon_mask(mask.shape, review.get('mirror_polygons', []))
            manual = polygon_mask(mask.shape, review.get('extra_exclusion_polygons', []))
            if mirror.any():
                mirror = cv2.dilate(mirror, cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (33,33)))
            keep = optical_keep(mask.shape, circles[cam]) & (remove == 0) & (mirror == 0) & (manual == 0)
            output = (keep.astype(np.uint8)*255)
            dest = final_dir / Path(r['filename']).with_suffix('.png')
            Image.fromarray(output).save(dest)
            row = {'camera':cam, **r, 'keep_fraction':float(keep.mean()),
                   'operator_fraction':float((remove != 0).mean()),
                   'mirror_fraction':float((mirror != 0).mean()),
                   'sha256':hashlib.sha256(dest.read_bytes()).hexdigest()}
            rows.append(row)
    write_json(args.work / ('candidate-audit.json' if args.candidate else 'mask-audit.json'), {
        'semantics':'255=keep; 0=excluded ray. Masked background is not automatically observed.',
        'shape':[3840,3840], 'operator_dilation_radius_px':args.dilation,
        'mirror_dilation_radius_px':16, 'optical_circle':circles,
        'review_annotations_sha256':hashlib.sha256(args.annotations.read_bytes()).hexdigest(),
```

build_masks.py (continued)

```
208          'frames':rows})
```

E10. Camera, overview and cutaway behavior

review/production/room-viewer.js | Exact source excerpt, lines 17-141
SHA-256 35b8e275baa7e40193d2a75810bb78b4a99df1a0d62af95f708b01cda8a11d64

room-viewer.js

```
17  try{renderer=new THREE.WebGLRenderer({canvas,antialias:selectedOnOpen==='osmo'?false:!phoneInput,alpha:true,powerPreference:'high-performance'});
18  renderer.setPixelRatio(Math.min(devicePixelRatio,phoneInput?1:1.25));
19  renderer.setClearColor(0x141b23,1);renderer.outputColorSpace=THREE.SRGBColorSpace;}
20  catch(error){rendererError=error;}
21  const scene=new THREE.Scene(),camera=new THREE.PerspectiveCamera(75,1,.025,100);
22  camera.up.set(0,1,0);
23  scene.add(new THREE.HemisphereLight(0xffffff,0x6c7890,2));
24  const light=new THREE.DirectionalLight(0xffffff,2);light.position.set(3,8,5);scene.add(light);
25  const keys=new Set(),velocity=new THREE.Vector3(),wheelVelocity=new THREE.Vector3();
26  const forward=new THREE.Vector3(),right=new THREE.Vector3(),euler=new THREE.Euler(0,0,0,'YXZ');
27  let data,model,mode='walk',yaw=0,pitch=0,speed=1.2,drag=null,ready=false,dirty=true;
28  let recorded=null,comparison=false,seeInside=false,shaded=false;
29  let orbit={target:new THREE.Vector3(),theta:0,phi:1,radius:8};
30  let framingPoints=new Float32Array();
31  let forceRender=false,frames=[],last=performance.now(),renderCount=0;
32  let manifest,manifestPromise,currentQuality,asset,selection,downloadAbort,phase='starting',loadStarted=0;
33  let downloadBytes=0,preparedMs=0,modelDiagnostics={},idleTimer,totalTimer,metadataReady=false;
34  const pointers=new Map(),touchMoves=new Map();
35  let gesture=null;
36
37  function invalidate(){dirty=true;}
38  function release(){keys.clear();touchMoves.clear();pointers.clear();gesture=null;velocity.set(0,0,0);wheelVelocity.set(0,0,0);drag=null;}
39  function focus(){canvas.focus({preventScroll:true});}
40  function angles(){camera.getWorldDirection(forward);yaw=Math.atan2(-forward.x,-forward.z);pitch=Math.asin(THREE.MathUtils.clamp(forward.y,-1,1));}
41  function orient(){euler.set(pitch,yaw,0,'YXZ');camera.quaternion.setFromEuler(euler);camera.up.set(0,1,0);}
42  function clearRecorded(){recorded=null;if(comparison)setCompare(false);$('views').selectedIndex=0;camera.fov=mode==='orbit'&&!nativeGaussian?42:75;
+ resize();}
43  function updateSurface(){
44    if(!model)return;
45    if(nativeGaussian){seeInside=false;$('see-inside').setAttribute('aria-pressed','false');invalidate();return;}
46    model.traverse(o=>{if(o.isMesh){for(const m of (Array.isArray(o.material)?o.material:[o.material])){
47      // Open the exterior by back-face culling. Never slice cabinets, frames or walls by height.
48      m.clippingPlanes=[];m.side=seeInside?THREE.FrontSide:THREE.DoubleSide;m.needsUpdate=true;
49    }}});invalidate();
50    $('see-inside').setAttribute('aria-pressed',String(seeInside));
```

room-viewer.js (continued)

```
}
function updateMode(){
  $('walk').setAttribute('aria-pressed',String(mode==='walk'));
  $('orbit').setAttribute('aria-pressed',String(mode==='orbit'));
  $('touch-controls').classList.toggle('walking',ready&&mode==='walk'&&!comparison);
  $('views').disabled=!ready||!data?.presets?.length;$('compare').disabled=!ready||!data?.presets?.length;
  $('hint').textContent=phoneInput?(mode==='walk'?'Drag to look \xb7 Hold arrows to walk \xb7 Pinch to move':'Drag to orbit \xb7 Pinch to zoom \xb7
 Two fingers to pan'):(mode==='walk'?'Drag to look \xb7 WASD move \xb7 Q/E down/up \xb7 Wheel step':'Drag to orbit \xb7 Right-drag to pan \xb7 Whee
l zoom \xb7 WASD pan');
}
function home(){
  if(!ready)return;release();recorded=null;comparison=false;$('source').hidden=true;stage.classList.remove('compare');
  camera.position.fromArray(data.startCamera.position);camera.quaternion.fromArray(data.startCamera.quaternion);
  if(data.homeTarget)camera.lookAt(new THREE.Vector3().fromArray(data.homeTarget));
  angles();pitch=0;orient();mode='walk';seeInside=false;updateSurface();updateMode();
  $('views').selectedIndex=0;$('compare').setAttribute('aria-pressed','false');camera.fov=75;resize();invalidate();
}
function orbitCamera(){
  const {radius:r,phi:p,theta:t,target}=orbit;
  camera.position.set(r*Math.sin(p)*Math.sin(t),r*Math.cos(p),r*Math.sin(p)*Math.cos(t)).add(target);
  camera.up.set(0,1,0);camera.lookAt(target);angles();invalidate();
}
function fittedOrbitRadius(aspect=camera.aspect){
  if(nativeGaussian)return orbit.radius;
  const b=data.bounds,center=new THREE.Vector3().fromArray(b[0]).add(new THREE.Vector3().fromArray(b[1])).multiplyScalar(.5);
  const direction=new THREE.Vector3(Math.sin(orbit.phi)*Math.sin(orbit.theta),Math.cos(orbit.phi),Math.sin(orbit.phi)*Math.cos(orbit.theta));
  const horizontal=new THREE.Vector3(Math.cos(orbit.theta),0,-Math.sin(orbit.theta));
  const vertical=new THREE.Vector3().crossVectors(direction,horizontal);
  const tangent=Math.tan(THREE.MathUtils.degToRad(camera.fov/2));let radius=.3;
  const point=new THREE.Vector3();
  for(let i=0;i<framingPoints.length;i+=3){
    point.fromArray(framingPoints,i).sub(center);const depth=point.dot(direction);
    radius=Math.max(radius,depth+1.12*Math.abs(point.dot(horizontal))/(tangent*aspect),depth+1.12*Math.abs(point.dot(vertical))/tangent);
  }
  return radius;
}
function setMode(next){
  if(!ready)return;release();clearRecorded();mode=next;
  if(next==='orbit'){
    camera.fov=nativeGaussian?75:42;camera.updateProjectionMatrix();
    if(nativeGaussian){
      // Preserve the current interior viewpoint. The video never observed the
      // apartment exterior, so the mesh dollhouse framing is not a valid GS start.
      camera.getWorldDirection(forward);orbit.target.copy(camera.position).addScaledVector(forward,2);
      const offset=camera.position.clone().sub(orbit.target);orbit.radius=offset.length();
```

room-viewer.js (continued)

```
      orbit.theta=Math.atan2(offset.x,offset.z);orbit.phi=Math.acos(THREE.MathUtils.clamp(offset.y/orbit.radius,-1,1));
    }else{
      const b=data.bounds;orbit.target.fromArray(b[0]).add(new THREE.Vector3().fromArray(b[1])).multiplyScalar(.5);
      orbit.theta=.7;orbit.phi=.85;orbit.radius=fittedOrbitRadius();
    }
    seeInside=true;orbitCamera();
  }else{camera.fov=75;camera.updateProjectionMatrix();if(camera.position.y>3.3||camera.position.y<.15)home();angles();orient();seeInside=false;}
  updateSurface();updateMode();invalidate();
}
function top(){setMode('orbit');orbit.phi=.001;orbit.radius=fittedOrbitRadius();orbitCamera();focus();}
function rotate(){
  clearRecorded();release();
  if(mode==='orbit'){const zoom=orbit.radius/fittedOrbitRadius();orbit.theta+=Math.PI/2;orbit.radius=fittedOrbitRadius()*zoom;orbitCamera();}
  else{yaw+=Math.PI/2;orient();invalidate();}focus();
}
function preset(index){
  const c=data.presets[index];if(!c)return;
  release();recorded=c;camera.position.fromArray(c.position);camera.quaternion.fromArray(c.quaternion);
  camera.up.set(0,1,0);angles();mode='walk';seeInside=false;updateSurface();updateMode();
  $('views').value=String(index);$('source-frame').src=c.source;
  $('source-label').textContent=c.label+' \xb7 '+c.name;resize();invalidate();
}
function setCompare(value){
  if(!ready||!data.presets.length)return;
  if(value&&!recorded){let nearest=0,distance=Infinity;data.presets.forEach((c,i)=>{const d=camera.position.distanceToSquared(new THREE.Vector3().f
+ romArray(c.position));if(d<distance){distance=d;nearest=i;}});preset(nearest);}
  comparison=value;stage.classList.toggle('compare',value);$('source').hidden=!value;
  $('compare').setAttribute('aria-pressed',String(value));updateMode();resize();invalidate();
}
function calibratedProjection(c){
  const [fx,fy,cx,cy]=c.intrinsics,w=c.imageWidth,h=c.imageHeight,n=camera.near;
  camera.projectionMatrix.makePerspective(-cx/fx*n,(w-cx)/fx*n,cy/fy*n,-(h-cy)/fy*n,n,camera.far);
  camera.projectionMatrixInverse.copy(camera.projectionMatrix).invert();
}
function viewport(){
  const w=canvas.clientWidth,h=canvas.clientHeight;
  if(recorded){const ratio=recorded.imageWidth/recorded.imageHeight,ww=Math.min(w,h*ratio),hh=ww/ratio;
    return {x:(w-ww)/2,y:(h-hh)/2,w:ww,h:hh};}
  return {x:0,y:0,w,h};
}
function resize(){
  if(!renderer)return;
  const w=canvas.clientWidth,h=canvas.clientHeight;if(!w||!h)return;
  if(ready&&mode==='orbit'&&!recorded){orbit.radius*=fittedOrbitRadius(w/h)/fittedOrbitRadius(camera.aspect);orbitCamera();}
  renderer.setSize(w,h,false);camera.aspect=w/h;camera.updateProjectionMatrix();
```

room-viewer.js (continued)

```
  if(recorded)calibratedProjection(recorded);invalidate();
}
const observer=new ResizeObserver(resize);observer.observe(stage);observer.observe(canvas);
window.addEventListener('resize',()=>{phoneInput=chooseQuality({...deviceSignals,requested:null,width:innerWidth})==='phone';document.body.classLis
t.toggle('phone',phoneInput);resize();updateMode();});
```

E11. Gestures, movement, reset and mode handlers

review/production/room-viewer.js | Exact source excerpt, lines 151-218
SHA-256 35b8e275baa7e40193d2a75810bb78b4a99df1a0d62af95f708b01cda8a11d64

room-viewer.js

```
canvas.addEventListener('pointermove',e=>{
  if(pointers.has(e.pointerId))pointers.set(e.pointerId,{x:e.clientX,y:e.clientY});
  if(pointers.size>1){
    const next=gestureState();
    if(gesture){zoom(pinchRatio(gesture.distance,next.distance));pan(next.x-gesture.x,next.y-gesture.y);}
    gesture=next;e.preventDefault();return;
  }
  if(!drag||drag.id!==e.pointerId)return;
  const dx=e.clientX-drag.x,dy=e.clientY-drag.y;drag.x=e.clientX;drag.y=e.clientY;
  if(mode==='walk'){
    if(drag.pan){right.set(Math.cos(yaw),0,-Math.sin(yaw));camera.position.addScaledVector(right,-dx*.003);camera.position.y+=dy*.003;}
    else{yaw-=dx*.003;pitch=THREE.MathUtils.clamp(pitch-dy*.003,-1.45,1.45);orient();}
  }else if(drag.pan){right.set(Math.cos(orbit.theta),0,-Math.sin(orbit.theta));orbit.target.addScaledVector(right,-dx*orbit.radius*.001);orbit.targ
et.y+=dy*orbit.radius*.001;orbitCamera();}
  else{orbit.theta-=dx*.004;orbit.phi=THREE.MathUtils.clamp(orbit.phi-dy*.004,.001,1.55);orbitCamera();}
  invalidate();
});
for(const event of ['pointerup','pointercancel','lostpointercapture'])canvas.addEventListener(event,e=>{pointers.delete(e.pointerId);drag=null;gest
ure=null;});
canvas.addEventListener('wheel',e=>{
  if(!ready)return;e.preventDefault();clearRecorded();
  const dy=THREE.MathUtils.clamp(e.deltaY*(e.deltaMode===1?16:1),-120,120);
  if(mode==='walk'){forward.set(-Math.sin(yaw),0,-Math.cos(yaw));wheelVelocity.addScaledVector(forward,-dy*.022*speed);}
  else{orbit.radius=THREE.MathUtils.clamp(orbit.radius*Math.exp(dy*.001),.3,80);orbitCamera();}invalidate();
},{passive:false});
const movement=new Set(['KeyW','KeyS','KeyA','KeyD','KeyQ','KeyE','Space','KeyC']);
window.addEventListener('keydown',e=>{
  if(!ready||e.ctrlKey||e.metaKey||e.altKey||e.target.closest?.('select,input,button,a'))return;
  if(e.code==='Escape'){release();return;}
  if(e.code==='KeyR'){e.preventDefault();home();return;}
  if(!movement.has(e.code)&&!['ShiftLeft','ShiftRight'].includes(e.code))return;
```

room-viewer.js (continued)

```
  e.preventDefault();if(!keys.size){angles();clearRecorded();orient();}keys.add(e.code);invalidate();
});
window.addEventListener('keyup',e=>{keys.delete(e.code);});
function step(dt){
  forward.set(-Math.sin(yaw),0,-Math.cos(yaw));right.set(Math.cos(yaw),0,-Math.sin(yaw));
  const touch=new Set(touchMoves.values());
  const f=Number(keys.has('KeyW')||touch.has('forward'))-Number(keys.has('KeyS')||touch.has('back')),s=Number(keys.has('KeyD')||touch.has('right'))
-Number(keys.has('KeyA')||touch.has('left'));
  const y=Number(keys.has('KeyE')||keys.has('Space')||touch.has('up'))-Number(keys.has('KeyQ')||keys.has('KeyC')||touch.has('down'));
  const desired=new THREE.Vector3().addScaledVector(forward,f).addScaledVector(right,s);desired.y=y;
  if(desired.lengthSq()>1)desired.normalize();desired.multiplyScalar(speed*(keys.has('ShiftLeft')||keys.has('ShiftRight')?2.7:1));
  velocity.lerp(desired,1-Math.exp(-22*dt));
  if(!keys.size&&!touchMoves.size&&velocity.lengthSq()<.00001)velocity.set(0,0,0);
  const delta=velocity.clone().add(wheelVelocity).multiplyScalar(dt);
  wheelVelocity.multiplyScalar(Math.exp(-12*dt));
  if(delta.lengthSq()>1e-10){
    if(mode==='walk')camera.position.add(delta);
    else{orbit.target.add(delta);orbitCamera();}invalidate();
  }
}
function render(){
  const v=viewport();renderer.setScissorTest(false);renderer.setViewport(0,0,canvas.clientWidth,canvas.clientHeight);renderer.clear();
  renderer.setViewport(v.x,v.y,v.w,v.h);renderer.render(scene,camera);
  // Verification waits for GPU completion; normal navigation stays asynchronous.
  if(forceRender)renderer.getContext().finish();
  renderCount++;
}
function loop(now){
  const dt=Math.min((now-last)/1000,.05);last=now;
  if(ready&&phase==='ready')step(dt);
  if(ready&&phase==='ready'&&(dirty||forceRender)){try{render();dirty=false;frames.push(dt*1000);if(frames.length>2400)frames.shift();updateDiagnos
tics();}catch(error){showFailure(error);}}
  requestAnimationFrame(loop);
}
requestAnimationFrame(loop);
$('reset').onclick=()=>{home();focus();};$('top').onclick=top;$('rotate').onclick=rotate;
$('walk').onclick=()=>{setMode('walk');focus();};$('orbit').onclick=()=>{setMode('orbit');focus();};
$('see-inside').onclick=()=>{seeInside=!seeInside;updateSurface();focus();};
$('speed').onchange=()=>{speed=Number($('speed').value);focus();};
$('views').onchange=()=>{if($('views').value!=='')preset(Number($('views').value));focus();};
$('compare').onclick=()=>{setCompare(!comparison);focus();};
```

E12. Zoom / pan, cancellation and hidden native controls

review/production/room-viewer.js | Exact source excerpt, lines 236-274
SHA-256 35b8e275baa7e40193d2a75810bb78b4a99df1a0d62af95f708b01cda8a11d64

room-viewer.js

```
function gestureState(){const [a,b]=[...pointers.values()];return {x:(a.x+b.x)/2,y:(a.y+b.y)/2,distance:Math.hypot(a.x-b.x,a.y-b.y)};}
function zoom(ratio){
  if(!ready)return;
  if(mode==='orbit'){orbit.radius=THREE.MathUtils.clamp(orbit.radius*ratio,.3,80);orbitCamera();}
  else{camera.getWorldDirection(forward);camera.position.addScaledVector(forward,(1-ratio)*2);invalidate();}
}
function pan(dx,dy){
  const factor=(mode==='orbit'?orbit.radius:3)*.0017;
  right.set(1,0,0).applyQuaternion(camera.quaternion);
  const up=new THREE.Vector3(0,1,0).applyQuaternion(camera.quaternion);
  const target=mode==='orbit'?orbit.target:camera.position;
  target.addScaledVector(right,-dx*factor).addScaledVector(up,dy*factor);
  if(mode==='orbit')orbitCamera();else invalidate();
}
for(const button of document.querySelectorAll('[data-move]')){
  button.addEventListener('pointerdown',e=>{if(!ready||mode!=='walk')return;e.preventDefault();button.setPointerCapture(e.pointerId);touchMoves.set
+(e.pointerId,button.dataset.move);invalidate();});
  for(const event of ['pointerup','pointercancel','lostpointercapture'])button.addEventListener(event,e=>{touchMoves.delete(e.pointerId);invalidate
+();});
}
$('zoom-in').onclick=()=>{clearRecorded();zoom(.82);};$('zoom-out').onclick=()=>{clearRecorded();zoom(1/.82);};
$('more').onclick=()=>{const open=$('extra').classList.toggle('open');$('more').setAttribute('aria-expanded',String(open));resize();};
function chooseAnotherModel(key){loads.cancel();downloadAbort?.abort();stopTimers();release();ready=false;disposeModel(model);model=null;const url=
+new URL(location.pathname,location.origin);url.searchParams.set('quality',key);url.searchParams.set('v','20261031-native-provisional');location.ass
+ign(url.href);}
$('quality').onchange=()=>chooseAnotherModel($('quality').value);
$('change-model').onclick=()=>chooseAnotherModel(currentQuality||selectedOnOpen);
$('retry').onclick=$('recover-phone').onclick=()=>chooseAnotherModel(currentQuality==='osmo'?'osmo':'phone');
$('cancel-load').onclick=()=>chooseAnotherModel(currentQuality||selectedOnOpen);
async function showGallery(){
  try{await loadMetadata();if(!data)applyCameraData(currentQuality||chooseQuality(deviceSignals));const c=data.photos[Number($('gallery-view').valu
+e)||0];if(!c)return;$('gallery-photo').src=c.source;$('gallery-caption').textContent=c.label+(c.name?' \xb7 '+c.name:'');if(!$('gallery').open)$('g
+allery').showModal();release();}
  catch(error){$('load-status').textContent='Photos could not open. Check the connection and try again.';}
}
$('photos').onclick=$('loading-photos').onclick=$('recovery-photos').onclick=showGallery;
$('gallery-view').onchange=showGallery;$('gallery-close').onclick=()=>{$('gallery').close();$('gallery-photo').removeAttribute('src');};

function controlsEnabled(value){for(const button of document.querySelectorAll('[data-3d],[data-move]'))button.disabled=!value;for(const id of ['sha
+de','see-inside','top']){$(id).hidden=nativeGaussian;if(nativeGaussian)$(id).disabled=true;}}
```

room-viewer.js (continued)

```
function updateDiagnostics(){
  const state={build:'20261031-native-provisional',phase,quality:currentQuality||'choosing',cameraSet:asset?.cameraSet||null,phoneInput,downloadBytes,fileBytes:asset?.fileBytes||0,
    representation:nativeGaussian?'gaussian':'mesh',splats:modelDiagnostics.splats||0,activeSplats:spark?.activeSplats||0,lodSplatBudget:modelDiagnostics.lodSplatBudget||0,triangles:modelDiagnostics.triangles||0,geometryMiB:modelDiagnostics.geometryMiB||0,textures:modelDiagnostics.textures||[],
    preparedSeconds:preparedMs/1000,mode,seeInside,heightClipping:false,renderCount,comparison,recorded:recorded?.name||null,
    position:camera.position.toArray(),activeMoves:[...touchMoves.values()],keys:[...keys],viewport:[canvas.clientWidth,canvas.clientHeight]};
  $('diagnostics').textContent=JSON.stringify(state,null,2);document.body.dataset.phase=phase;document.body.dataset.quality=state.quality;
```

E13. Renderer selection, LoD and startup acceptance

review/production/room-viewer.js | Exact source excerpt, lines 328-405
SHA-256 35b8e275baa7e40193d2a75810bb78b4a99df1a0d62af95f708b01cda8a11d64

room-viewer.js

```
async function loadAsset(quality){
  const request=loads.begin(),generation=request.id;downloadAbort?.abort();stopTimers();release();ready=false;controlsEnabled(false);
  disposeModel(model);model=null;framingPoints=new Float32Array();modelDiagnostics={};shaded=false;$('shade').setAttribute('aria-pressed','false');
  $('source').hidden=true;stage.classList.remove('compare');comparison=false;recorded=null;
  currentQuality=quality;asset=null;selection=null;data=null;downloadBytes=0;preparedMs=0;loadStarted=performance.now();
  if($('gallery').open)$('gallery').close();$('gallery-photo').removeAttribute('src');
  $('source-frame').removeAttribute('src');$('compare').setAttribute('aria-pressed','false');
  $('load-preview').hidden=true;
  $('quality').value=quality;$('loading').hidden=false;$('recovery').hidden=true;$('load-progress').hidden=false;$('load-progress').value=0;
  $('retry').hidden=true;$('cancel-load').hidden=false;$('cancel-load').textContent='Choose another model';$('load-title').textContent='Opening '+({phone:'iPhone Fast',native:'iPhone Full',osmo:'Osmo'}[quality]||'the model');
  setPhase('details','Opening room details\u2026');
  try{
    await loadMetadata();if(!loads.current(generation))return;applyCameraData(quality);
    if(!renderer)throw Error('3D graphics could not start on this browser. Try reloading the phone view or use Photos.');
    downloadAbort=request.controller;const controller=downloadAbort;
    let idleExpired=false,totalExpired=false;
    const resetIdle=()=>{clearTimeout(idleTimer);idleTimer=setTimeout(()=>{idleExpired=true;controller.abort();},45000);};
    resetIdle();totalTimer=setTimeout(()=>{totalExpired=true;controller.abort();},selection.downloadTimeout);
    setPhase('downloading','Downloading the room\u2026',`0 / ${(asset.fileBytes/1e6).toFixed(1)} MB`);
    let buffer;
    try{const response=await fetch(asset.url,{signal:controller.signal});
      buffer=await readModel(response,asset.fileBytes,{signal:controller.signal,format:asset.format||'glb',onProgress:(bytes,total)=>{
        downloadBytes=bytes;resetIdle();$('load-progress').value=Math.min(94,bytes/total*94);
        setPhase('downloading',`Downloading the room \xb7 ${Math.round(bytes/total*100)}%`,`${(bytes/1e6).toFixed(1)} / ${(total/1e6).toFixed(1)} MB`);
      }});
```

room-viewer.js (continued)

```
    }catch(error){if(idleExpired)throw Error('The connection stopped sending the room. Please retry this view.');if(totalExpired)throw Error('The d
ownload is taking too long. Retry this view or use Photos.');throw error;}
    stopTimers();if(!loads.current(generation))return;
    setPhase('preparing','Preparing the 3D room\u2026','The download is complete.');$('load-progress').value=96;
    await new Promise(resolve=>requestAnimationFrame(()=>requestAnimationFrame(resolve)));
    if(nativeGaussian){
      const policy=nativePolicy(asset,phoneInput);
      setPhase('preparing','Preparing the 3D room\u2026','The download is complete. Preparing the view for this device.');
      const {SparkRenderer,SplatMesh}=await withDeadline(import('../vendor/spark/dist/spark.module.js'),60000,'The 3D viewer could not download. Ch
oose a model to retry, or use Photos.');
      spark=new SparkRenderer({renderer,onDirty:invalidate,autoUpdate:false,lodSplatCount:policy.lodSplatCount,sortRadial:true,preBlurAmount:0,blur
Amount:.3,minSortIntervalMs:phoneInput?24:12});scene.add(spark);
      const candidate=new SplatMesh({fileBytes:buffer,fileType:nativeSparkFileType(asset.format),fileName:asset.url,lod:true,nonLod:false,extSplats
:false,editable:false,raycastable:false});candidate.userData.nativeGaussian=true;
      // Free the reference to the downloaded bytes as soon as the worker takes them.
      buffer=null;
      let abandoned=false;
      const parsing=candidate.initialized.then(result=>{if(abandoned||!loads.current(generation)){candidate.dispose();throw new DOMException('Super
seded','AbortError');}return result;});
      try{model=await withDeadline(parsing,selection.prepareTimeout,'The browser could not prepare this room. Choose a model to retry, or use Photo
s.');}
      catch(error){abandoned=true;candidate.dispose();scene.remove(spark);spark.dispose();spark=null;throw error;}
      if(!loads.current(generation)){disposeModel(model);return;}scene.add(model);
      if(asset.worldFromTraining){model.matrix.fromArray(asset.worldFromTraining);model.matrixAutoUpdate=false;}
      const [lo,hi]=data.bounds,points=[];for(const x of [lo[0],hi[0]])for(const y of [lo[1],hi[1]])for(const z of [lo[2],hi[2]])points.push(x,y,z)
;framingPoints=Float32Array.from(points);
      modelDiagnostics={splats:asset.splats,lodSplatBudget:policy.lodSplatCount,triangles:0,geometryMiB:0,textures:[]};
    }else{
      const [{GLTFLoader},{MeshoptDecoder}]=await withDeadline(Promise.all([import('three/addons/loaders/GLTFLoader.js'),import('meshoptimizer')]),
60000,'The 3D viewer could not download. Choose a model to retry, or use Photos.');
      const loader=new GLTFLoader().setMeshoptDecoder(MeshoptDecoder);
      let abandoned=false;
      const parsing=loader.parseAsync(buffer,'').then(gltf=>{if(abandoned||!loads.current(generation)){disposeModel(gltf.scene);throw new DOMExcept
ion('Superseded','AbortError');}return gltf;});
      let gltf;
      try{gltf=await withDeadline(parsing,selection.prepareTimeout,'The browser could not prepare this 3D view. Retry a lightweight view or use Pho
tos.');}
      catch(error){abandoned=true;throw error;}
      buffer=null;
      if(!loads.current(generation)){disposeModel(gltf.scene);return;}
      model=gltf.scene;scene.add(model);modelDiagnostics=inspectModel(model);
      if(selection.lite&&(modelDiagnostics.geometryMiB>selection.geometryLimit||modelDiagnostics.textures.some(([w,h])=>w>selection.textureLimit||h
>selection.textureLimit)))throw Error('The lightweight view exceeded its graphics budget. Please use Photos and report this message.');
    }
    ready=true;home();if(!nativeGaussian&&(selection.lite||params.get('mode')==='orbit'))setMode('orbit');resize();
```

room-viewer.js (continued)

```
    setPhase('drawing','Drawing the room\u2026');$('load-progress').value=98;
    await withDeadline(renderer.compileAsync(scene,camera),30000,'The browser could not finish drawing the room. Retry the phone view or use Photos
.');
    if(!loads.current(generation))return;
    if(nativeGaussian){
      const preparingSpark=spark;
      render(); // Establish the viewport before the first LoD/sort update.
      await waitForNativeDraw({
        signal:request.controller.signal,isCurrent:()=>loads.current(generation),
        update:async()=>{if(renderer.getContext().isContextLost())throw Error('3D graphics were interrupted.');scene.updateMatrixWorld(true);camera
.updateMatrixWorld(true);await preparingSpark.update({scene,camera});},
        draw:render,activeSplats:()=>preparingSpark.activeSplats,
      });
      preparingSpark.autoUpdate=true;
    }else render();
    if(renderer.getContext().isContextLost())throw Error('3D graphics were interrupted. Please reload the phone view.');
    preparedMs=performance.now()-loadStarted;phase='ready';controlsEnabled(true);$('loading').hidden=true;$('load-progress').value=100;
    $('stats').textContent=`${asset.label} \xb7 ${nativeGaussian?(modelDiagnostics.splats/1e6).toFixed(2)+'M splats':(modelDiagnostics.triangles/1e
6).toFixed(2)+'M triangles'} \xb7 ${(asset.fileBytes/1e6).toFixed(1)} MB`;
    $('quality-note').textContent=asset.note||'Approximate scale \xb7 reconstruction defects remain';
    updateMode();invalidate();updateDiagnostics();
  }catch(error){if(!loads.current(generation))return;disposeModel(model);model=null;showFailure(error);}
```

E14. Explicit model choice and generation cancellation - complete

review/production/choice-core.js | Complete file
SHA-256 17594a4bb57b6d9e2922b6cee372ea21459944811f66aba9f5ad531e19019d6f

choice-core.js

```
export function canonicalChoice(value){
  const aliases={'iphone-fast':'phone','iphone-full':'native','osmo-lite':'osmo'};
  value=aliases[value]||value;return ['phone','native','osmo'].includes(value)?value:null;
}

// Selecting a radio or arriving through a deep link is not download consent.
// Only open() may launch the viewer, once, after the explicit Open model press.
export function createChoiceGate({preselected=null,onLaunch=()=>{}}={}){
  let selected=canonicalChoice(preselected),launched=false;
  return {
    select(value){if(launched)return false;selected=canonicalChoice(value);return Boolean(selected);},
    open(){if(launched||!selected)return null;launched=true;onLaunch(selected);return selected;},
    state(){return {selected,launched};}
  };
```

choice-core.js (continued)

```
}

export function createLoadLifecycle(){
  let id=0,controller=null;
  return {
    begin(){controller?.abort();controller=new AbortController();return {id:++id,controller};},
  // A timeout aborts the current request but must still show its failure.
  // Supersession/cancel invalidate the token separately from AbortSignal.
  current(token){return token===id;},
    cancel(){controller?.abort();id++;}
  };
}
```

E15. Landing-page resource gate - complete

review/production/chooser.js | Complete file
SHA-256 aaad9e4e20359e2a1d7223e284f772f8c6f57fcf4d6ad585a04bcca02340673d

chooser.js

```
import {canonicalChoice,createChoiceGate} from './choice-core.js';
const $=id=>document.getElementById(id),params=new URLSearchParams(location.search);
const choices=[...document.querySelectorAll('input[name="model-choice"]')];
const initial=canonicalChoice(params.get('quality'));
let timer,selected=null;
const returnToChoices=()=>{const url=new URL(location.pathname,location.origin);if(selected)url.searchParams.set('quality',selected);url.searchPara
ms.set('v','20261031-native-provisional');location.assign(url.href);};
const fail=()=>{
  if(document.body.dataset.phase)return;
  clearTimeout(timer);document.body.dataset.phase='failed';$('load-title').textContent='The viewer could not start';
  $('load-status').textContent='Please check the connection, then choose a model again.';
  $('load-detail').textContent='No model download has started.';$('load-progress').hidden=true;
  $('retry').hidden=false;$('retry').textContent='Choose a model';$('loading-photos').hidden=true;
};
const gate=createChoiceGate({preselected:initial,onLaunch:key=>{
  selected=key;document.body.dataset.selectedModel=key;
  $('open-model').disabled=true;$('model-chooser').hidden=true;$('viewer-app').hidden=false;
  $('load-title').textContent='Opening '+({phone:'iPhone Fast',native:'iPhone Full',osmo:'Osmo'}[key]);
  $('load-status').textContent='Starting the viewer\u2026';$('load-detail').textContent='';
  $('retry').onclick=returnToChoices;$('cancel-load').onclick=returnToChoices;$('change-model').onclick=returnToChoices;
  $('loading-photos').disabled=true;$('photos').disabled=true;
  timer=setTimeout(fail,20000);
  const script=document.createElement('script');script.src='__BUNDLE__';script.defer=true;script.type='module';
  script.onerror=fail;document.body.appendChild(script);
```

chooser.js (continued)

```
}});
for(const input of choices){
  input.checked=input.value===initial;
  input.addEventListener('change',()=>{gate.select(input.value);$('open-model').disabled=!gate.state().selected;$('choice-status').textContent='Pre
ss Open model to start the selected download.';});
}
$('open-model').disabled=!initial;$('open-model').addEventListener('click',()=>gate.open());
if(initial)$('choice-status').textContent='Your linked model is selected. Press Open model to start its download.';
window.addEventListener('pagehide',()=>clearTimeout(timer));
```

Appendix F. Evidence, graph coverage and fingerprints

The report is self-contained; the following catalogue identifies the local artifacts behind its claims and figures. Hashes bind the inspected files. Original-media checksums mentioned in the narrative are retained import/provenance records, not a claim that the 1.16 GB OSV was newly rehashed during this report. Image crops and comparison sheets are derived, labelled diagnostic figures; underlying source/render hashes are catalogued.

Graphify inspected a curated 30-file code snapshot and produced 212 nodes, 385 graph edges and 25 communities. The initial Windows-stdin multiprocessing extraction was discarded and superseded by serial extraction. Graph health reports 166 dangling edges and 9 same-endpoint relation collapses. Consequently it is an incomplete static navigation aid, not a runtime proof. Direct source lines and recorded execution artifacts support the branch findings. Its AST pass used 0 external-LLM input/output tokens; that does not describe the token cost of this chat or report authoring. [E29]

Useful code groups: navigation modes (setMode, home, updateSurface), camera framing (fittedOrbitRadius, orbitCamera, resize), load lifecycle (loadAsset, controlsEnabled, setPhase), delivery policy (nativePolicy) and explicit choice (createChoiceGate). A static connection from loadAsset to these functions does not imply that the mesh and Gaussian branches execute the same behavior.

E01 | user-latest-osmo.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-ux-report-20261031\assets\user-latest-osmo.png
3,199,748 bytes | SHA-256 f9e588d59a2c7fda3ba784056cfca2624f5cff1f93fe5ba2eba029db6378726a

E02 | room-be410c75b8.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\production\public\room-be410c75b8.json
9,794 bytes | SHA-256 be410c75b8e4506e77bc936a8be542f66d85d38cfa0051abaf71d3754245738f

E03 | room-viewer.js
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\production\room-viewer.js
32,801 bytes | SHA-256 35b8e275baa7e40193d2a75810bb78b4a99df1a0d62af95f708b01cda8a11d64

E04 | native-delivery.js
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\production\native-delivery.js
2,802 bytes | SHA-256 85a419a301c77e1e3a236f0e39de09c198c1d391727c570426fefef7dde68fa0

E05 | clean-cameras.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-cameras.json
6,811 bytes | SHA-256 6c6c525abecb5b4cc63790686c274a895e84b9348117bf3b4a68b76fc21a6051

E06 | export-clean-cameras.py
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\export-clean-cameras.py
10,033 bytes | SHA-256 86196893006b5d16d65ebf814fa1ceebcf84888d3a7c394b21f2040c1a0bc0ff

E07 | config.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_OSMO360\clean-v3\explicit-full-v1\training\osmo-explicit-depth-v1\config.json
8,866 bytes | SHA-256 9603e38a6512b50c1a85e720e0e34f93867c31ebc2e4ba8202ba7676aa98df05

E08 | explicit-full-train-job.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\explicit-full-train-job.json
4,091 bytes | SHA-256 5df07fd8881af6c21934f4e9f5578365f4d1f9fac8ad42fd6a53c3ed81452cff

E09 | explicit-full-geometry-job.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\explicit-full-geometry-job.json
1,804 bytes | SHA-256 4d64e09c11b2ee1ce2914ae13afd1bf2c2562ffccc5cdeef75f8dbb8ab350d74

E10 | explicit-pinhole-audit.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_OSMO360\clean-v3\explicit-full-v1\dataset\explicit-pinhole-audit.json
505,733 bytes | SHA-256 12d341f179d4573da8cb21a369f48237e916a04d14b9beed2bacdf861c8cc008

E11 | depth-scale-audit.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_OSMO360\clean-v3\explicit-full-v1\dataset\depth-scale-audit.json
1,150,469 bytes | SHA-256 dc49c97d8fc0b41e644ea6d85604e1c6218a4ef387c0cac083837cdcc92a72f3

E12 | trusted-depth-and-seed-audit.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_OSMO360\clean-v3\explicit-full-v1\dataset\trusted-depth-and-seed-audit.json
246,702 bytes | SHA-256 5c4d0f88a8ba202481a0e891e571eb59536c705b9ad4f5ea5200af3a668ab640

E13 | fresh-sfm-preparation-audit.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\geometry\fresh-sfm-preparation-audit.json
165,557 bytes | SHA-256 71eb80d205038bab263d4c46398d92218d40e522c10a75ae95b3b029a408cb37

E14 | frame-selection.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\frame-selection.json
272,815 bytes | SHA-256 bf38c1a17a2aa5eb49a54b529d613797b3ba68ed1042d6c8bf48b9df4412f37c

E15 | masks-ready.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\masking\masks-ready.json
134 bytes | SHA-256 b189aba9f459495899343df4b1410bee21366b334afab902cfb01d1e9dc1c9ef

E16 | clean-matched-review.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-matched-review.json
8,528 bytes | SHA-256 bb75e4c7341b52569f20fc16b36e7ac32e9316dea2691ce7540fd9f2f95da283

E17 | valid-native-evaluation.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_OSMO360\clean-v3\explicit-full-v1\analysis\valid-native-evaluation.json
256,426 bytes | SHA-256 dfa32376aee74318cbf9845dbe7955fb8a2144af449d037b70a6c07da65d28e4

E18 | clean-final.sog.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-final.sog.json
2,500 bytes | SHA-256 731ad1685cfc7d8d895d8aa14511e6a9e600a3d498f31351d50682f838fccbf4

E19 | full-final-review.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\full-final-review.json
2,218 bytes | SHA-256 26b4800f9c28fd02464d3609b61fe0362472109598589568e044390f9d5cb26c

E20 | public-browser-verification.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\production\deployment\public-browser-verification.json
2,976 bytes | SHA-256 13ac485fe66c29a325d75392b8f7679cf55abedd5ad07d638c5bd34d33634820

E21 | final-browser-verification.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\production\final-browser-verification.json
2,668 bytes | SHA-256 1c082ec198676bca5633ca66ff6891e519c019e15dc38b60f0b65b5b7bcce505

E22 | release-approval.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\production\release-approval.json
3,075 bytes | SHA-256 b5759a9ec8a0dd36dd304c3d069d14078f3f1b21e957cdba0147edfcebaa2a93

E23 | phone-asset-report.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\mobile-20261029\phone-asset-report.json
973 bytes | SHA-256 a5ad6d7f37fed36d0be95addf07ee089243ee970458f240c4af59fcf3233ba8d

E24 | allviews-raw-verification.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261027_IPHONE_WIDE\recovery-v2\analysis\allviews-raw-verification.json
8,023 bytes | SHA-256 1affbf77d03352bcee9728e38245e8c75e5c82a4a25b5a73a50c0495c0e75e7a

E25 | all-usable-coverage.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261027_IPHONE_WIDE\recovery-v2\analysis\all-usable-coverage.json

62,364 bytes | SHA-256 0c7f3caab8e2749787cf6925c176665e57e6c72f1002ef60ced54c9a02151f95

E26 | Rumah_Full_Height_Verification_2026-10-29.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\outputs\Rumah_Full_Height_Verification_2026-10-29.json
7,224 bytes | SHA-256 b3223b89da3c3141533cd87a70985c5cc3c3f6a9dd7436532ee3526ee8b03eb4

E27 | build_recovery_report.py
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\recovery-report\build_recovery_report.py
27,799 bytes | SHA-256 68d10cc6c6e0dbc2d7adaf29b98d82699fd837dbb59c48f5d23035d14d2fcba9

E28 | manifest.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\report\reference_pack\manifest.json
34,463 bytes | SHA-256 b966c04ee0ed64c51f915b60378baadb507ce6edc5274cb766b8efb26ec3b801

E29 | summary.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-ux-report-20261031\graphify-out\summary.json
6,222 bytes | SHA-256 639f4614218b5d39f1de337f2d772d7c262af1631adf8795862fb61926d3f0de

E30 | bath-pilot-review.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\bath-pilot-review.json
1,264 bytes | SHA-256 fedcf0c3156d9a6a671b0cedb6b156a6a0c3c6f5edac02f62155663e08667414

E31 | visibility-experiment-result.json
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\visibility-experiment-result.json
484 bytes | SHA-256 ca34ea755147cf32880dbc09968a5573f2b653235db46584486258e00c16ab3e

E32 | Rumah_Full_Height_2026-10-29.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\outputs\Rumah_Full_Height_2026-10-29.jpg
56,581 bytes | SHA-256 642605a837651ef0ec2deaafd53177977132aa9cf64dd8a2f1292a1da5ce908c

E33 | Rumah_Phone_View_2026-10-29.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\outputs\Rumah_Phone_View_2026-10-29.jpg
22,648 bytes | SHA-256 38e8ad0e67021d5bb465daca244e21c8eee0c32cb96529aead5c42dbec7f9fce

E34 | iPhone_Wide_Rebuilt_Overview.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\outputs\iPhone_Wide_Rebuilt_Overview.png
817,176 bytes | SHA-256 7d0b90f245adc8726ab2bdbc4b45760ba172c9b37c72a0ab54fa94ae9ac2c20b

E35 | F04-codex-nikon-034.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\report\reference_pack\images\F04-codex-nikon-034.png
1,005,331 bytes | SHA-256 0f572436ca48a2b637bcc1ad3a5fe960ef601a1bed3da232d1c62398e23144f2

E36 | F03-samsung-accepted.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\report\reference_pack\images\F03-samsung-accepted.png
83,432 bytes | SHA-256 754927e828fb747f2b39d48e0719621f3a820db68781c09220839f7c233aef74

E37 | full-rebuild-bathroom-before-after.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\full-rebuild-bathroom-before-after.jpg
201,365 bytes | SHA-256 291e4422589990079ad470ed4af46935fad8af1b80ed4fde091c62e433b0a491

E38 | full-eval-six-worst-valid.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\full-eval-six-worst-valid.jpg
534,782 bytes | SHA-256 a367b7f31cc1c0bf43edd1101cf447756914143fd4352451f18776f3a53bf17a

E39 | full-native-eval-zero.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\full-native-eval-zero.jpg
41,141 bytes | SHA-256 4e6ea41919513855c06ac38ad6cbe919a9fb20f32b75bd61befa700624a756e3

E40 | source-qa-seven.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\source-qa-seven.jpg

141,957 bytes | SHA-256 5f8657027bbef63bef35d7888f5afdbd67fd1aeb9ff5cc0ab23051403276240e

E41 | fresh-sfm-preparation-audit-continuity.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\geometry\fresh-sfm-preparation-audit-continuity.png
126,552 bytes | SHA-256 5d0dd6d305ca62132de583a25a4794f4b8f0225cbeed9cac396903bd72999e04

E42 | second-mirror-cam1-outline-review.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\geometry\second-mirror-cam1-outline-review.jpg
265,686 bytes | SHA-256 4f962e12f6c0a0e04192fc063e50b5f9904c42896380b170f437b12a1ce7198a

E43 | fresh-01-cam0-frame_000027_train.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-cameras-photos\fresh-01-cam0-frame_000027_train.jpg
229,008 bytes | SHA-256 9e81f90b1ecf0a1df6ef9db87526c5caafafea7d1b0d7c1efcc283c95ff72e50

E44 | clean-fullNative-pose-1.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullNative-pose-1.png
766,402 bytes | SHA-256 eb9feb81ccb30267141bdf312f437652021d81e0b3d27e8deab09ccd37fa1b78

E45 | clean-fullSog-pose-1.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullSog-pose-1.png
765,958 bytes | SHA-256 cf2cfbe3bcbf5a79648f1db720ec13dfd69b493a7873ded38bc95ebc2b996606

E46 | fresh-02-cam0-frame_000357_train.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-cameras-photos\fresh-02-cam0-frame_000357_train.jpg
150,097 bytes | SHA-256 7ba9efc2de120c0ee802b34f475f6e8ffa8c562832e3a26012da935f3eaec136

E47 | clean-fullNative-pose-2.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullNative-pose-2.png
691,362 bytes | SHA-256 b16f4ec42adf3268501e55b3dfd4f339ed5ff07d13906654d7941f83edeee02f

E48 | clean-fullSog-pose-2.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullSog-pose-2.png
689,828 bytes | SHA-256 2e90125049a5db2c3b51516777a70f172a1bc764b362804acae639f3e5ae57d1

E49 | fresh-03-cam1-frame_000600_train.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-cameras-photos\fresh-03-cam1-frame_000600_train.jpg
201,368 bytes | SHA-256 ac531ed57965e2e98da36cea69c27c541854471e99ee5e2919fe26d7f351b7d0

E50 | clean-fullNative-pose-3.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullNative-pose-3.png
695,208 bytes | SHA-256 e50d1522f6362c29ec5563965d4aaae72044489bba92cc65c7c203f3a6f87535

E51 | clean-fullSog-pose-3.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullSog-pose-3.png
694,937 bytes | SHA-256 52340c64695e385375659614aa819035db2192147ee7b4c18424a93e93176ccc

E52 | fresh-04-cam0-frame_000750_eval.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-cameras-photos\fresh-04-cam0-frame_000750_eval.jpg
204,651 bytes | SHA-256 7fd2d87684d829d07fa14ea36e767a151626260a5fc60ba454bdb0bbc7a30d40

E53 | clean-fullNative-pose-4.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullNative-pose-4.png
659,483 bytes | SHA-256 ef365d12a81a62e736a3950def7a779c597e7ce88b8e2db2c177045176558fd9

E54 | clean-fullSog-pose-4.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullSog-pose-4.png

657,230 bytes | SHA-256 f78aecd78c36e3fb97e9cc2c3366e0daf6d59a6f6f50626ac428f1d593c9df21

E55 | fresh-05-cam0-frame_001197_train.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-cameras-photos\fresh-05-cam0-frame_001197_train.jpg
216,164 bytes | SHA-256 6b6a89da28a0c164aea8c5892f085db604a1b8eacc44e6c8023c898b4eab1e6f

E56 | clean-fullNative-pose-5.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullNative-pose-5.png
744,368 bytes | SHA-256 277c3cf6debc35b34d0124b25f58438adfa5ac575ef9038ca97684844e908487

E57 | clean-fullSog-pose-5.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullSog-pose-5.png
743,850 bytes | SHA-256 a3552383c4908a43815a4a5640cd59c22f422418d29e1222c98ecd86fd518c74

E58 | fresh-06-cam0-frame_001557_eval.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-cameras-photos\fresh-06-cam0-frame_001557_eval.jpg
217,098 bytes | SHA-256 961d8b109cc151d9f64cd45eb56bc553d39d21a98185abb9c8dcc7a70df69c2b

E59 | clean-fullNative-pose-6.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullNative-pose-6.png
723,963 bytes | SHA-256 77683f704d6d21c8183ff64719d735119ce180b6dace4396640a45816171d18e

E60 | clean-fullSog-pose-6.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullSog-pose-6.png
722,839 bytes | SHA-256 0fc30bb8334e9797648a134d6d5a70f7c685f7f418268e0925d00558c5ba0763

E61 | fresh-07-cam0-frame_001800_train.jpg
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\clean-cameras-photos\fresh-07-cam0-frame_001800_train.jpg
239,869 bytes | SHA-256 006bb010285d48fc68e20065dcd969cdabb07e07d298a71efd5691caab8bbeb4

E62 | clean-fullNative-pose-7.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullNative-pose-7.png
771,127 bytes | SHA-256 a7e7d9207cd55e172eb37862ed5b7341242ff75570b0df736fa828d9a5c6f877

E63 | clean-fullSog-pose-7.png
C:\Users\bombe\Documents\Codex\2026-10-27\files-pasted-by-the-user-system\work\osmo-clean-20261031\review\renders\clean-fullSog-pose-7.png
770,560 bytes | SHA-256 933a35d3949f9d7457843a165a1d5807fcaea891cf334a389ff7f419b43954d5

E64 | cameras.txt
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_OSMO360\clean-v3\dataset\sparse\0\cameras.txt
375 bytes | SHA-256 7bdaf6ef05f03429e7a0da908f58fe79e2b7f42ddfefb683069354fd09df912b

E65 | metrics.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_OSMO360\clean-v3\explicit-full-v1\training\osmo-explicit-depth-v1\metrics.json
11,202 bytes | SHA-256 3a12c3837304a0ccbebff26e6d851238f52f83b984123fb84b9641d581e64377

E66 | input-verification.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_OSMO360\clean-v3\explicit-full-v1\analysis\input-verification.json
1,194 bytes | SHA-256 46d0544852214029082298678 5d2e74ff113ca17fa8338cb21e7f0555309f65f

E67 | scene_transform.json
C:\Users\bombe\OneDrive\Documents\ChatGPT\Ruang Kamare\roomscan-reconstruction\data\20261031_OSMO360\clean-v3\explicit-full-v1\training\osmo-explicit-depth-v1\scene_transform.json
5,447 bytes | SHA-256 a51d0dcad4fd7e6763b3884bbca0ecbba3001a8191ea25f5def8f2b2f2243920

Appendix G. Primary technical references

R1. Spark: Splat RGBA-XYZ SDF editing
https://sparkjs.dev/docs/splat-editing/
Documents region-based per-splat RGBA edits, including plane and box SDFs. Supports a potential cutaway mechanism; it does not supply a finished architectural dollhouse or validate our model.

R2. Spark: Level-of-Detail
https://sparkjs.dev/docs/lod-getting-started/
Documents LoD detail/performance trade-offs, packed versus extended storage, prebuilt trees and paged RAD. Platform guidance is a starting point, not a measurement on this user's phones.

R3. 2D Gaussian Splatting for Geometrically Accurate Radiance Fields, SIGGRAPH 2024
https://surfsplatting.github.io/
Surface-oriented Gaussian disks and geometric consistency are relevant to a proxy/mesh pilot. This method was not executed for the current release.

R4. Mip-Splatting: Alias-free 3D Gaussian Splatting, CVPR 2024
https://niujinshuchong.github.io/mip-splatting/
Explains smoothing and Mip filtering trade-offs under changes of image sampling. It does not prove that filtering is the dominant source of this model's blur.

R5. 3D Gaussian Splatting for Real-Time Radiance Field Rendering, SIGGRAPH 2023
https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/
Primary context for sparse-point initialization, anisotropic Gaussian optimization and radiance-field rendering; it is not evidence that a textured architectural mesh or cutaway was built here.

References checked 31 October 2026. External references support mechanisms and proposed experiments. Facts about the actual reconstruction and viewer come from local code, recorded settings, hashes, images and test results.

Terminology used in this report

Term	Meaning here
Dollhouse	An outside, whole-room architectural overview with a usable ceiling/exterior cutaway.
Gaussian / splat	An optimized anisotropic opacity/colour primitive, not a triangle face.
LoD	Runtime selection of a lower-detail representation within a draw budget.
SfM	Camera pose and sparse-point reconstruction from overlapping images.
Depth prior	A learned geometry estimate used as guidance; not measured LiDAR or surveyed truth.
Cold start	New optimization from an audited seed rather than a contaminated Gaussian checkpoint.
PSNR	Pixel-error metric; does not by itself establish sharpness, geometry quality or control usability.
Held out	Excluded from Gaussian appearance training/guidance; the same video may still contribute to SfM camera estimation.

Final assessment: technical delivery is working; product-quality acceptance failed. The evidence supports a viewer-parity correction and controlled blur diagnosis before another full reconstruction cycle.